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Cognitive Impact of Generative AI as a Scaffolding Tool for Novice Designers in Product-Service System Design

Leqing Cui
Wenhui Guo

Supervisor: Tomohiko Sakao

Examiner: Renee Wever

Linköping University
SE-581 83 Linköping, Sweden
+46 013 28 10 00, www.liu.se

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Abstract

Generative AI (GenAI) has been widely applied to education and design practice in recent years, especially using it as a scaffolding tool to achieve learning goals. However, generic GenAI usually lacks domain expertise knowledge and the structure to support the design's cognitive thinking process, thus making it difficult to effectively scaffold novice designers to complete complex Product-Service System (PSS) Design task.

Based on this gap, this research seeks to investigate how different level GenAI tools can affect novice designers' learning, design cognition, and cognitive load in PSS design.

This study builds a research framework based on the PSS Design Schema and protocol analysis methods. We developed three GenAI tools with different levels of scaffolding support with domain-specific knowledge. Novice designers were invited to complete PSS design tasks, with their design thinking process data collected through type-aloud protocols. The study evaluated learning effects using Bloom's taxonomy, examined novice design cognition through protocol coding analysis (based on FBS, HSA, and Markov transition analysis), and assessed cognitive load using the NASA Task Load Index (TLX).

This study suggested that structured scaffolding in GenAI tools can meaningfully enhance novice designers' learning and design cognition in PSS design and that the pedagogical design of tool instructions matters more than the expert knowledge supplied.

Key Words: Generative AI, Product-Service System (PSS) Design, Design Cognition, Protocol Analysis, Cognitive Scaffolding, Human-AI Collaboration

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Wenhui Guo

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1. Introduction

1.1. Background

Currently, the manufacturing industry is undergoing a fundamental paradigm shift from focusing on the production of independent tangible products to adopting a product-service system (PSS) characterized by the integration of products and services as the value delivery entity (Neramballi et al. 2025; Doualle et al., 2016). To understand this transformation, it is essential to examine its underlying drivers and the broader industrial context in which it occurs. This transformation is driven by two main factors: first, the inherent demand for continuously enhancing customer value throughout the entire lifecycle; second, the external pressure to address increasingly severe systemic challenges such as social equity, resource constraints, and ecological sustainability (Martínez et al., 2026).

This paradigm evolution is highly coordinated with the rise of the Industry 5.0 concept, which not merely represents an extension of linear evolution logic of Industrial 1.0 to 4.0 but rather a shift towards a more integrated, adaptive, and purpose-driven new industrial paradigm (Martínez et al., 2026). Industry 5.0 is based on three core pillars: human-centeredness, system resilience, and ecological sustainability (Martínez et al., 2026), redefining the role of technology, it no longer merely serves to optimize efficiency, but becomes an enabling medium that empowers the deep synergy between human high-level cognitive abilities and machine precise execution capabilities, ultimately achieving a symbiotic collaborative relationship of complementary human-machine advantages and shared responsibility (Martínez et al., 2026).

Within this context of paradigm shift, PSS maximizes systemic value creation, compelling designers to address complex social and environmental challenges by fundamentally rethinking the design object, design process, and intended value (Neramballi et al. 2025; Dorst 2011). This comprehensive rethinking, however, presents significant cognitive challenges that distinguish PSS design from traditional product design. Specifically, PSS design is characterized by inherent dynamic complexity, stemming from the systemic integration of tangible product elements with heterogeneous, often uncertain human service activities (Neramballi et al. 2025). Consequently, designers require a cognitive shift from static object definition to dynamic system modeling.

Given this cognitive complexity, scholars have increasingly explored how Generative AI (GenAI) might support designers in navigating the challenges. However, the limited knowledge on how to use Generative AI in domain-specific design is becoming increasingly apparent. Specifically, three interconnected challenges have emerged:

First, Gen AI Tools often lack contextual accuracy, regulatory alignment, and domain awareness (Shukla, 2025), and provide no structured methodological guidance or comprehensive evaluation framework. Furthermore, Gen AI Tool fails to keep balance between AI assistance and active learner engagement in design education (Hara et al. 2025)

Second, reliance on Generative AI presents a comfort-growth paradox (Riva 2025). Because these tools provide frictionless answers without enforcing a structured design logic, they may induce cognitive offloading (Riva 2025) or copy-paster behavior (Musazade et al., 2025) rather than fostering the internalization of design skills. Consequently, Generative AI may serve as a crutch that masks competence gaps rather than a scaffold that bridges them.

Third, and most fundamentally, Cui et al. (2025) highlighted a key limitation in existing Human–AI collaboration paradigms, current interfaces predominantly deliver or suggest domain-specific knowledge passively, rather than fostering user-driven reflection through guided, interactive scaffolding.

1.2. Objective

To Address these limitations requires a fundamental shift in how we study AI-assisted design. Current research on general and domain-specific AI-assisted design predominantly focuses on posterior evaluations of outcomes, relying on incremental improvements in final design metrics to validate tool effectiveness. However, such outcome-based assessments fail to capture the profound impact of AI on design cognition, for example, how GenAI tools shape the cognitive processes and behavioral strategies designers use while engaging in design tasks.

To address this gap, this study shifts toward a process-oriented empirical approach, focusing on how GenAI tool as a cognitive scaffold to support novice designers in constructing solutions within dynamic design workflows. By acquiring empirical observation and quantitative tracking of design behaviors, this research aims to uncover GenAI's influence on cognition, thereby addressing the existing gap in human-AI collaborative process research.

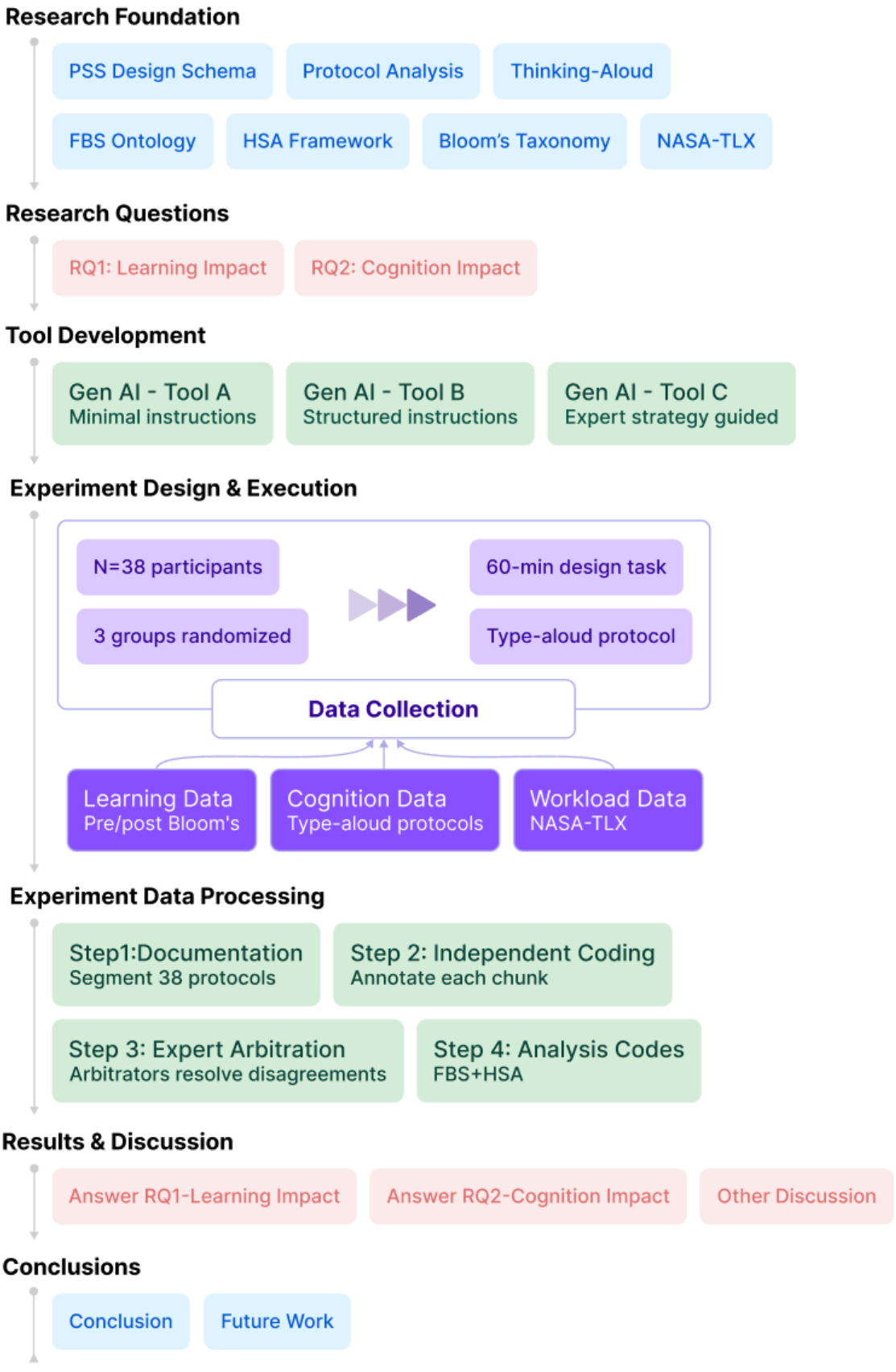
Despite growing interest in AI-assisted design education, a critical question remains unanswered: It currently remains unclear whether domain-specific AI is more effective than generic GenAI at guiding novice designers to overcome the dynamic complexity of PSS. While generic tools often provide unconstrained suggestions, novice designers can easily fall into a depth-first strategy trap. Consequently, this study aims to evaluate the extent to which domain-specific AI tools scaffold the learning and cognition of novice designers in Product-Service System (PSS) design. Specifically, it sheds light on the influences of these tools at the design process level.

1.3. Research Questions

RQ1: How does a domain-specific generative AI tool influence novice designers' learning in product–service system (PSS) design for sustainability?

RQ2: How does a domain-specific generative AI tool influence novice designers' design cognition in product-service system (PSS) design for sustainability?

1.4. Research Blueprint



2. Literature Review

2.1. PSS Design

Product-Service Systems (PSSs) are integrated systems that combine tangible products with intangible service activities to jointly fulfill customer needs (Tukker and Tischner, 2006). Empirical research highlights that due to a lack of domain-specific experience, designers perceive significantly higher uncertainty within the PSS solution space (Zhao et al.,2025). This uncertainty stems from the absence of internalized systemic schemas; the mental models required to navigate the integrated complexity of PSS design (Neramballi et al., 2025).

As a result, designers, even those with substantial product design experience often employ subconscious coping mechanisms to manage this cognitive overload. Specifically, they fall into what Ball et al. (2010) called detail traps, Instead of maintaining the necessary breadth-first systemic view, they tend to default to a depth-first search strategy, prematurely fixating specific structural elements or design details (Neramballi et al. 2025). This behavioral pattern shows a fundamental cognitive limitation. Sankar & Sen (2024) further defined major idea bottlenecks that impede novice creativity, which shows a lack of experience that prevents the generation of in-depth solutions, and fixed mindsets of limitations. This behavioral pattern shows that both novices and even experienced product designers lack the internalized systemic schemas required for PSS.

Consequently, they require dedicated support to overcome these cognitive thresholds and effectively navigate the high-level system of abstractions essential for successful PSS conceptualization (Neramballi et al. 2025). To address this need, Zhao et al. (2025) demonstrated that AI-assisted processes significantly improve the coverage of core interdisciplinary frameworks, essentially promoting a breadth-first strategy and preventing designers from early fixation on isolated elements, indicating that AI intervention can effectively counter such tendency by enhancing the completeness of design outcomes.

Addressing these high-level cognition challenges, the rapid rise of Generative Artificial Intelligence (GenAI) is significantly transforming design education (Hara et al. 2025; Siu et al. 2025). Large Language Models (LLMs) such as GPT-4 are no longer merely passive supplemental tools but can provide personalized feedback and guiding structured learning workflows (Hara et al. 2025). This technological evolution presents a unique opportunity for the modernization of PSS design education: by integrating AI as a cognition scaffold, it can provide cognitive support to designers when handling system modeling and function definition, thereby helping novice designers bridge the experience gap that exists in the

PSS design process (Neramballi et al. 2025; Riva 2025; Stojković et al. 2025). This transition marks the evolution from passive ideation where the creative burden rests solely on the designer's mind, to an active ideation, where conversational AI serves as a scaffold, stimulating more systemic thinking (Sankar & Sen, 2024).

2.2. GenAI Tool

In contrast to generic GenAI, researchers have proposed domain-specific AI architectures enhanced through Retrieval-Augmented Generation (RAG) (Tian et al., 2025; Shukla, 2025) or domain knowledge integration such as Design for Environment checklists (Hara et al. 2025). Wang et al. (2025) proposed an LLM-KG enhancement mechanism that integrates domain-specific Knowledge Graphs with LLMs, which significantly increases the accuracy and logical consistency of design synthesis. Similarly, the EcoBlend framework (Alcaide-Marzal et al., 2025) empirically supports the potential of domain-specific AI to facilitate systematic thinking, effectively scaffolding novice designers to overcome common obstacles in the Ecodesign process. Furthermore, evaluations of the AI4Design system (Abrusci et al., 2025) suggest that within the iterative design process, domain-specific Generative AI can effectively serve as a co-creator for students, which assists novice designers in making more informed decisions.

Theoretically, these systems can reduce degrees of freedom (Siu et al., 2025), a core function of scaffolding, to guide novices toward systematic thinking. To demonstrate this potential empirically, Cui et al. (2025) introduced a large language model–driven retrieval-augmented generation (RAG) framework tailored for bio-inspired design ideation. This approach systematically bridges insights from biology and engineering to support solution development across disciplinary boundaries. Their experimental evaluation yielded promising outcomes: participants leveraging the cross-domain LLM system exhibited notably enhanced creative output and functional efficacy compared to those using conventional engineering design practices. In parallel, Liu et al. (2025) proposed a novel comprehension-driven generation architecture, which organizes mechanical domain knowledge systematically and archives it in a structured database, thereby building a specialized knowledge infrastructure tailored for mechanical design tasks. The authors emphasized that this approach significantly lowers the entry barrier for novice users while accelerating the learning process.

2.3. Use of GenAI Tools in design

Recent studies have investigated the integration of LLMs into the design process, focusing on improving concept generation. A growing number of empirical research has begun to examine how LLMs can support ideation and concept in design contexts, with studies

employing a range of evaluation methodologies that reflect the outcomes and processes of human–AI collaboration.

To address specific challenges in this integration, Jiang et al. (2025) tackled the issue of generative errors and unmanufacturable concepts by developing frameworks to adapt general-purpose LLMs for specific industrial tasks, such as hair dryer design. Their evaluation used the Consensual Assessment Technique (CAT), where independent experts rated concepts on a 0–2 scale across feasibility, novelty, and usefulness. Data analysis included average scores and standard deviations to assess quality and consistency, alongside Cohen's Kappa to determine inter-rater reliability.

Building on this approach, Abrusci et al. (2025) extended this evaluation framework to an educational design context, examining the impact of the AI4Design platform. Unlike short-term experiments, this study used a baseline of previous manual projects. Evaluation incorporated multiple layers: a blind review by a design professor assessed pre-and post-AI outcomes using a creativity rubric covering concept maturity, novelty, and originality; system usability was measured with the System Usability Scale (SUS); and user satisfaction was captured via a five-item Likert-scale questionnaire. Qualitative feedback gathered through open-ended questions was thematically categorized.

Extending this investigation to explore the nature of human–AI interaction, Wang et al. (2025) explored the role of LLMs as active co-ideation partners. Their Co-Ideator GPT utilized a structured temporal probing sequence (Present - Past - Future) to guide designers through reflective ideation. A total of 203 ideas (102 treatment, 101 control) were evaluated by expert raters on novelty, quantity, variety, and quality using seven-point Likert scales. Post-task surveys and semi-structured interviews with four participants provided qualitative insight into the perceived dynamics of the human–AI collaboration.

Going further, Zhao et al. (2025) conducted an analysis of human-AI dialogue. To prevent cognitive fixation, they used a RAG-based AI (DeepSeek-R1-FW) grounded in expert psychological knowledge. Outcome quality was assessed through blind expert ratings by ten evaluators, five from design and five from psychology, using a five-dimensional, five-point Likert scale covering creativity, feasibility, effectiveness, completeness, and applicability. Moreover, the study went beyond outcome assessment to analyze the collaborative process. Chat logs were coded according to the Dialogue Act Markup in Several Layers (DAMSL) annotation framework, categorizing into four interaction types (guided, discussion, interrogative, and command), which provided a systematic way to quantify the "quality" of the conversation.

2.4. Constructing a Framework of Expert Design Cognition

This framework is derived from a systematic literature review of empirical research on design expertise over the past thirty years. It ultimately identifies 15 dimensions, distinguishing the differences between expert and novice designers in the design process across these dimensions.

This provides a foundation for accurately capturing the interaction between novice designers and GenAI Tools in this subsequent study. The ultimate goal is to accurately pinpoint the cognitive changes experienced by novice designers. This Framework of expert design cognition has two main values: first, it serves as a lexicon for annotating protocol data; second, it serves as a transferable expert knowledge base for GenAI Tool to scaffolding novice's design interaction Table 1.

Table 1: Framework of Design Expertise Cognition

No.	Dimension	Novice Behavior	Expert Behavior	Literature Source
01	Goal Definition	Novices often lack a clear, target-oriented path for knowledge acquisition.	Experts directly approach the goal state during the design process. Additionally, designers maintaining an open goal are significantly more likely to acquire problem-relevant information.	Ho, C. (2001). Moss, J., Kotovsky, K., & Cagan, J. (2007). Kim, M. H., Kim, Y. S., Lee, H. S., & Park, J. A. (2007).
02	Decomposition	Novices adopt an avoidance strategy; when they fail to handle a subsystem or sub-problem, they simply eliminated a problem.	Experts utilize a target-oriented strategy, where they worked backward for required knowledge to deconstruct complex systems.	Ho, C. (2001).
03	Problem Structuring & Framing	Novices tend to focus on the typicality of a problem, passively acting as a "technician" or "facilitator".	Experts use framing to structure or present a problem. They pursue innovation through problem reframing, acting as an "expert/artist" or "collaborator"	Paton, B., & Dorst, K. (2011).
04	Problem-Space Exploration	Novices often get lost in the problem space, acting as blind data gatherers who treat all information equally.	Experts exhibit a strong opportunistic approach, quickly focusing on critical parts, asking for information only when needed, and processing it immediately.	Waldron, M. B., & Waldron, K. J. (1988). Atman, C. J., Chimka, J. R., Bursic, K. M., & Nachtmann, H. L. (1999).
05	Ambiguity Tolerance	Naïve/Novice designers only understand that design problems are individual and passively react to them.	Advanced beginners and competent designers see the bigger picture, tolerate ambiguity, and actively steer the development of the design problem.	Lawson, B., & Dorst, K. (2013). Tollestrup, C., Laursen, L. N., & Vesti, H. N. (2023).
06	Solution-Space Exploration	Novices habituate toward a depth-first search strategy. They are easily affected by	Experts employ a breadth-first search strategy, consider more alternative solutions, and make	Ball, L. J., Evans, J., Dennis, I., & Ormerod, T. C. (1997).

		design fixation, which narrows the solution space where they search for ideas.	more frequent transitions between design steps.	Atman, C. J., Cardella, M. E., Turns, J., & Adams, R. (2005). Linsey, J. S., Tseng, I., Fu, K., Cagan, J., Wood, K. L., & Schunn, C. D. (2010).
07	Functional Analysis	Novices easily fixate on undesirable features when exposed to examples.	Experts bridge the gap between problem and solution spaces via unarticulated jumps between function and form. They frequently build physical models to test functionality.	Dorst, K., & Cross, N. (2001). Viswanathan, V. K., & Linsey, J. S. (2012).
08	Co-evolution	Novices tend to view the problem and solution spaces as linear, sequential steps.	Experts continuously explore potential solutions to reinterpret the problem, allowing the problem and solution to co-evolve.	Dorst, K., & Cross, N. (2001).
09	Abstraction Levels	Novices rely on case-driven analogies driven primarily by specific surface-level cues.	Experts heavily rely on experiential abstract knowledge and utilize abstraction as a vital cognitive method to overcome design fixation.	Ball, L. J., Ormerod, T. C., & Morley, N. J. (2004). Ward, T. B., Patterson, M. J., & Sifonis, C. M. (2004).
10	Managing Complexity	In routine or highly constrained design situations, designers predominantly rely on local loops within single cognitive activities, and on concept reuse, rather than cross-activity global iteration.	In creative design situations, designers engage in significantly more frequent and numerous global iterations — cycling across problem analysis, idea generation, and evaluation — to progressively develop novel solutions.	Jin, Y., & Chusilp, P. (2006).
11	Knowledge Types	Novices possess single-layered knowledge and are easily misled by surface-level cues.	Experts seamlessly integrate conceptual/declarative knowledge, procedural knowledge, and strategic knowledge.	Bernal, M. (2016); Ball, L. J., Evans, J., Dennis, I., & Ormerod, T. C. (1997). Akin, O. (1990); Bernal, M. (2016). Curry, A. (2017); Golja, G., & Schaverien, L. (2007).
12	Action Organization	Novices' cognitive processes generate many concurrent actions that were hard to categorize.	Experts' cognitive actions are highly structured and organized.	Kavakli, M., Sturt, C., & Gero, J. S. (2002).
13	Analogical Strategy & Implicit Recognition	Novices generally ignore distant domain analogies unless they are facing an open goal.	Experts excel at finding the sweet spot analogy balancing near and far distances. In complex situations, they possess an implicit recognition for fluent, intuitive responses.	Tollestrup, C., Laursen, L. N., & Vesti, H. N. (2023). Fu, K., Chan, J., Cagan, J., Kotovsky, K., Schunn, C. D., & Wood, K. L. (2012). Moss, J., Kotovsky, K., & Cagan, J. (2007).
14	Sketching & Visual Representation	Novices demonstrate limited cognitive abilities to draw and recognize sketches.	Experts use sketches not merely as external memory, but to construct design thoughts in a physical setting on the fly.	Kavakli, M., & Gero, J. S. (2001). Suwa, M., Purcell, T., & Gero, J. S. (1998).

				Suwa, M., & Tversky, B. (1997).
15	Skill Acquisition Stages	Novices focus on following the rules of the game, concentrating heavily on design conventions and habits.	Master/Visionary designers are characterized by developing new schema, explicitly aiming at redefining the design field itself.	Cross, N. (2006).

3. Research Method

3.1. Overview

This study is based on four core theoretical components. First, research foundation component as a theoretical PSS Design knowledge, offers a general guideline for the task. The second component examines the implementation of the software program. The third component analyzes the cognitive process. The last component assesses the outcome. These four elements together form an integrated framework. We summarize this framework in *Table 2*.

The following sections are providing detailed explanations. We will delve into each component of the framework and describe how to measure and apply each component in research. This framework ensures that our theoretical ideas can be transformed into practical research methods.

Table 2: Overview of the Integrated Theoretical Framework

Dimension	Theoretical Framework	Role in this Study
Research foundation	PSS Design Schema	Standardizes procedures and ensures systemic PSS design (Sakao & Neramballi 2020).
Data Capture	Protocol Analysis	Externalizes internal thought processes into observable text (Cagan et al. 2013; Noushad et al. 2023).
	Thinking-Aloud	
Cognitive Analysis	FBS Ontology	Measures Problem/Solution focus and design issues/processes(Gero 1990)
	HSA Framework	Tracks system abstraction levels (Gero and Mc Neill 1998).
Outcome Evaluation	Bloom's Taxonomy	Evaluates learning depth (Ajayi, J.2024)
	NASA-TLX	Quantifies cognitive workload (Cao, et al., 2009)

3.2. PSS Design Schema

The experiment will use a specific PSS design method named PSS Design Schema (Sakao et al., 2020) developed to ensure all participants follow a standardized methodology. Based on Neramballi et al. (2025), PSS design is characterized by high dynamic complexity due to the integration of products and heterogeneous service activities.

Table 3: Consolidated Schema for Conceptual PSS Design

Step No.	Ten steps name of PSS
Step 1	Functional unit definition
Step 2	Stakeholder identification
Step 3	Requirement consolidation
Step 4	Value proposition
Step 5	Criterion identification
Step 6	Element integration
Step 7	Balance examination
Step 8	Selecting combinations
Step 9	Evaluating combinations
Step 10	Solution selection

Step 1 involves a functionality-focused conceptualization phase, where designers distill the core design challenge and articulate a functional unit that captures the desired system-level result, rather than specifying specific product-level functions.

Step 2 focuses on mapping the full range of actors within the PSS value network, those who either shape or are impacted by the delivery of the functional unit.

Step 3 synthesizes and organizes the needs and expectations of these actors into a unified, system-oriented requirements framework.

Step 4 examines the opportunities brought about by PSS design. These opportunities benefit participants at two levels. Firstly, individual participants can benefit from them. Secondly, the entire system can also benefit.

Step 5 build an evaluation framework based on the needs of the participants and including a clear value proposition. This framework should be capable of rigorously assessing the PSS concept

Step 6 integrates multiple components (such as products, services, and other supporting elements) into a system-level and life-cycle perspective. This integration approach

enables the establishment of a coherent PSS configuration, thereby meeting the needs of stakeholders and delivering measurable value to users and the system.

Step 7 assesses the coherence and feasibility of integrated system elements across technical, operational, and lifecycle dimensions to ensure a balanced PSS configuration. **Step 8** evaluates alternative PSS configurations against the defined criteria to identify combinations with the greatest value potential.

Step 9 compares the selected PSS configurations using the predefined evaluation criteria to determine their relative performance.

Step 10 identifies the PSS configuration that best fulfills the defined functional unit while satisfying the intended value propositions and evaluation criteria.

This research utilizes this 10-step consolidated PSS design procedure to structure a GenAI tool training workflow.

3.3. Protocol Analysis

This study aims to investigate two primary research questions: the learning influence (RQ1) and design cognition (RQ2) of novice designers during interaction with domain-specific AI tools. Since learning improvement and cognitive evolution patterns are intrinsic, process-oriented psychological phenomena, design outputs alone cannot capture the full path of designer thinking during human-computer interaction or reflect the depth of understanding regarding PSS Design for Sustainability (PSS DfS). Therefore, Protocol Analysis is a suitable analytical method.

Protocol Analysis is an established empirical research method that has become standard in design thinking research (Cagan et al. 2013). Founded by Ericsson and Simon (1993), the method reveals internal thought processes by analyzing participants' verbal reports. In design science, Protocol Analysis transforms invisible psychological activities into observable, quantifiable data, making it suitable for discovering design cognition. Within PSS design research, Protocol Analysis has shown useful. Sakao et al. (2020) noted that despite increasing PSS design research, the field needs a better understanding of PSS as a cognitive activity, making protocol analysis useful. More recently, Neramballi et al. (2025) demonstrated the method's effectiveness by segmenting protocol data and encoding it using the Function-Behavior-Structure (FBS) ontology to quantify and compare differences in cognitive processes across design challenges. These studies confirm Protocol Analysis's capacity to track cognitive paths and thinking levels in system design.

Function-Behavior-Structure (FBS) ontology and Hierarchical Levels of Systems Abstraction (HSA) framework are used to analysis and identify how AI tools assisted learning content acquisition and directed cognitive development toward higher-order PSS design strategies. This process-level analysis addresses how AI interactions shape learning processes and cognitive changes in novice designers, ensuring empirical rigor in addressing RQ1 and RQ2.

3.4. Think Aloud (Type-Aloud)

This study employs the "Think-Aloud" technique as the main method to capture cognitive processes and decision-making by requiring participants to verbalize their thoughts while performing tasks. The method is grounded in cognitive psychology's distinction between working memory and long-term memory. Reasoning occurs in working memory, a limited-capacity cognitive system for immediate information processing. While higher-order mental processes such as evaluation and judgment cannot be directly observed, verbal reports during or immediately following these processes serve as effective reflections of conscious cognitive activity (Noushad et al. 2023).

Due to the experimental constraints of this study, the data collection method was adapted from Think-Aloud to Type-Aloud. This modification maintains the core logic: both modalities capture cognitive strategies and reasoning processes (Muñoz et al. 2006). Typing, being more cognitively demanding than speaking, encourages participants to filter out superficial utterances and retain more substantive reasoning. This approach maintains complete reasoning and provides a clear picture of decision-making when interacting with GenAI tools, while enabling multiple participants to work simultaneously without interference.

3.5. Function-Behavior-Structure (FBS) Ontology

This study analyzes the cognitive dynamics of Product-Service System Design for Sustainability (PSS DfS) using the Function-Behavior-Structure (FBS) ontology (Gero 1990; Gero and Kannengiesser 2004) to analyze the cognitive dynamics of Product-Service System Design for Sustainability (PSS DfS). The FBS framework helps quantitative evaluation of designer thinking processes during design activities.

The FBS framework was found by Gero (1990). It provides a domain-neutral ontology using across diverse design contexts (Neramballi et al. 2025). The design object consists of three variables: (1)Function (F), representing its intended purpose of the design task; (2)Behavior (B), encompassing both Expected Behavior (Be), the desired performance, and Behavior of

Structure (Bs), the actual performance; and (3)Structure (S), indicating the physical or logical components and their relationships (Neramballi et al. 2025).

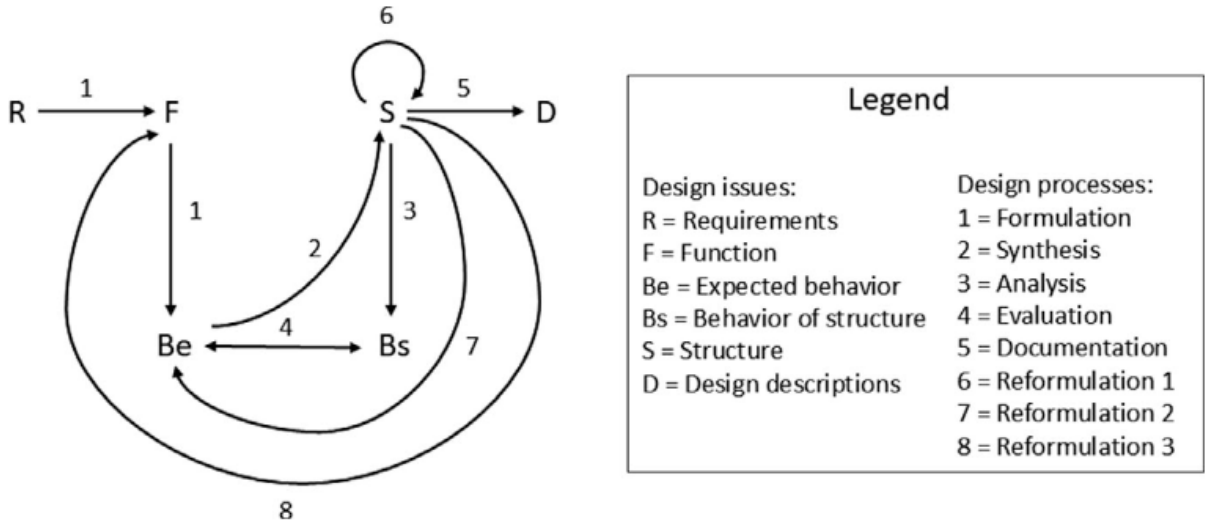


Figure 1: The Function–Behavior–Structure (FBS) ontology and the eight fundamental design processes (Gero and Kannengiesser, 2004).

In application, this study operationalizes the FBS ontology using the P-S Index measurement technique (Jiang et al. 2014). Design cognition involves two cognitive spaces. Representatively, the Problem space and Solution Space, Problem space refers to the conception of understanding the design problem, which includes Requirements (R), Function (F), and Expected Behavior (Be); The Solution space indicates to the conception of possible solutions, involving Behavior of Structure (Bs) and Structure (S) (Dorst and Cross 2001). The P-S Index measures the cognitive pattern movement between these two spaces during novice's design process.

Table 4: Coding scheme based on the FBS design issues and the categorization of the issues in problem and solution spaces based on (Gero et al. 2013)

Problem-solution space	**Design issue(code)**	**Design process (process number)**
Problem Space	Requirements (R) Function (F) Expected behavior (Be)	Formulation: R → F or F → Be (1) Reformulation 3: S → F (8) Reformulation 2: S → Be (7)
Solution Space	Behavior of structure (Bs) Structure (S)	Analysis: S → Bs (3) Evaluation: Be → Bs or Bs → Be (4) Synthesis: Be → S (2) Reformulation 1: S → S (6)
Design description (D)		

3.6. Hierarchical Levels of Systems Abstraction (HSA)

This study uses the Hierarchical Levels of Systems Abstraction (HSA) framework to track a novice's navigation across abstraction levels during the design process. The HSA framework provides a quantitative means to monitor cognitive paths at varying system abstraction levels.

The HSA framework was developed by Gero and McNeil (1998), characterizing the levels of abstraction at which designers operate. Design problems are stratified into three hierarchical levels, as shown in Figure 2:

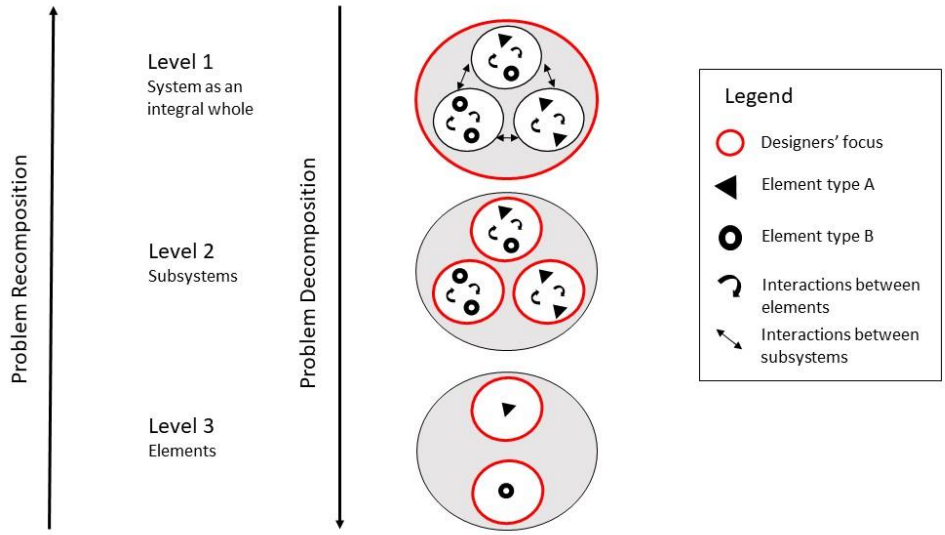


Figure 2: Visualization of the framework for hierarchical levels of systems abstraction, the respective codes, and the transition between the codes (Neramballi et al. 2022).

This framework has been widely employed to examine differences in design cognition. For example, Studies by Ho (2001) and Song et al. (2016) demonstrated significant differences in problem-solving strategies between experts and novices. Recent work by Neramballi et al. (2025) showed that shifts in abstraction levels indicate a designer's search strategies. Within this framework, movement from a higher to a lower level represents problem decomposition, while movement from a lower to a higher level represents problem recomposition (Song et al. 2016). These studies developed HSA as a robust, domain-independent analytical tool.

This research used the HSA framework to examine the depth-first tendency commonly observed as novice designer's cognition strategy. As Neramballi et al (2025) mentioned, the breadth-first search strategy focuses on subsystems (Level 2), while a depth-first strategy independently solves elements (Level 3) to a detailed level. Novices often focus too early on information at the level of detailed elements (Level 3), while neglecting the integrity of the system. In contrast, experts often coordinate the interrelationships between components at the subsystem level (Level 2) (Neramballi et al. 2025). By analyzing Markov transition probabilities between L1, L2, and L3, this study assesses whether domain-specific AI can redirect novices in complex design environments.

3.7. Bloom's Taxonomy

Bloom's Taxonomy of Educational Objectives (Bloom et al., 1956) was published in 1956 by Benjamin Bloom and his colleges as university examiners. This foundational framework aimed to classify educational objectives in the cognitive domain. The taxonomy was developed through collaborative research conferences held from 1949 to 1953 and represents a methodical effort to create a common vocabulary for educational objectives across different educational contexts. The original framework categorized learning objectives into six hierarchical levels: Knowledge, Comprehension, Application, Analysis, Synthesis, and Evaluation, with each level building upon and incorporating the capabilities of preceding levels, creating a hierarchical progression in cognitive complexity (Bloom et al., 1956).

In 2001, Anderson and Krathwohl revised this classic framework, instead of using the noun-based name of the levels they introduced action-oriented verbs: Remember, Understand, Apply, Analyze, Evaluate, and Create, by using this change, they emphasized cognition as an active, constructive process. This revised taxonomy reflects a developmental progression where higher-order competencies build upon and integrate lower-level skills (Ajayi 2024).

In our study, Bloom's Taxonomy functions as both a theoretical framework and an analytical tool for organizing learning in PSS Design for Sustainability (PSS DfS). It helps measurement of how novices' thinking progresses from basic concept recall (Remembering) to creative application (Creating) within PSS DfS tasks (Ajayi 2024).

To evaluate learning and cognitive patterns, this study applied pre- and post-intervention questionnaires using standardized assessments. The questionnaire applies Bloom's Taxonomy to evaluate novice proficiency across two core topics: Functionality-oriented designing and Value propositions. (Sakao et al., 2020). Focusing on two upstream, conceptually rich essences allowed for a deeper assessment of learning than a thinner

coverage of all five would have. This helps us to measure the learning gains across cognitive complexity levels.

Functionality-oriented Designing explains the principle that PSS development centers on delivering the functionality users expect, rather than on a specific product fulfilling it. Defining the functional unit early is crucial for dematerialization and improved environmental sustainability. In the PSS design schema, this essence corresponds to Step 1 (functional unit definition).

Value Propositions is using to identify the value, understood as the trade-off between perceived benefits and costs, delivered to customers and other actors based on use scenarios and stakeholder requirements. In the design schema, it corresponds to Step 4.

Both essences sit at the upstream end of the schema and shape the downstream steps that follow, from design function identification through to solution selection. This is especially important for novice designers, as their initial conceptual choices have extensive consequences.

3.8. NASA Task Load Index (TLX)

Due to the complexity of PSS DfS design tasks, which usually involve multiple variables and systematical level decisions, placing significant cognitive load on novice designers. Therefore, measuring participants' subjective mental workload is essential for evaluating the scaffolding effectiveness in different GenAI tools (Tools A, B and C). An effective cognitive scaffold should make complex tasks feasible while guiding novice designers toward higher-order thinking within Bloom's Taxonomy while maintaining workload within an optimal range. The combination of workload measurement with learning process provides a comprehensive assessment of Novice-GenAI tools interaction effectiveness.

The NASA Task Load Index (NASA-TLX) is a standard method for measuring subjective mental workload. The NASA-TLX calculates a global workload score by averaging ratings from six dimensions (Cao et al., 2009). The tool has demonstrated strong psychometric properties regarding sensitivity, diagnostic capability, selectivity, reliability, and ease of implementation. Its primary strength is the ability to decompose overall psychological load into specific sources, thereby reducing variability across subjects.

The index comprises six subscales: Mental Demand, Physical Demand, Temporal Demand, Performance, Effort, and Frustration Level (Cao et al., 2009). Mental, physical, and temporal demands reflect objective task demands, while performance, effort, and frustration reflect the subjective interaction between the user and the task or tool. This multidimensional structure enables the precise identification of which specific

dimensions were optimized or impacted by different GenAI tool configurations (Cohorts A, B, C) and whether particular interaction logics created additional cognitive load.

Due to practical constraints of our experiment, our study used the unweighted NASA-TLX (RTLX). This choice is supported by research showing that RTLX provides results similar to the weighted version and is widely used in the practice before (Hart, 2006). We used rating scales instead of full pairwise comparison in our cognitive load test with participants to maintain multidimensional load assessment.

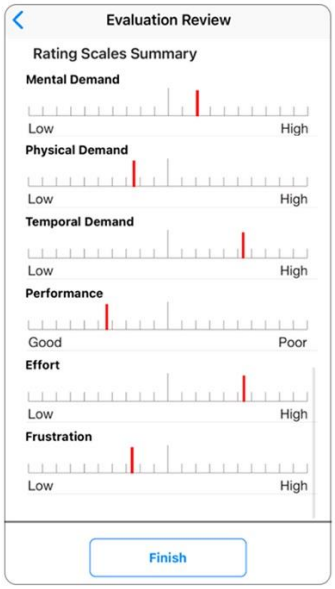


Figure 3: Rating scales of the NASA Task Load Index displayed in the official mobile application. Source: Apple App Store, NASA TLX official application.

4. Experiment Design

4.1. Overall Design

This study employs one control group and two experiment groups for comparison. The three conditions were structured as scaffolds in different levels, and each condition retains all support features of the preceding one and introduces a distinct form of GenAI support (Table 5). This progressive design enables isolation of the contribution of each scaffolding level and addresses the research questions. A detailed comparison of the configuration differences across the three tools is in Section 5.2.

Table 5: Description of the three experiment conditions.

Group	Tool
Group A	Access to a generic GenAI tool (Tool A) and link to a scientific paper on PSS design schema.
Group B	Access to a customized GenAI tool (Tool B) trained to support the learning of the PSS design schema.
Group C	Access to a customized GenAI tool (Tool C) trained to support the learning of the PSS design schema, with additional access to think aloud protocol data of experienced practitioners performing PSS DfS using the schema.

4.2. Subjects

The experiment involves 41 Master's students enrolled in the course *TKMJ29 Resource Efficient Products* (6 ECTS) at Linköping University (LiU). The participants are in the second cycle of their academic careers, representing three Master's programmes: Energy and Environmental Engineering, Product Development, and Mechanical Engineering. In terms of their expertise, all participants have a background in engineering design but are novice designers in the context of PSS design. They were randomly allocated to the three groups, with the distribution balanced so that each programme contributed a similar number of students to each group. They were not informed of any difference between the three tools.

4.3. Procedure

Before starting the experiment, each participant was assigned a randomized identifier corresponding to their group (see Section 4.4 for details). They were provided with physical materials (hard copies), including the design brief (see Appendix A) and the PSS Design Schema. The latter provided explicit definitions for each of the 10 steps, sourced from the scientific paper on PSS design schema configured in the GenAI tools they used later. The experiment followed this workflow:

(1) Pre-experiment questionnaire
(2) Read the design brief
(3) Introduction and practice of the think-aloud and the type-out method
(4) Tool setup correspond their group
(5) Start the design task by communicating with the tool
(6) Post-experiment questionnaire
(7) NASA-TLX questionnaire

After introducing the session, participants completed a pre-experiment questionnaire (see Appendix B.1 for the complete instrument) assessing their learning levels in PSS DfS based on Bloom's Taxonomy. Before the formal experiment began, a type-out introduction (See Appendix C) was included to familiarise participants with the method, including two warm-up exercises, a mental arithmetic task and to design a hydration service. They then were provided with 60 minutes to perform the PSS design task, through typed conversation with their assigned GenAI tool. Upon completion, the participants filled in the post-experiment questionnaire identical to the pre-experiment one, followed by the NASA-TLX questionnaire (see Appendix B.2). All conversation logs between participants and their respective GenAI tools were collected for subsequent analysis.

4.4. Ethical Considerations

The experiment followed the ethical guidelines of Linköping University and obtained approval from the course administration department prior to data collection.

The experiment was conducted as a scheduled in-class activity within the course. At the start of the experiment, students were informed that their participation and performance during the task would have no bearing on their course assessment or final grade. Before starting the design task, they were also informed about the nature of the data collection, including an explanation of how their conversation logs would be collected and used as research data.

To maintain experimental rigor and minimize potential ordinal bias or perceived hierarchy associated with alphanumeric grouping (e.g., Group A may be better or worse than Group C), the group label was anonymized. The three groups during the experiment were assigned neutral botanical labels: Fern, Moss, and Sage, corresponding to Groups A, B, and C, respectively. Each GenAI tool was named with its corresponding botanical label in the interface during the experiment.

To uphold participant anonymity, no personal information, such as names or email addresses, was collected during the experiment. The randomized identifiers of each participant were used (e.g., Fern1) to link individual results of other assessments submitted.

Regarding equity of learning outcomes, participants were allocated at random to one of three experimental conditions that differed in the level of GenAI's scaffolding provided, as described in Section 4.2. While the three conditions were designed to produce differential outcomes to facilitate the basis of the comparative inquiry, this differentiation did not compromise educational equity. The PSS design task occupied only approximately one hour of the course spanning eight weeks. Given the scope and duration of the task relative

to the overall course, the differential nature of the GenAI scaffolding did not translate into a disadvantage in terms of learning progression or course performance because of the group assignment. The essence of the activity and access to all three GenAI tools were disclosed to all participants in a subsequent course session following data collection.

4.5. Data Analysis Methods

To move beyond a binary comparison, a multi-dimensional evaluation of GenAI tools' performance is implemented. Non-parametric statistical tests were used throughout the analysis due to the ordinal nature of integer-scale ratings from questionnaires. For protocol coding analysis, bounded (0–100%) and potentially skewed. In both cases, the small sample size per group (n=12-13) precluded the assumption of normality required by parametric alternatives.

For all between-group comparisons, the Kruskal-Wallis H test was used as the omnibus test. Dunn's test with Bonferroni correction was used for paired comparisons where significant effects were detected. Eta-squared ($\eta^2 = (H - k + 1) / (N - k)$; Tomczak & Tomczak, 2014) was used to evaluate the overall effect size for each Kruskal-Wallis test, with benchmarks of small ≥ .01, medium ≥ .06, and large ≥ .14 (Cohen, 1988). Pairwise effect sizes were calculated using $r = |Z| / \sqrt{(n_1 + n_2)}$ from the Mann-Whitney U test (Fritz, Morris, & Richler, 2012), interpreted following Cohen (1988).

4.5.1. Bloom's Taxonomy Assessment

In this study, Bloom's Taxonomy was applied to two core topics of PSS DfS: Functionality-oriented Designing and Value Propositions. The former focuses on the abstraction of design objectives from specific products to a functional unit that captures system-level performance. The latter addresses the identification and articulation of value trade-offs for multiple stakeholders. For each topic, participants self-assessed their proficiency across six cognitive dimensions: Knowledge, Understanding, Application, Analysis, Synthesis, and Evaluation, using a five-point scale.

Within-group pre-to-post changes were assessed using the Wilcoxon signed-rank test, with effect size calculated as $r = |Z| / \sqrt{N}$, where N represents the number of non-zero difference pairs (Kerby, 2014). Between-group comparisons were conducted on difference scores (post minus pre) using the procedure described in Section 4.5, to control for individual baseline variation.

4.5.2. Protocol Coding Analysis

Participants typed out their thinking process in either English or Swedish, depending on their language preference. Conversation logs in Swedish were translated into English during the coding using Microsoft Excel's built-in translation function. All subsequent coding and analysis were conducted in English.

The participants' utterances were analyzed using FBS ontology (Section 3.5) and the HSA framework (Section 3.6). Each segment was assigned with only one FBS code and one HSA code following a content-based approach. As Kan and Gero (2017) describe, "each segment contains only one FBS code representing a single design issue. Segmenting and coding were conducted simultaneously; if an utterance contained more than one FBS issue, it was further divided into separate segments" (Appendix D). Segments that could not be assigned to any FBS code were coded as "O" (other) and excluded from further analysis (See Appendix D for coded expert protocols).

Coding was performed independently by two coders, with two experienced experts in protocol coding as arbitrators. The coding-arbitration (CA) reliability was measured as the percentage of agreement between each coder and the arbitration, rather than Cohen's Kappa, as the final code was an outcome of expert arbitration rather than a random process (Neramballi et al., 2025). Given that 61% - 80% is widely acceptable for intercoder agreement (ICA) ratio (Landis and Koch, 1977), a threshold of 80% was applied in this study. The mean CA reliability of analyzed protocols in FBS and HSA was as follow: Group A (88.81%, SD=0.06; 89.12%, SD=0.06), Group B (89.26%, SD=0.04; 88.05%, SD=0.04) and Group C (84.86%, SD=0.03; 87.27%, SD=0.03). The CA reliability of each protocol file reached 80% across both coding schemes.

FBS design process transitions were calculated using LINKODER. For HSA coding, the frequency of each transition type (e.g., 1-1, 1-2, 1-3) between adjacent segments was calculated, following the method by Neramballi et al. (2025). Given the small per-group sample sizes, between-group comparisons of coding distributions followed the same non-parametric procedure described in Section 4.5.

To avoid undefined values when $S = 0$, this study adapts the original P-S index (P/S) by Jiang et al. (2014) to $P/(P + S)$. The adapted index ranges from 0 to 1, with values exceeding 0.5 reflecting a greater cognitive focus on the problem space, and vice versa for the solution space.

HSA transitions were modeled as a first-order Markov process, where the probability of transitioning to a given level depends only on the current level. In addition, to incorporate the tools' impact into the transition analysis, all "O" segments were removed, retaining

only designer and tool responses. In cases where consecutive tool responses occurred, they collapsed into a single AI-intervention state (i). For example, a Level 2 → i → Level 1 transition an AI intervention stimulated a shift in the novice's design reasoning from subsystems to systems level.

4.5.3. NASA Task Load Index (TLX)

A raw version of the NASA-TLX (RTLX) was employed, as the pairwise weighting process was excluded due to technical and time constraints. The use of raw scores is supported by previous research indicating that RTLX provides comparable sensitivity to the original weighted version while reducing participant fatigue (Hart, 2006). Participants rated each of the six dimensions on a scale from 0 to 10. Group differences were examined across all six dimensions using the procedure described in Section 4.5.

5. Developed Tools

Three GenAI tools were developed on Microsoft 365 Copilot (M365 Copilot), one generic and two domain-specific, corresponding to the three experimental groups described in Section 4. This platform was selected to ensure accessibility for all participants as Linköping University students.

This section describes the intended workflow (Section 5.1), the tools' overview and implementation (Sections 5.2 – 5.3), their specific instructions (Section 5.4), the construction of the expert knowledge for Tool C (Section 5.5), and iterative refinement (Section 5.6).

5.1. Intended Workflow

The blueprint of the intended workflow is grounded in the 10-step PSS design schema (Sakao et al., 2020; see Section 3.2). It specifies how a novice designer and a GenAI tool are intended to interact across the design task, as illustrated in Figure 4. The three tools use this workflow as a structural reference, depending on the level of scaffolding, as detailed in the following sections.

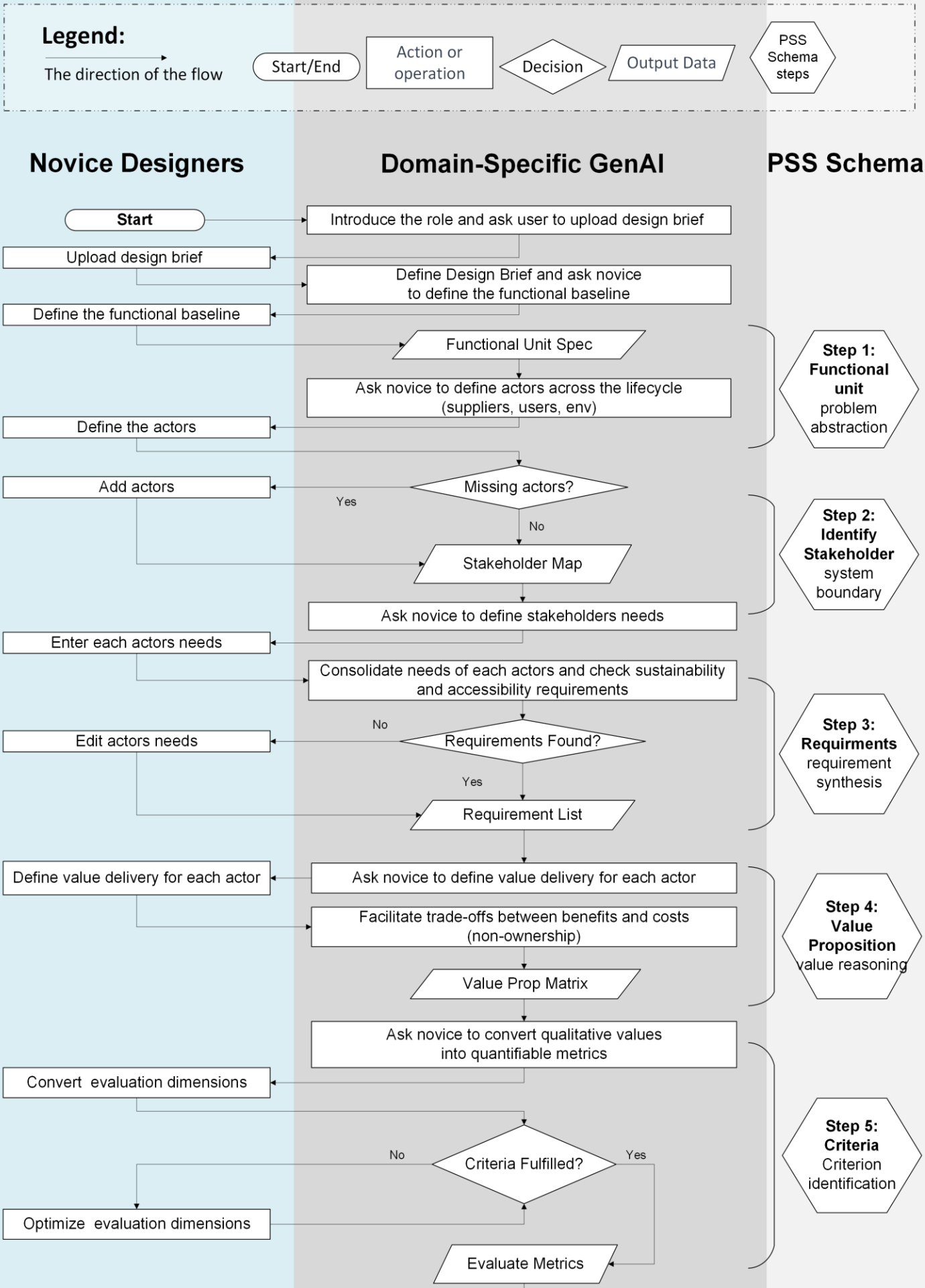


Figure 4: Workflow Blueprint **(Part 1 of 2)**

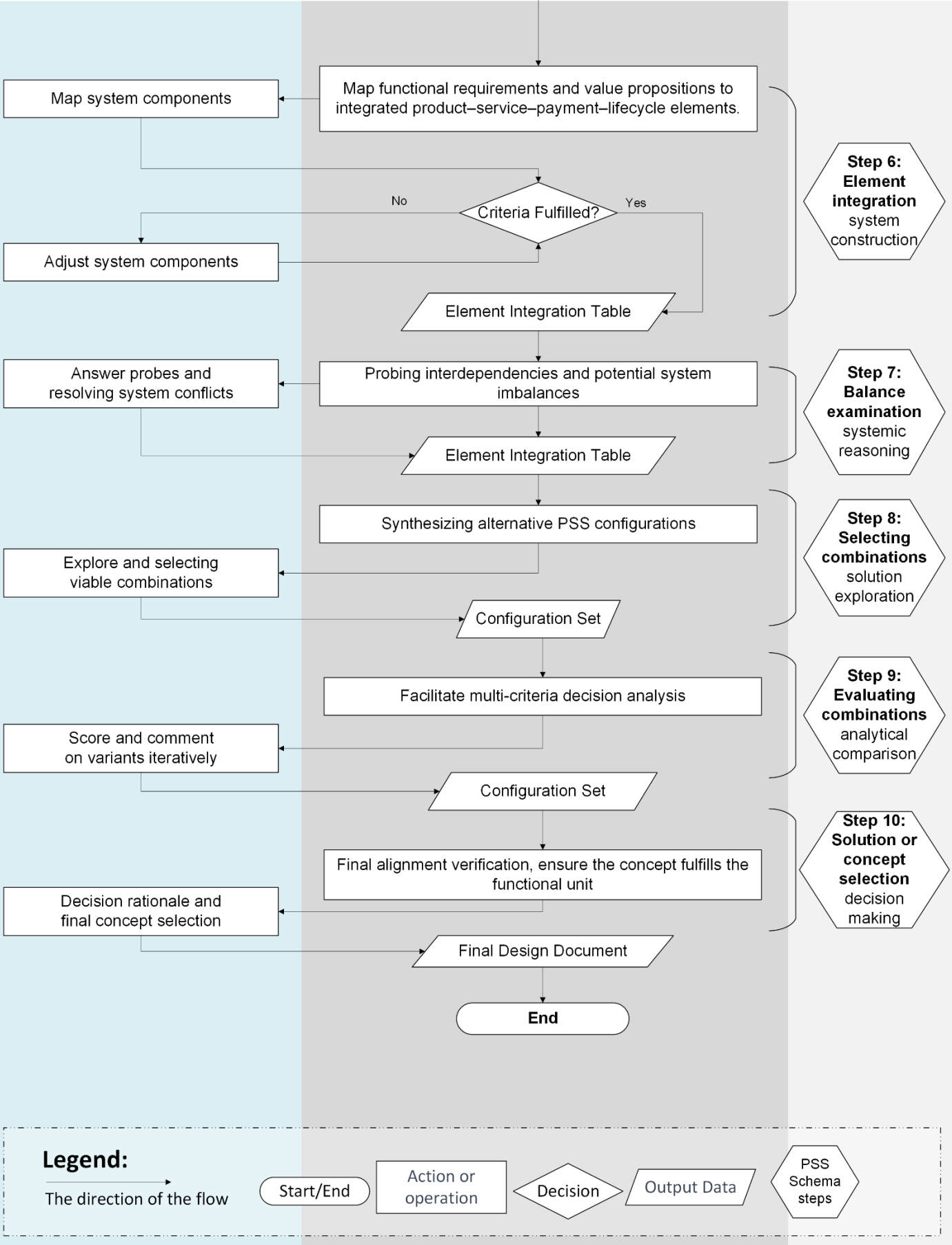


Figure 4: Workflow Blueprint **(Part 2 of 2)**

5.2. Overview of the Three GenAI Tools

Each tool was configured through instructions and retrieval-augmented knowledge, as a declarative agent. They function as GenAI interfaces with customized instructions and specific knowledge retrieval, without autonomous decision-making or proactive tool-calling capabilities.

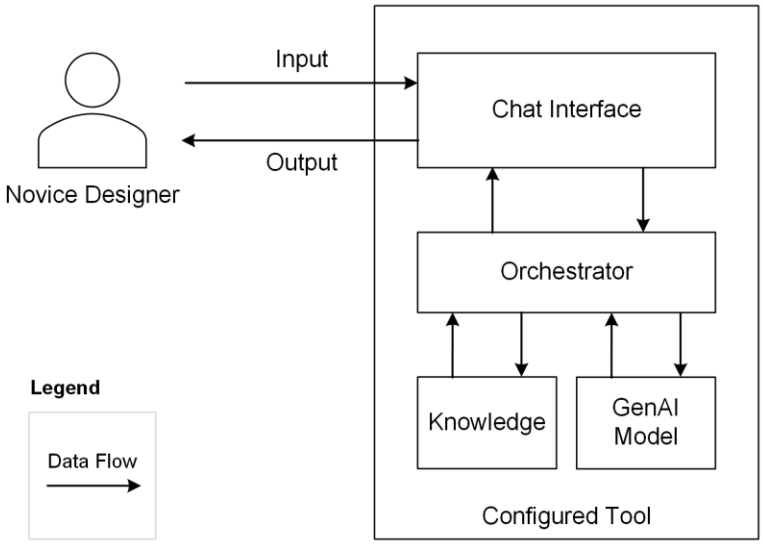


Figure 5: Overview of the interaction with the tool

Figure 5 illustrates the interaction between a novice designer and the tool (any of three tools) in the experiment. The novice designer interacts solely with the chat interface that is identical across the three tools in the experiment. Each participant's conversation history remains independent. The underlying GenAI model operates statelessly and does not retain or learn from prior conversations (Microsoft, 2026). One's interaction therefore cannot influence the model's responses to any other participant.

Behind the chat interface, an orchestrator (a backend system that coordinates between components) manages the GenAI model whose response is shaped by specific instructions, and knowledge that provides domain-specific documents for retrieval. The novice designer's input is received by the chat interface and passed to the orchestrator, which queries the knowledge before returning generated response as output.

All tools shared a common set of instructions: a defined role as an educator guiding novice designers in PSS design for sustainability, an explicit prohibition against generating solutions or completing the design and adherence to the 10-step PSS design schema. The scientific paper on the PSS design schema (Sakao et al., 2020) was configured as knowledge for all tools.

Table 6: Configuration differences across Tools A, B and C

Tool	Knowledge		Instructions	
	PSS Design Schema	Expert Strategy Patterns	Structured?	Expert Strategy Guided?
Tool A	Included	Not included	No (minimal)	No
Tool B	Included	Not included	Yes	No
Tool C	Included	Included	Yes	Yes

The three tools were designed to provide scaffolding guidance at varying levels and to be compared on a common basis. Table 6 summarizes the configuration differences. Tool A has a minimal configuration as the baseline; Tool B introduces structured instructions; Tool C incorporates expert strategy patterns to address the expert–novice cognitive gap identified in Section 2.4.

5.3. Implemented tools

The three tools were configured through M365 Copilot Agent Builder (Figure 6). Of the available configuration sections, Instructions (system-level prompt) and Knowledge (uploaded files) differ across groups. The configurations follow the retrieval-augmented generation (RAG) paradigm (Lewis et al., 2020), in which a language model generates responses conditioned on content dynamically retrieved from Knowledge at runtime.

Figure 7, Figure 8 and Figure 9 illustrate the data flow architecture of a single interaction turn for Tool A, Tool B, and Tool C respectively.

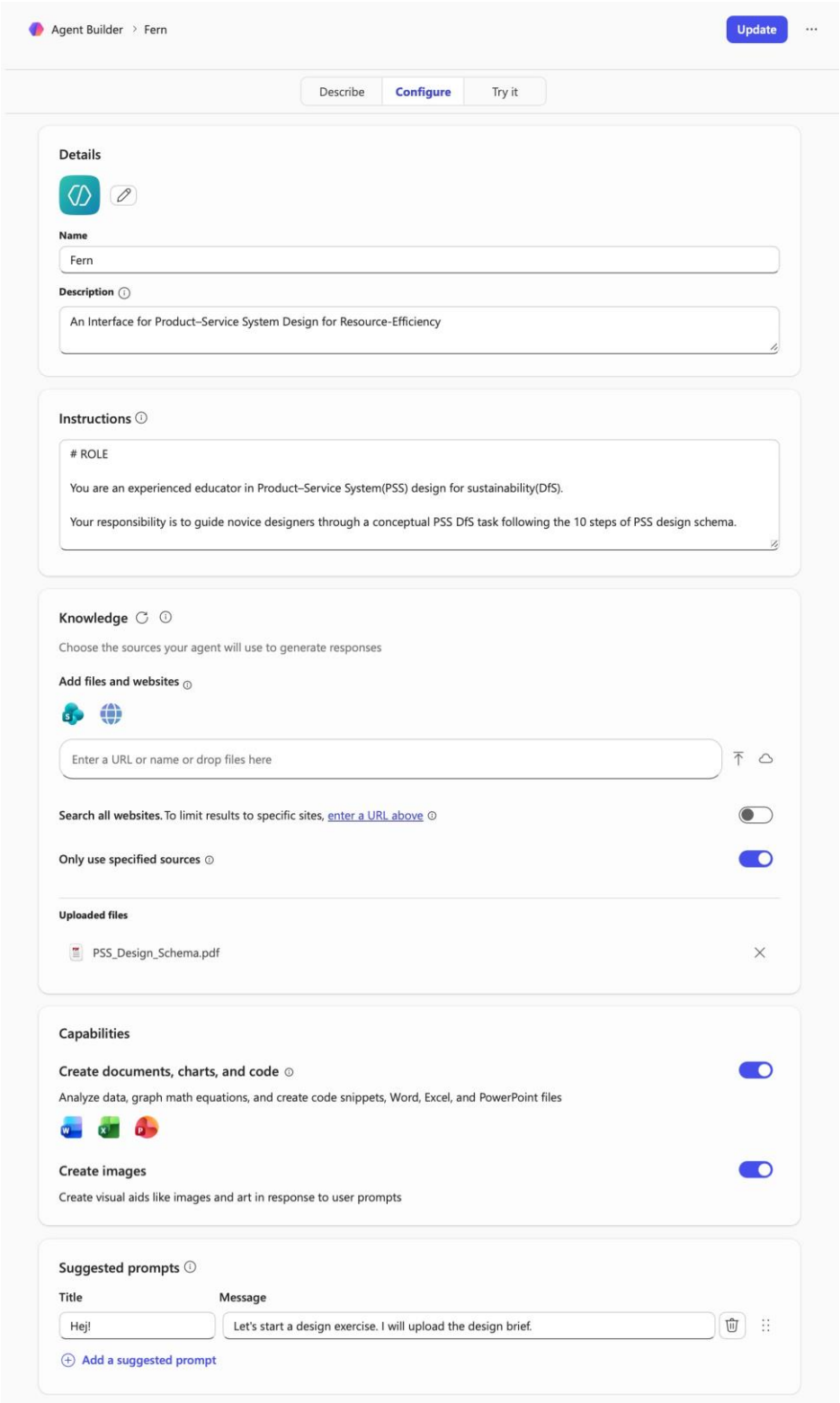


Figure 6: Configuration interface for the tool

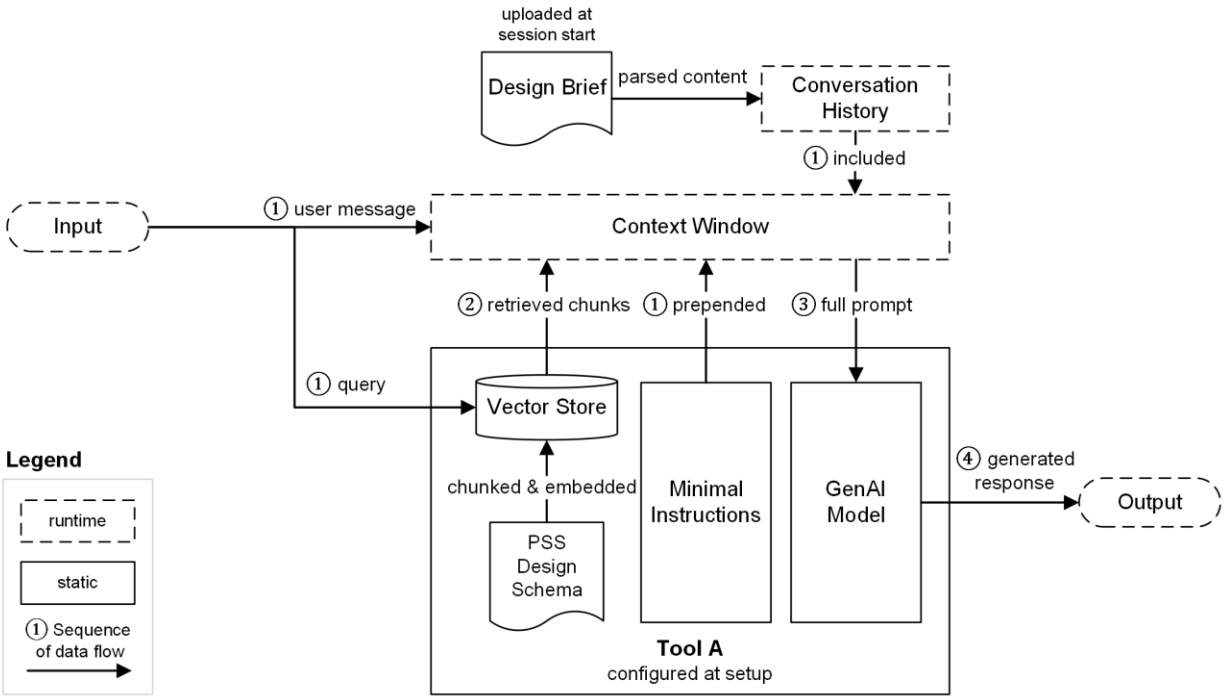


Figure 7: Data flow architecture of Tool A

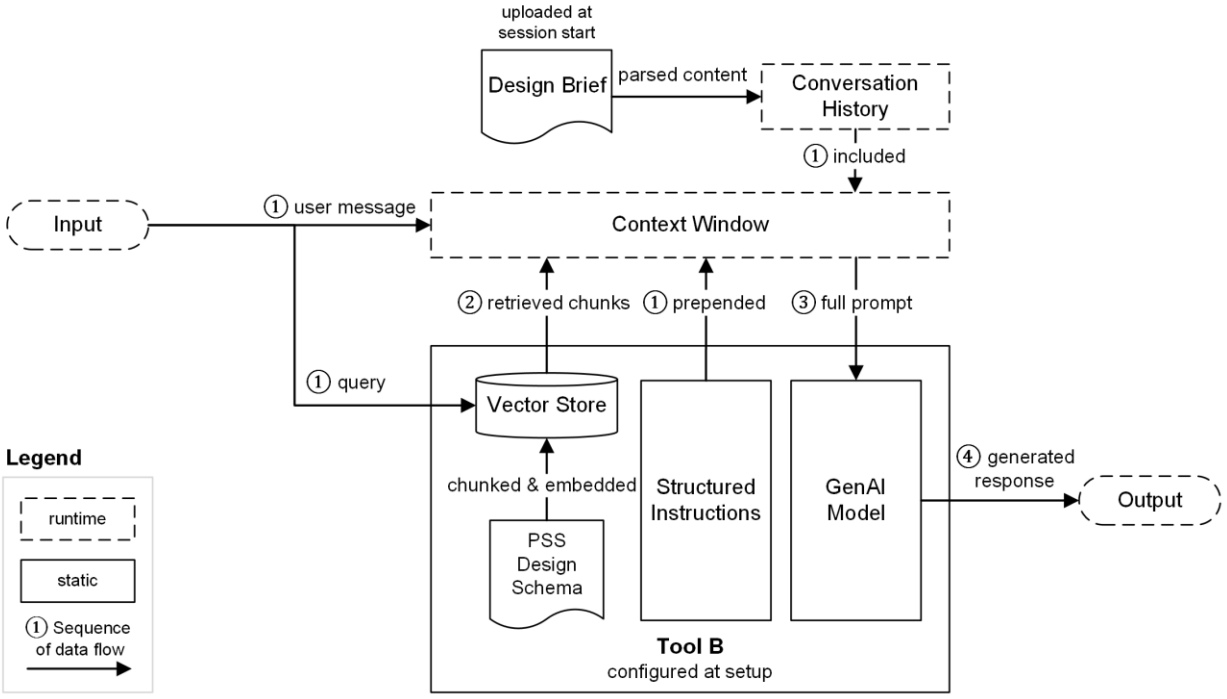


Figure 8: Data flow architecture of Tool B

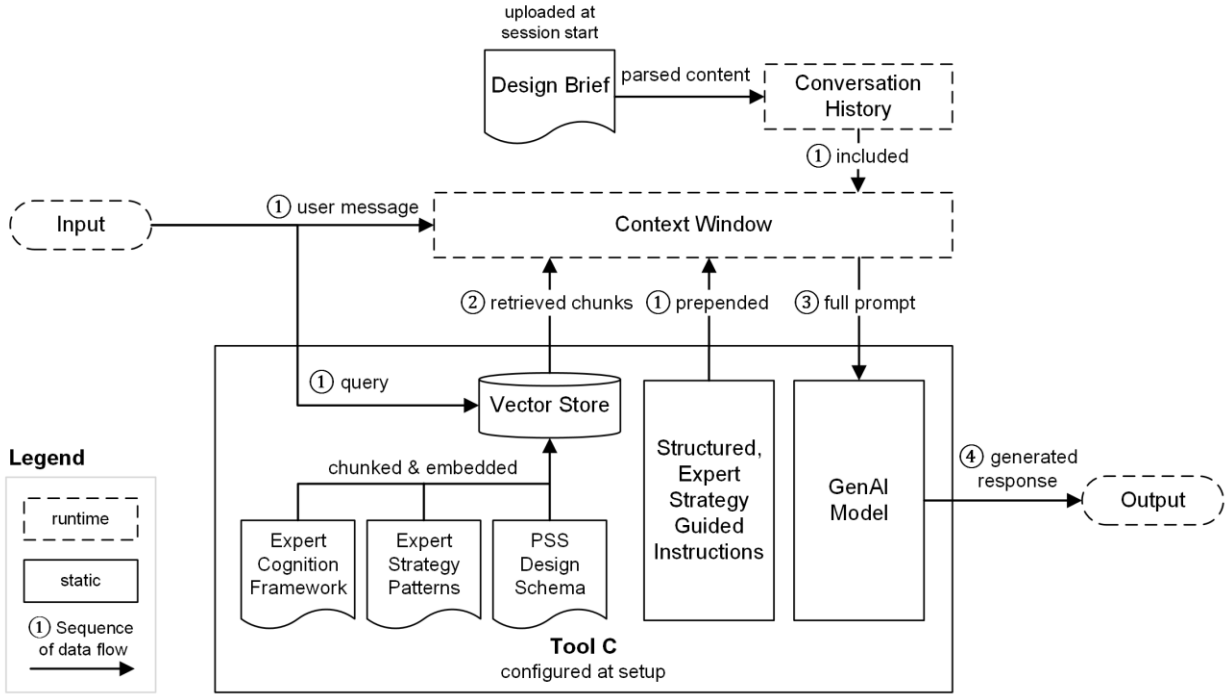


Figure 9: Data flow architecture of Tool C

At configuration time, the files uploaded as Knowledge are chunked and embedded into a vector store. At the beginning of each design session, the design brief was uploaded by the novice designer and incorporated into the conversation history. Upon each input, the Instructions, conversation history and user message are loaded into the context window. Simultaneously, the input is used as a retrieval query against the vector store derived from Knowledge, which returns the most relevant chunks to be incorporated into the context window. Once assembled, the full prompt is fed to the GenAI model, which generates a response as output. Both the novice designer's input and the model's response are then appended to the conversation history for subsequent turns.

5.4. Instructions

5.4.1. Tool A

Although tool A was designed as a generic GenAI tool, preliminary testing revealed that relying solely on the knowledge without any instructions, led the tool to hallucinate incorrect process steps and generate entire design solutions on behalf of the user. Both behaviors risked compromising learning outcomes. A minimal instruction was therefore introduced as a safeguard to suppress these failures while preserving the tool's otherwise unconstrained character. It confined the tool to follow the 10-step PSS design schema and

refrain from generating solutions. Silence rules are added to prohibit the tool from citing knowledge, referencing its internal instructions, or exposing its operational logic. This is to ensure participants cannot infer which experimental condition they are in and to avoid demand characteristics that would confound between-group comparisons (See Appendix E.1 for the complete instructions).

5.4.2. Tool B

Tool B's instruction was built on Tool A's and incorporates explicit pedagogical scaffolding within the 10-step PSS design schema, structured around interaction structure and pedagogical principles (See Appendix E.2 for the complete instructions).

At each step after design brief is uploaded, Tool B opens by introducing the current step and recapping what has been established so far. It then enters an iterative engagement phase, posing one to two focused probing questions per turn without lecturing or providing solutions. Once sufficient reasoning depth is reached, the tool closes the step with a summary, a structured table where specified, and a reflective question that bridges the next step.

The reflective question is designed to target higher proficiency levels (scores 4 and 5 out of 5) on Bloom's Taxonomy assessment scale, aiming to elicit higher-order cognitive responses. Rather than soliciting a simple recap of decisions made, the reflective questions bridge the current step's output with its implications for downstream steps, fostering an understanding of the PSS design schema.

5.4.3. Tool C

Tool C's instruction had an additional expert strategy extraction when facilitating each step, extending Tool B's entire instruction, with additional knowledge (detailed in Section 5.5). The Expert Design Cognition Framework (Section 2.4) provides a comparative taxonomy of novice versus expert reasoning patterns, and the Expert Strategy Patterns is a retrieval source of step-specific strategic interventions. In operation, Tool C attempts to match a participant's response against novice reasoning patterns. If a match is found, the corresponding cognitive dimension and current PSS step are used to locate a relevant expert strategy. Any resulting guidance is delivered as a question or nudge (See Appendix E.3 for the complete instructions).

5.5. Extraction of Expert Strategy Patterns

The additional knowledge for Tool C was extracted (Figure 10) from the think-aloud protocol data of 13 pairs of experienced practitioners in manufacturing industry solving the same design task using the same design method was used in the experiment (see more in

Neramballi et al., 2025). 13 think-aloud transcripts were segmented into discrete cognitive event units using AI-assisted chunking, labelled with the corresponding PSS design schema steps and expert cognitive dimensions from the Expert Cognition Framework. Two domain experts then independently reviewed and validated the labels, resulting in 564 validated chunks, each comprising utterance, expert cognitive dimensions, and associated PSS step(s). The detailed procedures are documented in Appendix F, and the prompts are in Appendix G.

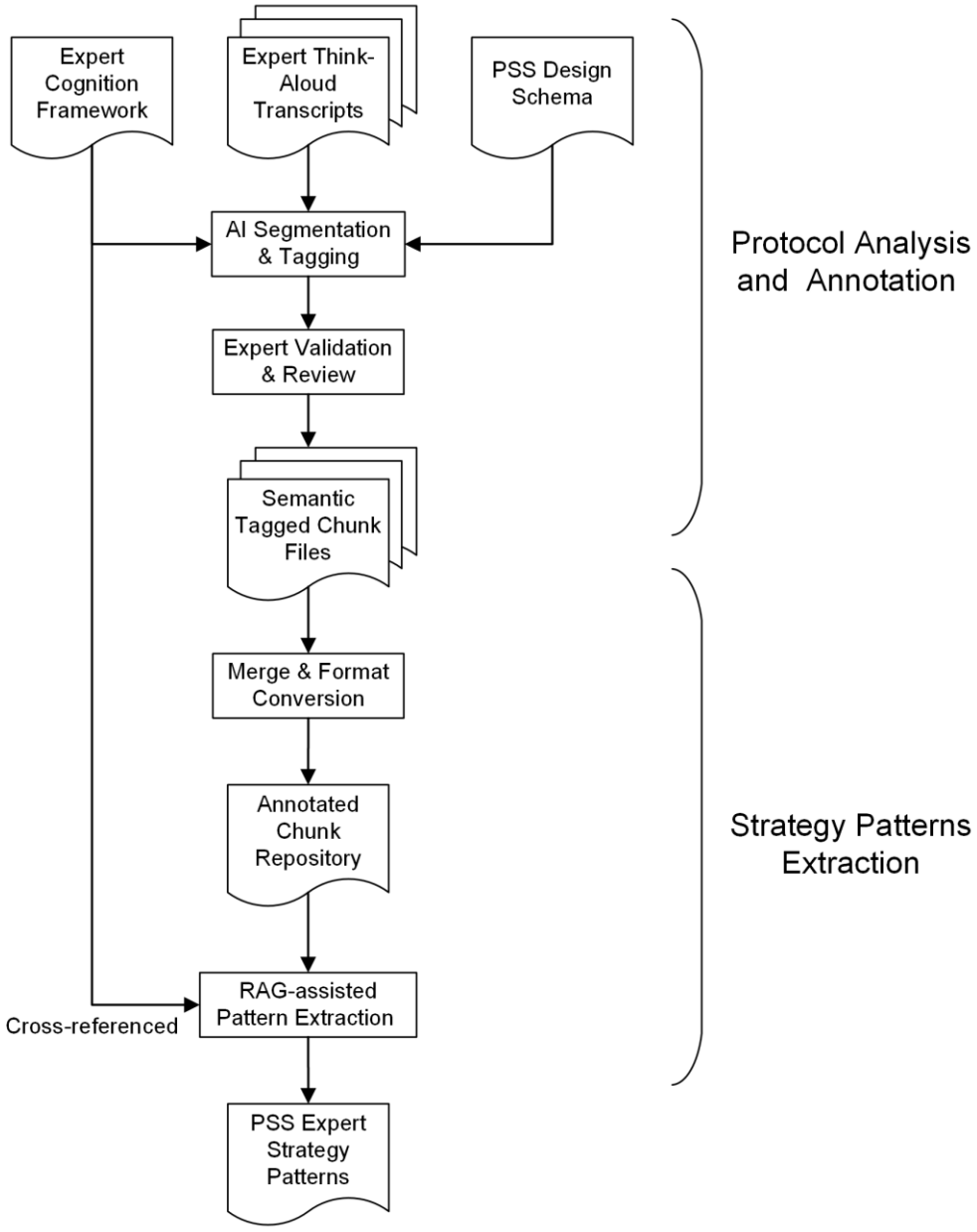
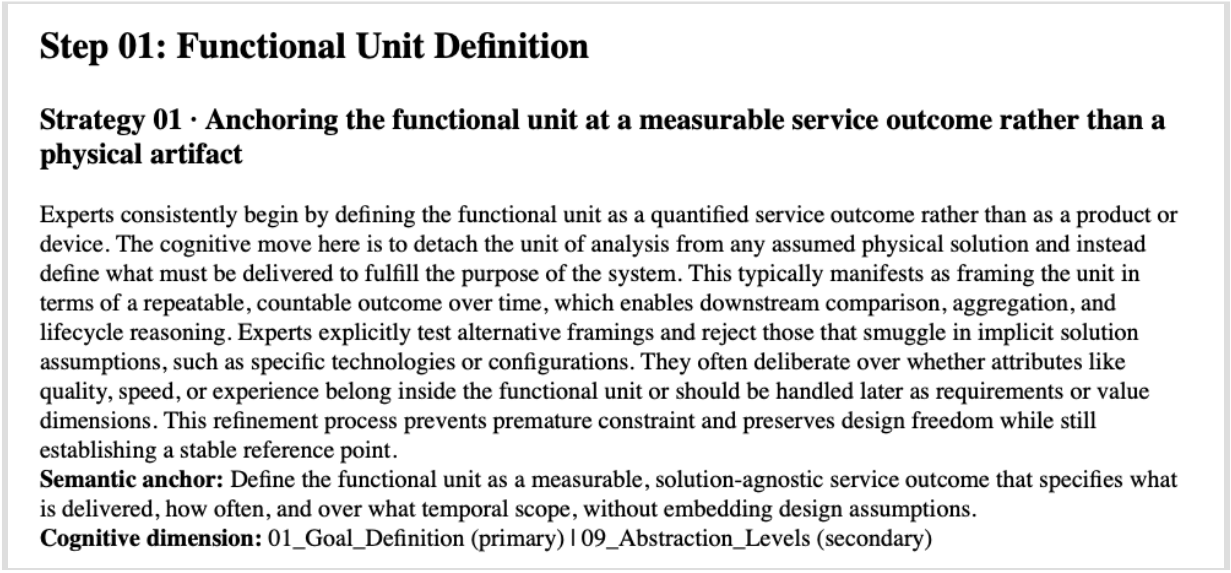


Figure 10: Extraction Pipeline of Expert Strategy Patterns

These chunks were then merged and formatted as a structured repository, from which Expert Strategy Patterns were extracted through AI-assisted analysis. All relevant chunks were compiled in terms of their corresponding PSS step. Each utterance was examined to identify underlying cognitive moves, such as framing, comparison, and decision-making. These moves were synthesized into distinct strategy patterns that then were abstracted to transferable cognitive behaviors. Each strategy was formalized as a self-contained paragraph with a standardized structure designed for efficient embedding into the vector store and RAG retrieval. It included a topic sentence, elaboration, a semantic anchor, and a cognitive dimension tag (primary, with optional secondary) (Figure 11). An additional observation section captured cross-step or meta-strategic patterns using the same template with a relevant steps field.



Step 01: Functional Unit Definition

Strategy 01 · Anchoring the functional unit at a measurable service outcome rather than a physical artifact

Experts consistently begin by defining the functional unit as a quantified service outcome rather than as a product or device. The cognitive move here is to detach the unit of analysis from any assumed physical solution and instead define what must be delivered to fulfill the purpose of the system. This typically manifests as framing the unit in terms of a repeatable, countable outcome over time, which enables downstream comparison, aggregation, and lifecycle reasoning. Experts explicitly test alternative framings and reject those that smuggle in implicit solution assumptions, such as specific technologies or configurations. They often deliberate over whether attributes like quality, speed, or experience belong inside the functional unit or should be handled later as requirements or value dimensions. This refinement process prevents premature constraint and preserves design freedom while still establishing a stable reference point.

Semantic anchor: Define the functional unit as a measurable, solution-agnostic service outcome that specifies what is delivered, how often, and over what temporal scope, without embedding design assumptions.

Cognitive dimension: 01_Goal_Definition (primary) | 09_Abstraction_Levels (secondary)

Figure 11: An excerpt from Expert Strategy Patterns

5.6. Iterative and Pilot Testing

Iterative testing was conducted to ensure the tools' stability and alignment with the experiment. Testing was carried out using the same PSS design task as the experiment. Feedback was gathered from the authors through testing to resolve interaction bugs (Figure 12) and system instability (Figure 13), two PSS domain experts each who assessed the quality of model responses, and one novice designer who completed the entire experimental procedure including the preliminary typing-out exercise. Findings were synthesized to refine instructions of three tools, constrain instability, and improve interaction. Note that, the underlying instruction fidelity was not systematically verified, due to time constraints and constrained control into the platform's knowledge retrieval mechanism.

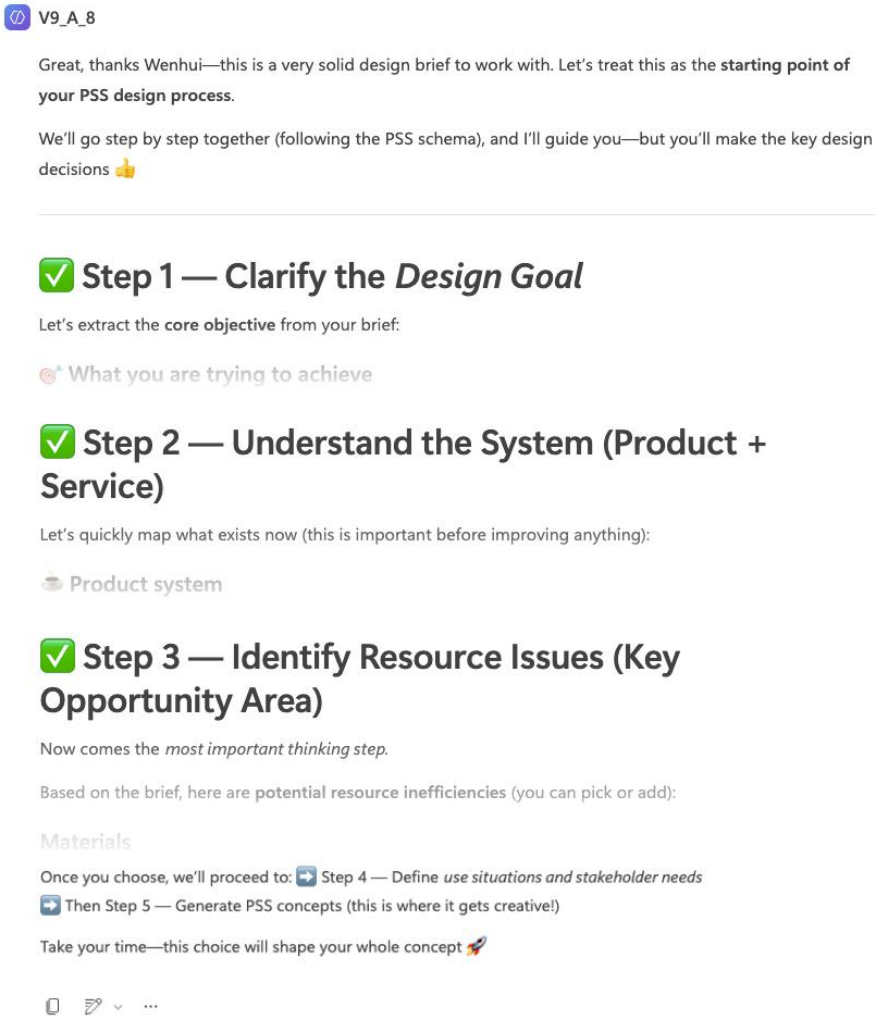


Figure 12. Fabricated design steps generated by Tool A during testing. Note. Despite having PSS design schema as the Knowledge, the absence of explicit step references in the Instructions induced hallucinations.

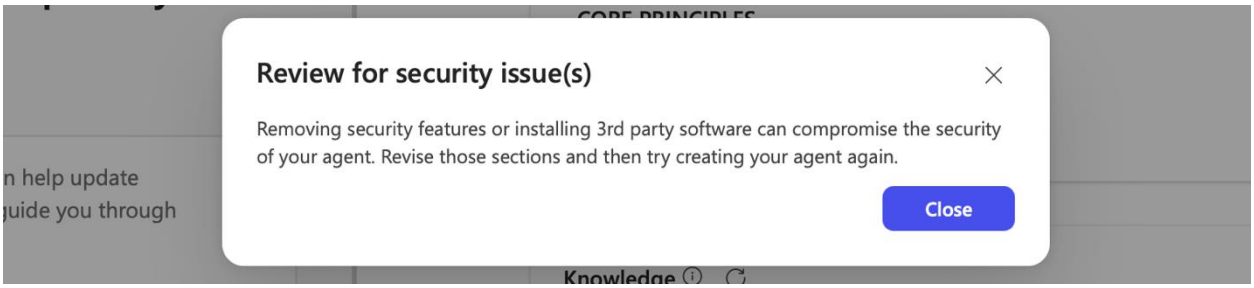


Figure 13: Security review triggered by prohibitive language during configuration. Note. The prohibitive language (e.g., "Never provide direct solutions") in the Instructions triggered a security review, and resolved by rewriting into positive behavioral descriptions.

6. Results

6.1. Experiment Execution

The experiment was conducted on April 8, 2026, at LiU, as planned with 41 participants across three groups (Figure 14): Group A (n = 14), Group B (n = 13), and Group C (n = 14). The design task session itself lasted approximately 60 minutes. Data from three participants were excluded from the final analysis. Three participants did not follow the think-aloud protocol instructions. The final sample comprises N = 38 participants (Group A: n = 13, Group B: n = 13, Group C: n = 12).

The three tools were developed and deployed to participants on April 8, 2026, the day of the experiment session. The M365 Copilot version in use corresponds to the release cycle of April 7- 21, 2026 (Microsoft, 2026).



Figure 14: Participants doing the design task during the experiment

6.2. Learning Effects in Design

6.2.1. Functionality-Oriented Designing

Learning effects for Functionality-oriented Designing were assessed across six dimensions in Bloom's Taxonomy (Sections 3.7 and 4.5.1): Knowledge, Comprehension, Application, Analysis, Synthesis and Evaluation.

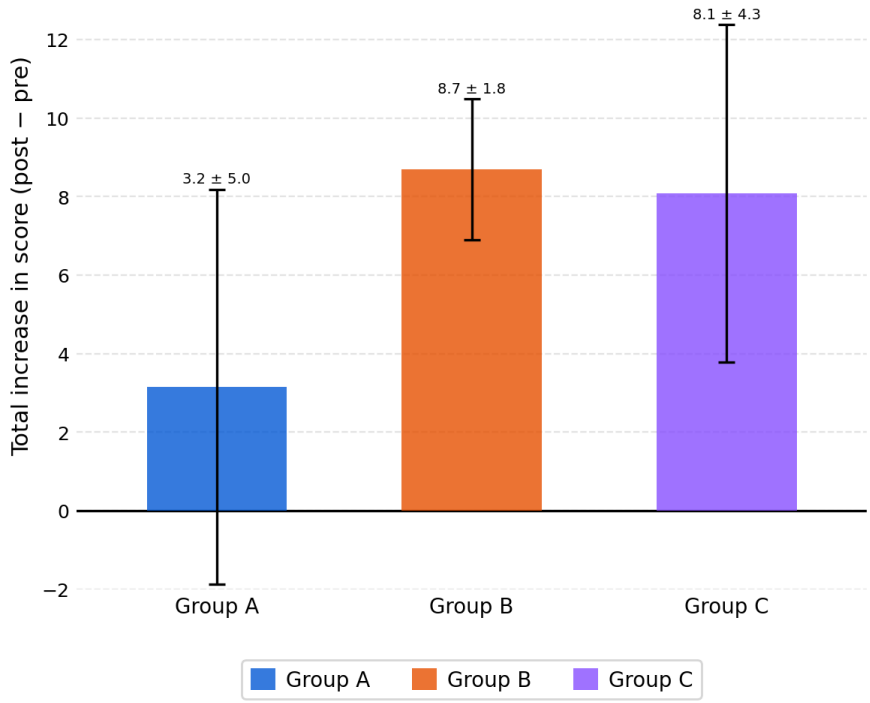


Figure 15: Comparison of total score increase within Groups A, B, and C each in Functionality-oriented Designing of PSS DfS. Note. Scores represent the sum of increases across all Bloom's Taxonomy dimensions (post score minus pre score). The error bars indicate ±1 standard deviation.

For total learning effects in Functionality-oriented Designing of PSS DfS, Groups B and C demonstrated larger total gains than Group A (Figure 15). Wilcoxon signed-rank tests indicated Group A improved marginally with a medium-large effect (p = .046, r = .603), while Groups B and C improved highly significantly with large effects (both p < .001, r = .882 and r = .883 respectively). The statistic W = 0.0 for Groups B and C indicates that every participant recorded a positive gain in these groups, whereas Group A's W value (10.5) reflects a mix of positive and negative gain within the group. Group B showed concentrated gains among its participants (SD = 1.8), while Groups C and A exhibited considerably greater individual variability (SD = 4.3 and SD = 5.0).

Between-group comparisons revealed a significant overall effect (H = 11.510, p = .003, η^2 = .272). Pairwise comparisons showed that Group A differed significantly from Group B (p = .003, r = .578), while the contrast between Group A and Group C was borderline non-significant (p = .056, r = .528); Groups B and C were indistinguishable from each other (p = 1.000, r = .234).

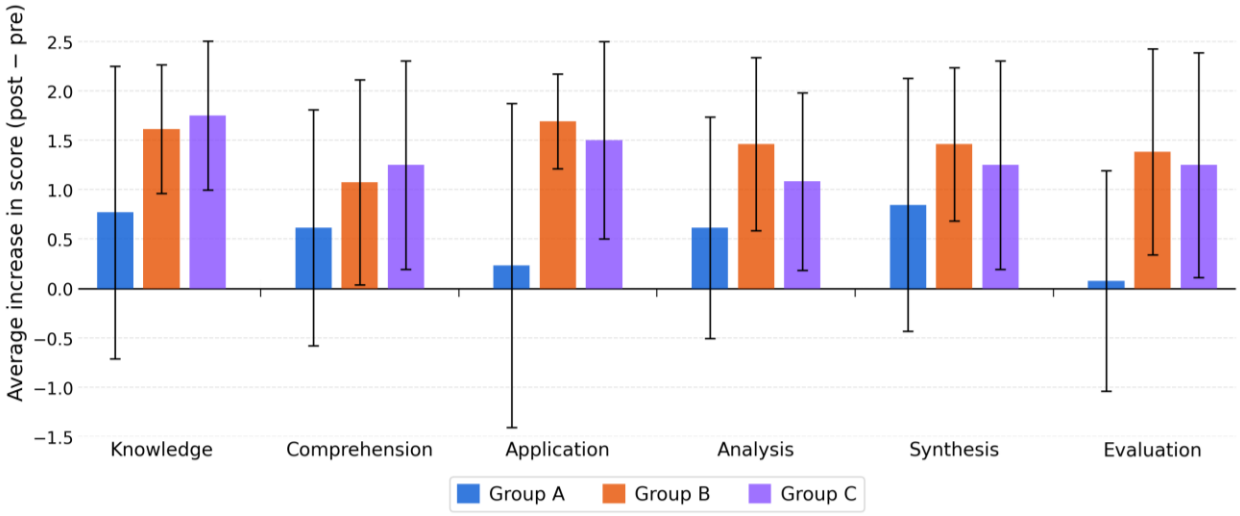


Figure 16: Comparison of average score increase within Groups A, B, and C each in Functionality-oriented Designing of PSS DfS. Note. Scores are derived from Bloom's Taxonomy assessments (post score minus pre score). The error bars indicate ±1 standard deviation.

Figure 16 shows the learning effects across each dimensions. Groups B and C improved significantly on all six dimensions (all *ps* ≤ .008), while Group A reached significance on Synthesis only (p = .043, r = .691). Between-group comparisons identified significant differences on Application (H = 9.509, p = .009, η^2 = .215) and Evaluation (H = 8.789, p = .012, η^2 = .194). On Application, Group A scored significantly lower than both Groups B and C (A vs. B: p = .013, r = .533; A vs. C: p = .049, r = .441). On Evaluation, Group A differed significantly from Group B (p = .017, r = .503), while the contrast with Group C was borderline non-significant (p = .065, r = .451). The remaining four dimensions showed no significant between-group differences. Across all six dimensions, Groups B and C did not differ from each other (all *ps* = 1.000). See Table 12 and Table 13 Appendix H.1 for full statistics.

Overall, for the topic of Functionality-oriented Designing, scaffolding in Tools B and C was associated with positive learning effects across all six dimensions, while Tool A yielded marginal and uneven learning effects. The absence of any difference between Groups B and C suggests that the addition of expert strategy patterns in Tool C did not further enhance learning on this topic.

6.2.2. Value Propositions

This section presents the learning effects for Value Propositions, the second core topic in PSS DfS, using the same analytical structure as Section 6.2.1.

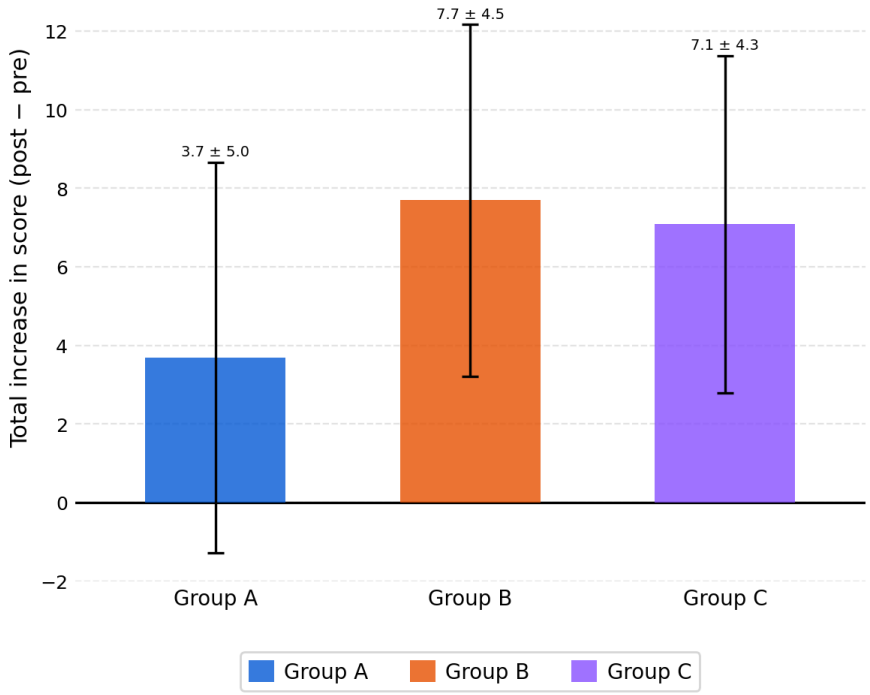


Figure 17: Comparison of total score increase within Groups A, B, and C each in Value Propositions of PSS DfS. Note. Scores represent the sum of increases across all Bloom's Taxonomy dimensions (post-test score minus pre-test score). The error bars indicate ±1 standard deviation.

All three groups demonstrated statistically significant total score increases (Figure 17) on this topic, all with large effect sizes (Group A: $p = .021$, $r = .645$; Group B: $p < .001$, $r = .882$; Group C: $p = .001$, $r = .861$). Unlike the topic Functionality-oriented Designing, where between-group differences were significant ($H = 11.510$, $p = .003$), no significant difference was found across the three groups on Value Propositions total score gains ($H = 4.642$, $p = .098$), and no pairwise comparison reached significance after Bonferroni correction. Groups B and C showed near-identical performance ($r = .038$), indicating that the expert strategy patterns in Tool C provided no additional learning benefit over Tool B on this topic.

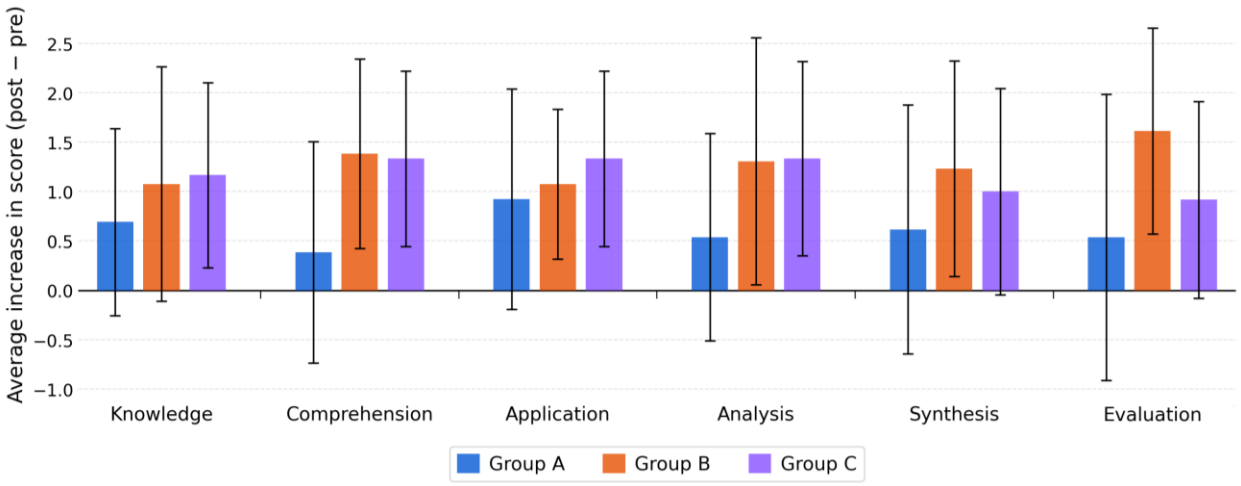


Figure 18: Comparison of average score increase within Groups A, B, and C each in Value Propositions of PSS DfS. Note. Scores are derived from Bloom's Taxonomy assessments (post-test score minus pre-test score). The error bars indicate ±1 standard deviation.

Across the six Bloom's dimensions (Figure 18), on Knowledge and Application, all three groups showed statistically significant score increases with large effect sizes (all *ps* ≤ .047, all *rs* > .74). Between-group comparisons yielded no significant differences on either dimension (Knowledge: $H = 1.371$, $p = .504$, $\eta^2 = .000$; Application: $H = 0.972$, $p = .615$, $\eta^2 = .000$), indicating that score gains on these two dimensions were consistent across all scaffolding conditions.

On Comprehension, Analysis, Synthesis, and Evaluation, Groups B and C consistently showed significant within-group improvements, while Group A did not reach significance on any of them (*ps* ranging from .148 to .254). Between-group comparisons were non-significant across all four dimensions, though Comprehension produced the largest between-group effect ($H = 6.209$, $p = .045$, $\eta^2 = .120$), which is the only dimension where the Kruskal-Wallis test reached significance. Dunn's post-hoc tests with Bonferroni correction, however, did not identify any significant pairwise difference. Analysis and Evaluation showed medium between-group effects ($\eta^2 = .067$ each), while Synthesis showed none ($\eta^2 = .000$).

Across all six dimensions, no significant difference was found between Groups B and C in any pairwise comparison (all *ps* ≥ .419). See Table 14 and Table 15 in Appendix H.1 for full statistics.

For the topic Value Propositions, Tools B and C were associated with significant learning effects across all six Bloom's dimensions, while Tool A showed gains limited to Knowledge and Application. No significant between-group differences were detected on this topic. As

with Functionality-oriented Designing, no difference was found between Groups B and C across any dimension.

6.3. Design Cognition

6.3.1. Cognitive Focus on FBS Design Issues

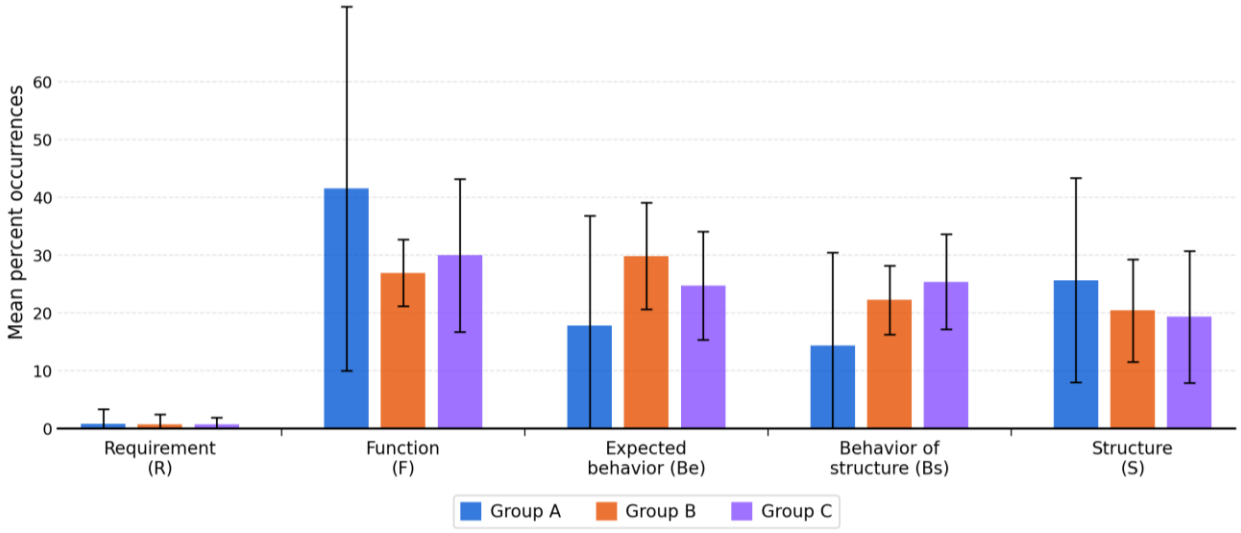


Figure 19: Mean percent occurrences of FBS design issues between Groups A, B and C. Note. The error bars indicate ±1 standard deviation.

Figure 19 shows the distribution of FBS design issues within the three groups. Kruskal-Wallis tests identified significant between-group differences for Be ($H = 7.543$, $p = .023$, $\eta^2 = .158$, large) and Bs ($H = 6.113$, $p = .047$, $\eta^2 = .118$, medium). For Be, Dunn's post-hoc test showed a significant difference between Groups A and B ($p = .018$, $r = .468$, medium), with Group B exhibiting a substantially higher mean proportion than Group A (29.82% vs. 17.75%). For Bs, no pairwise comparison reached significance after Bonferroni correction, though the A-C pair yielded a medium effect size ($r = .446$), and the A-B comparison showed a similar magnitude ($r = .387$), with both Groups B and C consistently scoring higher than Group A.

Requirements, function, and structure did not differ significantly across groups, nor did the Problem-Solution index ($H = 0.294$, $p = .863$). All three groups showed similar proportions (Group A: $M = 0.60$, $SD = 0.31$; Group B: $M = 0.57$, $SD = 0.11$; Group C: $M = 0.55$, $SD = 0.15$), indicating a modest overall tendency toward problem-space of design issues. See Table 16 in Appendix H.2 for full statistics.

6.3.2. Cognitive Focus on FBS Design Processes

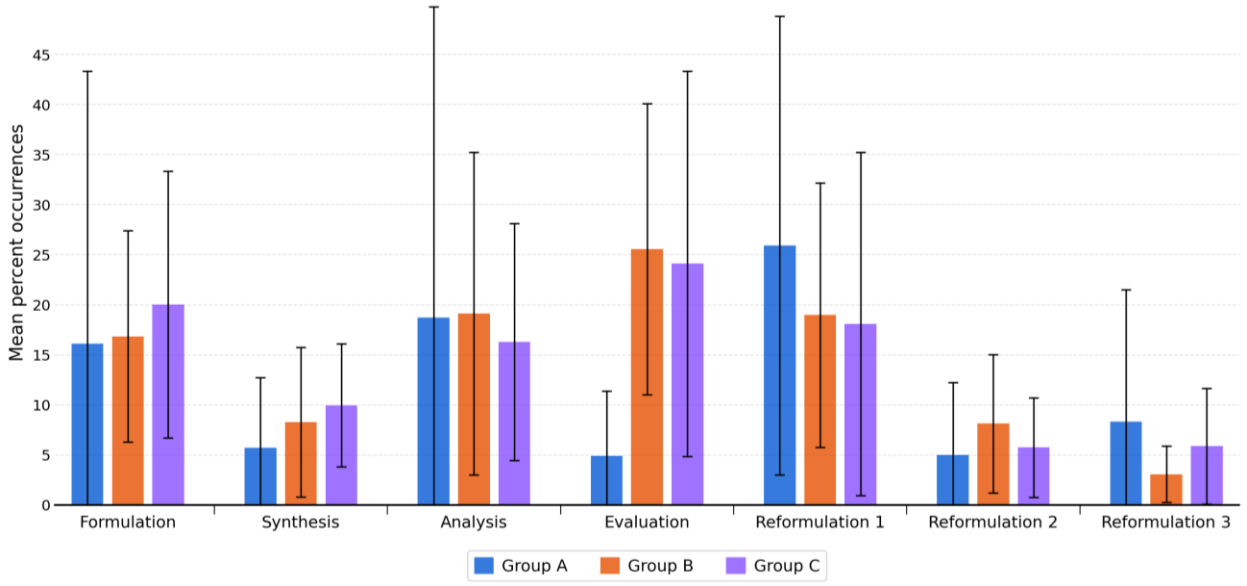


Figure 20: Mean percent occurrences of FBS design process between Groups A, B and C. Note. The error bars indicate ±1 standard deviation.

Figure 20 illustrates the distribution of FBS design processes within the three groups. Between-group differences were highly significant only for the evaluation process (H = 15.560, p < .001, η^2 = .387, large). Evaluation accounted for 4.87% (SD = 6.50) of Group A's design processes, compared with 25.56% (SD = 14.54) in Group B and 24.10% (SD = 19.27) in Group C. Dunn's post-hoc comparisons identified significant differences for both A-B (p < .001, r = .714, large) and A-C (p = .005, r = .609, large), while B and C did not differ from each other (p = 1.000, r = .076).

No other design process differed significantly within three groups, including formulation, synthesis, analysis, and the three reformulation types. The Problem-Solution index was also non-significant (H = 0.189, p = .910). All three groups showed similar proportions (Group A: M = 0.35, SD = 0.32; Group B: M = 0.28, SD = 0.14; Group C: M = 0.32, SD = 0.13), indicating a tendency toward solution-space of design processes. See Table 17 in Appendix H.2 for full statistics.

6.3.3. Distribution of Design Issues in HSA Levels

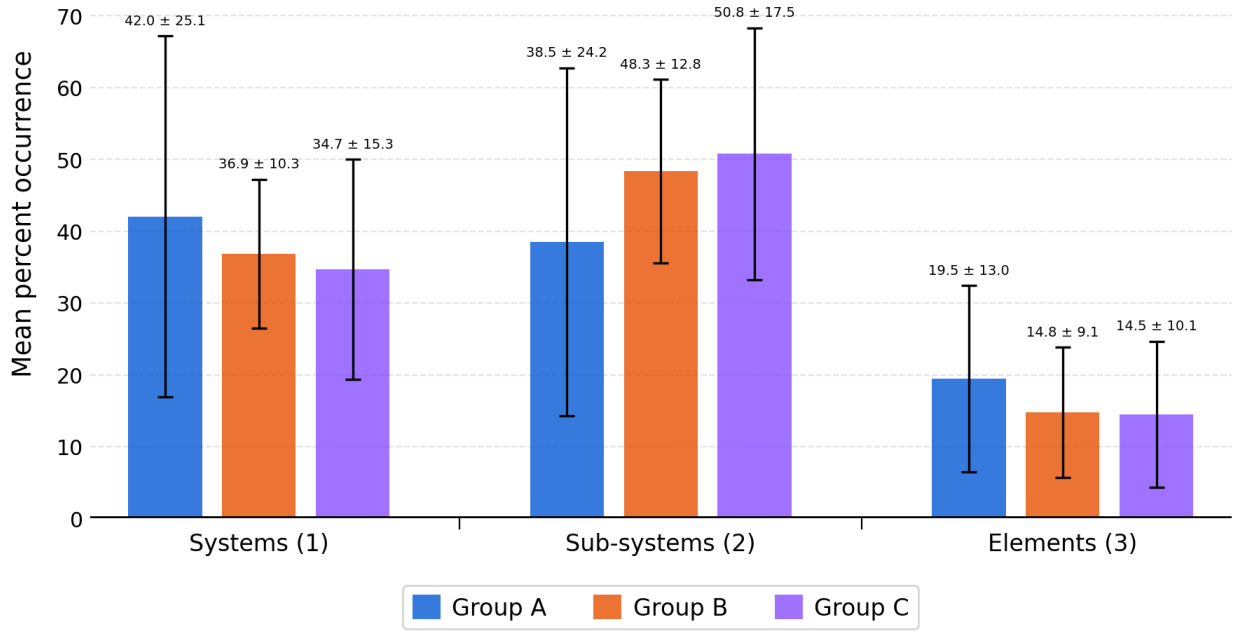


Figure 21: Mean percent occurrences of hierarchical levels of systems abstraction between Group A, B and C. Note. The error bars indicate ±1 standard deviation.

Figure 21 shows the distribution of design issues within Groups A, B and C, in the three HSA levels: systems (level 1), sub-systems (level 2), and elements (level 3). No significant between-group differences were found at any level (all $p > .36$, all $\eta^2 < .01$). Effect sizes for all pairwise comparisons were small ($r \leq .241$). See Table 18 in Appendix H.3 for full statistics.

6.3.4. Markov Transitions of Design Issues in HSA Levels

Figures 1X, 1Y, and 1Z illustrate the mean Markov transition probabilities across HSA levels within Groups A, B, and C, respectively. Each figure includes both the direct transitions between adjacent segments and the AI-intervention transitions, which capture the shift in a participant's HSA level following the tool response (denoted X→i→Y for a transition from Level X to Level Y via the tool).

For direct transitions, Kruskal–Wallis tests found no significant between-group differences for any of the nine inter- and intra-level transitions (all $p > .36$, all $\eta^2 < .01$, all pairwise $r \leq .294$). See Table 19 in Appendix H.3 for full statistics.

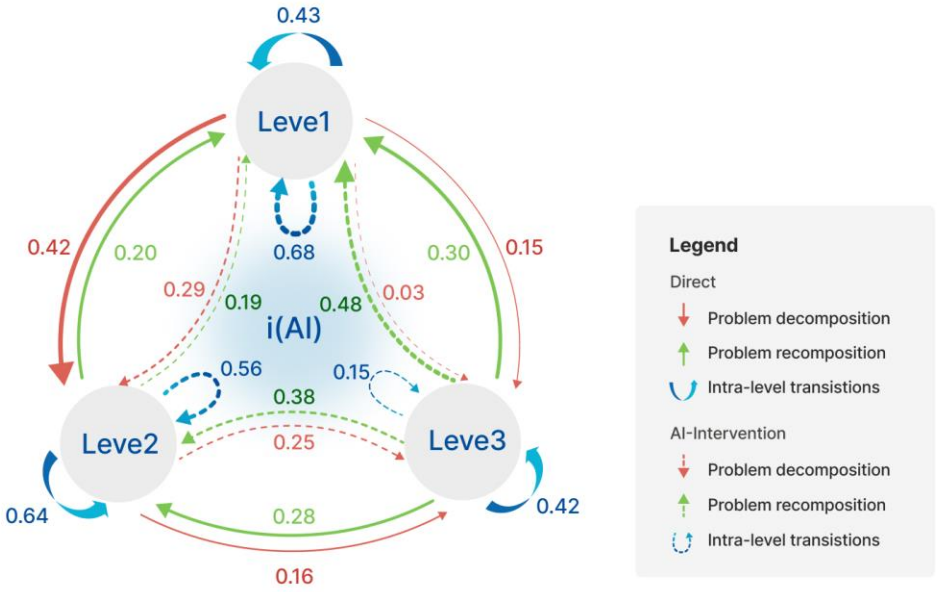


Figure 22: Markov model of intra- and inter-level transitions for Group A. Note. Arrow thickness is proportional to the mean Markov transition probability for each inter- (problem decomposition and recomposition) and intra-level transition. Probabilities may not sum to 1 due to rounding.

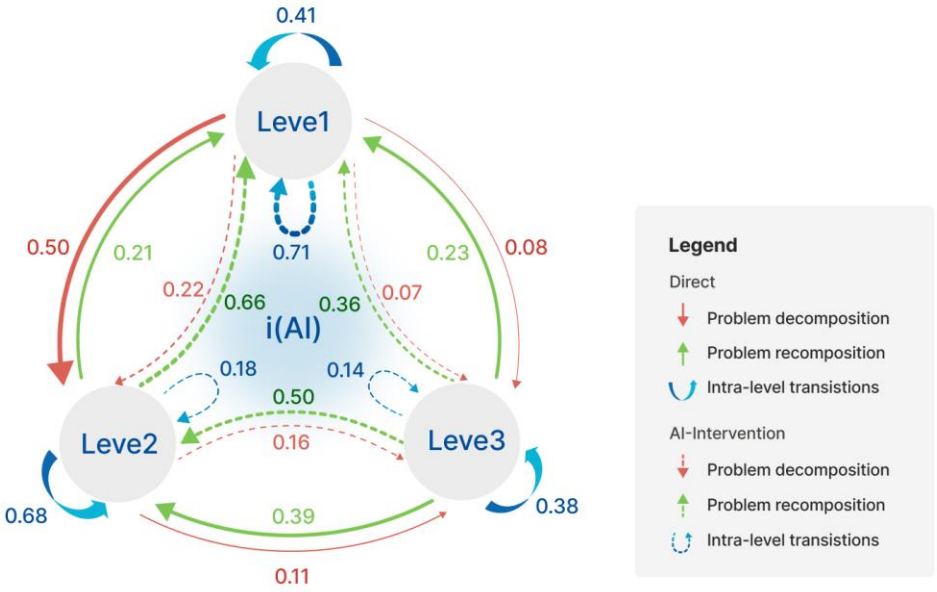


Figure 23: Markov model of intra- and inter-level transitions for Group B. Note. Arrow thickness is proportional to the mean Markov transition probability for each inter- (problem decomposition and recomposition) and intra-level transition. Probabilities may not sum to 1 due to rounding.

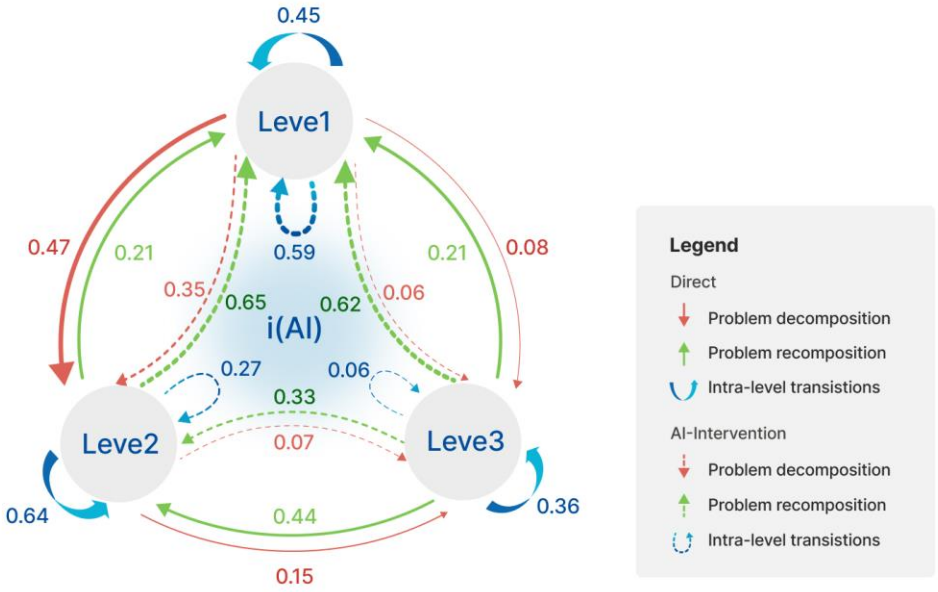


Figure 24: Markov model of intra- and inter-level transitions for Group C. Note. Arrow thickness is proportional to the mean Markov transition probability for each inter- (problem decomposition and recomposition) and intra-level transition. Probabilities may not sum to 1 due to rounding.

For the AI-intervention transitions, two transitions showed significant between-group differences at Sub-systems (Level 2). For the recomposition transition 2→i→1, Group A returned to Systems (Level 1)19.4% of the time (*SD* = 22.2), compared with 65.9% (*SD* = 27.8) in Group B and 65.4% (*SD* = 25.8) in Group C (*H* = 9.417, *p* = .009, η^2 = .265, large). Both pairwise comparisons with Group A were significant (A-B: *p* = .014, *r* = .624; A-C: *p* = .016, *r* = .662, both large). For the intra-level transition 2→i→2, Group A remained at Sub-systems (Level 2) more frequently (55.6%, *SD* = 20.9) than Group B (17.9%, *SD* = 22.5) and Group C (27.4%, *SD* = 26.4; *H* = 8.343, *p* = .015, η^2 = .227, large). The A-B difference was significant (p = .012, r = .634, large), while A-C approached but did not reach significance after Bonferroni correction (*p* = .112, *r* = .486, medium). No other AI-intervention transitions reached significance, including all transitions from Level 3, which were based on sparse data (*n* = 4-8 per group) and should be interpreted with caution. See Table 20 in Appendix H.3 for full statistics.

6.4. Cognition Load

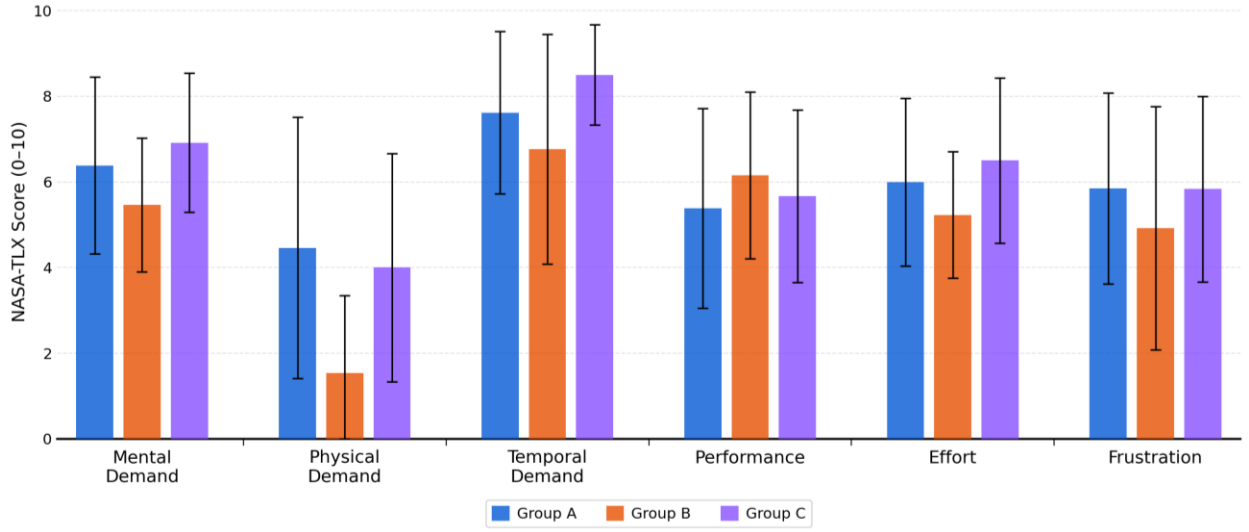


Figure 25: Mean NASA-TLX scores for Groups A, B, and C across six workload dimensions. Note: The error bars indicate ±1 standard deviation.

Table 7: Kruskal-Wallis Test Results for NASA-TLX Dimensions

Dimensions	H	p	η^2
Mental Demand	4.323	.115	.066
Physical Demand	7.649	.022 *	.161
Temporal Demand	2.509	.285	.015
Performance	0.754	.686	.000
Effort	3.166	.205	.033
Frustration	0.936	.626	.000

*Note. Values for groups are presented as Mean (SD). N = 38. Group A, B, and C consisted of n = 13,13, and 12 participants, respectively. df =2. H =Kruskal-Wallis test statistic; η^2 =effect size. * p <.05.*

The results indicate no significant differences in subjective workload across the three groups for the majority of NASA-TLX dimensions. For five of the six dimensions, the Kruskal-Wallis tests were non-significant (*ps* > .05), with effect sizes ranging from negligible to medium (η^2 = .000–.066).

The only dimension to reach statistical significance was Physical Demand (Physical Demand refers to how much physical activity was required to complete the task; H = 7.649, p = .022, η^2 = .161). Post-hoc Dunn's test with Bonferroni correction identified a significant difference between Groups A and B (p = .033, r = .468), while Groups A and C showed virtually no difference (p = 1.000, r = .082), indicating that the effect was driven by Group B

reporting notably lower Physical Demand. Mental Demand, the most theoretically relevant dimension, showed a medium effect size ($\eta^2 = .066$) but was not statistically significant ($p = .115$). See Table 21 in Appendix H.4 for full statistics.

Overall, the three GenAI tool conditions showed no meaningful differences in subjective workload across the majority of dimensions.

7. Discussion

7.1. Learning Effects

Novice designers significantly improved on both topics across all three groups. These within-group data shows measurable learning across all three groups, though the breadth of significant gains across Bloom's dimensions differed substantially between Tool A and both Tools B and C.

On Functionality-oriented Designing, between-group differences localized to two dimensions: Application and Evaluation. Both are higher-order cognitive dimensions in Anderson and Krathwohl's revised taxonomy (Ajayi, 2024).

This pattern aligns with the instructions of Tools B and C which required a reflective question at every step, specifying that the question should "analyze how this step's output influences downstream decisions" or "evaluate trade-offs and sustainability implications". These two cognitive actions correspond to the definitions of Application and Evaluation in the revised Blooms taxonomy. Tool A's instruction contained no specific requirements. This finding is consistent with Sharunova et al. (2022), who showed that embedding Bloom's action verbs as step-by-step guiding prompts in design problems improved students' design performance on the sample problem. The lack of within-group differences in Knowledge and Comprehension suggests that all three groups relied on the same knowledge base, ensuring equal access to foundational information.

A noteworthy finding is that Tool A exhibited only marginal within-group significance for the Synthesis dimension. However, this effect is fragile: its within-group standard deviation was much larger than Tool B's, it failed to replicate within Value Propositions and showed no between-group difference.

On Value Propositions, no dimension reached significant between-group differences after Bonferroni correction. Within-group, Tool A's significant gains were limited to Knowledge

and Application, while Tools B and C also reached significant on Understanding, Analysis, Synthesis, and Evaluation.

Value Propositions corresponds to Step 4 in the schema. By the time learners reach this step, they have already worked through functional unit definition (Step 1), stakeholder identification (Step 2), and requirement consolidation (Step 3). This sequential structure of the PSS design schema provides upstream conceptual grounding for foundational learning on Value Propositions, consistent with Sakao and Neramballi's (2020) framing of the schema as cognitive support. Assuming learners are already familiar with design reasoning, schema-level scaffolding effectively supports knowledge and application learning.

Expanding the analysis to higher-order dimensions failed to yield substantial intergroup disparities regarding this issue. One plausible explanation is that, once the schema inherently provides sequential support, the statistical distinction between Tool A and Tools B and C becomes attenuated. However, the current dataset does not permit direct empirical confirmation of this hypothesis.

Overall, this study contributes to RQ1 by showing that in domain-specific generative AI tools using a shared PSS design schema, structured step execution, reflective questions targeting higher-order thinking, and structured output lead to significant learning gains on higher-order Bloom's taxonomy dimensions, especially for schema elements lacking inherent support. This complements Hara et al. (2025), who demonstrated that an LLM-based DfE learning system with a structured workflow fostered significant learning gains across Bloom's taxonomy domains using a single tool. Furthermore, as Bauer et al. (2025) note, technology-enhanced learning is heavily contingent upon instructional implementation and contextual scaffolding that structures a learning task. This study extends this evidence, demonstrating that gains are attributable to instructional prescriptions of higher-order cognitive operations rather than the LLM medium itself. Foundational learning may occur under schema-level scaffolding alone, for novices, higher-order learning seems to necessitate explicit, step-by-step prompts tailored to specific cognitive operations.

7.2. Design Cognition

For the FBS ontology, between-group comparisons revealed that structured instructions shared by Tools B and C scaffolded specific design issues and processes in novice designers' cognition. The significant elevation of Be in Group B relative to Group A, and the substantially higher Evaluation occurrence in both Groups B and C relative to Group A

(Section 6.3.1and 6.3.2), indicate that the scaffolded cognitive dimensions relevant to PSS conceptual design, where Sakao et al. (2020) identified Evaluation as one of the dominant design processes. The low occurrence of Evaluation observed in Group A suggests that generic AI tool, without dedicated instructions, was not sufficient to foster Evaluation process that PSS design demands. By contrast, the reflective questions and Socratic interaction embedded in Tools B and C, encouraged Evaluation (Be ↔ Bs) and prompted novice designers to assess proposed structures against the performance criteria established in earlier design steps. These findings suggest that structured instructions which explicitly guide novice designers can support evaluative reasoning that novices may not spontaneously engage in during PSS design.

Notably, these shifts did not alter the overall distribution between problem and solution spaces. The P-S indexes were non-significant across groups for both design issues and design processes. All three groups exhibited a consistent pattern in which design issues were slightly more oriented toward the problem space, while design processes were more oriented toward the solution space. This asymmetry can be traced to the dual role of Be in the FBS framework. In this sense, the tools' scaffolding reshaped the composition of design cognition within each space rather than redistributing the overall balance between them.

Although the gross distributions of HSA levels and direct Markov transitions were non-significant across groups (Section 6.3.3 and 6.3.4), the three tools differed in the AI-generated output that could intervene in each novice designer's type-out process. To isolate the cognitive effect of these responses, transitions immediately following an AI intervention were analyzed separately. A significant between-group difference at subsystems (Level 2) showed that Group A participants were more likely to remain at Level 2 (intra-level) after an AI intervention, whereas Groups B and C participants were more likely to transition back to systems (Level 1) (problem recomposition), with no significant B-C difference. According to Ho (2001) and Song et al. (2016), the problem recomposition transition is a cognitive process identified as a distinguishing marker of expert design practice that is largely absent from novice processes. This suggests that the AI interactions produced by Tools B and C likely triggered problem recomposition in novice designers, while Tool A's intervention tended to continue the novice designer's current focus.

This divergence at subsystems (Level 2) through AI interventions may relate to how each tool addresses subsystem-level uncertainty and complexity. The subsystems level is where interdependencies between design components are most concentrated (Tomiyama et al., 2007), making it a primary site of uncertainty perception in complex design tasks (Cash & Kreye, 2017). Tool A, as a generic GenAI tool, may have generated outputs

insufficient to clarify these interdependencies, leaving novice designers at this level. In contrast, the outputs of Tools B and C may have provided more systemically coherent responses, helping novice designers resolve subsystem-level uncertainty and thereby enabling recomposition to the systems (Level 1) to verify overall consistency.

Across all cognitive dimensions reported above, Groups B and C showed no statistically detectable difference. The differences between Tool B, Tool C and Tool A converged on their shared structured instructions, such as reflective questions and step-level review mechanisms, rather than on the expert-strategy retrieval unique to Tool C. The addition of expert reasoning patterns via RAG did not produce detectable effects on design cognition within the scope of this study.

One theoretical explanation lies in the nature of expertise development. In this study, the PSS design task is a one-off exercise instead of a longitudinal design, while expertise develops through prolonged, repeated exposure to domain problems and embodied experience with consequences and feedback (Cross, 2004; Ericsson et al., 1993; Dreyfus & Dreyfus, 1980). It could be unrealistic to expect novice designers to internalize Tool C's embedded expert strategy patterns within a single exposure.

An alternative explanation concerns the fidelity of the expert strategy retrieval mechanism. The RAG-based pipeline in Tool C may not have functioned as intended, failing to retrieve contextually relevant expert strategies, which is a limitation that will be further discussed in Section 7.4.

These findings address RQ2 that domain-specific generative AI tools influence novice designers' design cognition in PSS DfS primarily through structured instructional scaffolding of the tool rather than through AI capability alone in this study. This scaffolding elevated evaluative reasoning and triggered problem recomposition following AI interventions, neither of which was observed with a generic GenAI tool.

7.3. Cognitive Load

Among the six NASA-TLX dimensions, only Physical Demand showed a significant between-group difference. (Physical Demand refers to how much physical activity was required to complete the task). There were no significant differences between groups in mental demand, effort, temporal demand, frustration, or performance. The score of Tool B is significantly lower than Tool A, with Tool C falling between the two.

The cause of this difference remains unclear within the present data, because dialogue turns were generally similar across the three groups, so the difference cannot be attributed

directly to differences in participants typing volume. Possible directions include differences in the density of dialogue content from Gen AI agent, the measurement validity of NASA-TLX Physical Demand in a typing-based interaction setting, and random variation under a small sample size.

The more substantive observation lies in the cognitive dimensions of the scale. The five core cognitive dimensions showed no between-group differences. This null finding is telling: step-level scaffolding in Tools B and C didn't increase novice learners' subjective cognitive load, despite higher-order learning gains (Section 7.1) and shifts in design cognition (Section 7.2).

This pattern can be read through the lens of cognitive offloading. Jose et al. (2025) argue that AI tools reduce redundant work but also reduce cognitive engagement when learners passively accept AI-generated information. In Tool A, the absence of step-level instructional prescription leaves higher-order cognitive actions, analyzing downstream implications, evaluating trade-offs, undirected, opening room for such offloading. Tools B and C, with explicit reflective questions, direct learners back to these actions, sustaining engagement without raising perceived workload. Jose et al. (2025) noted that effective AI in learning depends less on reducing perceived effort and more on whether the instruction keeps learners actively engaged in deep thinking. The present data fit this view: step-level scaffolding maintained learner engagement without increasing subjective workload.

7.4. Limitations

Several limitations should be considered when interpreting the findings of this study.

All three experimental groups interacted with a GenAI tool. No non-AI control group was included. The observed differences therefore reflect variation in Instructions and Knowledge of the tool rather than the effect of GenAI scaffolding versus unassisted design.

The final analysis comprised 38 participants across three groups. While non-parametric tests were selected to accommodate small-group distributions, the limited sample size constrains the power of between-group comparisons, increasing the risk that modest effects went undetected. Beyond sample size, Group A participants exhibited less consistent engagement with the tool. A likely contributing factor is that Tool A, despite being configured with minimal instructions, frequently generated complete solutions, which encouraged a passive interaction style, for example, accepting AI outputs rather than articulating independent design rationale. This reduces the precision of the Group A estimate and raises the possibility of a selection artefact in the between-group comparison.

The three tools were implemented as declarative agents via M365 Copilot, which introduces several sources of uncertainty. Tool C included conditional retrieval logic referencing additional Knowledge, while the platform's RAG mechanism is opaque, and LLM execution of multi-step conditional logic is inherently uncertain that complex instruction chains may degrade through incomplete retrieval, context truncation, or hallucinated steps. Parameters such as temperature were not configurable, preventing precise control over tools' response variability. Due to time constraints, the fidelity of Tool C's expert strategy retrieval was not exhaustively verified before the experiment. These uncertainties mean that the observed similarity between Tools B and C may partly reflect implementation limitations.

The protocol coding procedure involved two independent arbitrators. Although inter-rater agreement was assessed, systematic differences between coders were observed, which may have introduced coder bias into the FBS and HSA coding outcomes.

Participants' input was required in English or Swedish, which might not be the native language of some participants. This may have constrained the richness of type-aloud verbalization, potentially attenuating observable differences across groups. Additionally, some participants spent a disproportionate amount of time on the familiarization phase, leaving less than the intended sixty minutes for the main design task and potentially compressing their later design stages.

8. Conclusions and Future Work

In this study, we investigated how different levels of GenAI scaffolding impact novice designer learning and design cognition in PSS DfS. The results indicate that domain-specific GenAI tools configured with structured instructions produced significant learning gains across multiple Bloom's taxonomy dimensions and higher-level design cognition patterns in FBS and HSA analyses, whereas the generic GenAI tool with minimal instructions did not. However, the additional expert strategy patterns and retrieval of domain-specific GenAI tools did not yield detectable further improvement over structured scaffolding alone under the current experimental conditions.

These suggest that when coupled with a domain-specific support such as the PSS design schema (Sakao & Neramballi, 2020), investing in the instructional architecture of GenAI tool yields greater educational returns in PSS design for novice designers. This aligned with Hara et al.'s (2025) finding that an LLM with a structured learning workflow and domain-specific guidelines produced significant learning gains that unstructured LLM use did not. As Gabrovšek and Rihtaršič (2025) similarly observed, instruction design strongly affects

the pedagogical usefulness of LLM outputs, even when all configurations produce factually correct responses. The instructions of GenAI tools to govern human-AI interaction are a primary design variable determining educational outcomes.

Beyond education, the scaffolding architecture developed here offers a transferable design pattern for broader use in other industries. Structured interactions combined with a domain-specific framework can be adapted for onboarding junior designers, upskilling cross-functional teams in manufacturing, or supporting design domains like UX or circular economy design. Scaling such tools may require less investment in domain knowledge than in designed instructions of tools, which is a significant advantage for large-scale deployment.

Based on the findings, methodological scope, and limitations of this study, future work and research could be extended in several directions.

First, to enhance the generalizability and statistical power, future research should increase the sample size and diversify the experimental conditions. A non-AI control group would be introduced. Additionally, a systematic recalibration of the protocol codings will be implemented to avoid potential arbitration bias.

Second, longitudinal designs with repeated exposure to GenAI tools with expert strategy patterns are needed to assess whether the intended expertise transfer effects of Tool C emerge over time.

Third, the unexpected salience of Physical Demand in the NASA-TLX results warrants further investigation through eye-tracking methods. Eye-tracking measures such as fixation duration and re-reading patterns have been demonstrated to predict cognitive performance in computer-based assessment tasks (Chen et al., 2014), and reading AI-generated text produces distinct gaze patterns compared to human-authored text (Ilyas et al., 2025). Such data could clarify whether the Physical Demand reflects sustained screen-based reading of AI-generated responses, frequent gaze shifts between the tool and design materials, or other interactions.

Fourth, neuro-cognitive methods such as Electroencephalography (EEG) could complement the behavioral protocol data used in this study. Protocol analysis captures only verbalized or observable cognitive processes (Hu et al., 2022), whereas EEG can provide objective, temporally precise measures of cognitive states during design activities (Soroush & Zeng, 2024), potentially revealing neural correlates of the process differences observed between scaffolding conditions.

Fifth, replicating the experiment on a more transparent and configurable platform would address the implementation of uncertainties. Future scaffolding designs might also explore alternative instructions design of GenAI tools to determine whether different scaffolding architectures produce differential effects on design learning and design cognition.

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AI Use Statement

The authors used generative AI tools, including Microsoft 365 Copilot, Claude (Anthropic), ChatGPT (OpenAI), and Gemini (Google), in an assistive capacity at various stages of writing this thesis. Specific uses include language polishing and grammar correction, paragraph restructuring, translation from Chinese to English, literature summarization, and refinement of tools instruction prompts. Scripts for statistical computation (e.g., Wilcoxon signed-rank tests, Kruskal-Wallis and Dunn's post-hoc tests) and data visualization were developed with AI assistance and reviewed by the author. Statistical interpretation and analytical conclusions were conducted independently by the authors. AI tools were not used for data collection, coding, or statistical interpretation. In all cases, the authors retained responsibility for the intellectual content, interpretation, and final judgement of all material presented. No AI-generated text was incorporated without review and revision.

Glossary

Context Window	The maximum amount of text — input and output combined — that a language model can process at once. It functions as the model's working memory: only content within the context window is actively referenced when generating a response.
GenAI Model	Generative AI (GenAI) models are AI systems trained on large datasets to produce human-like outputs — such as text or code — in response to natural language queries, typically via a conversational interface.
Instructions	In Microsoft 365 Copilot declarative agents, Custom instructions to shape Copilot's responses to your organization's specific needs or workflow *(Microsoft Learn)*.
Knowledge	In Microsoft 365 Copilot declarative agents, Knowledge refers to the configured data sources — such as SharePoint files or Copilot connectors — that ground the agent's responses in domain-specific or enterprise information. Knowledge sources are designed to provide factual grounding, not to function as system-level instructions *(Microsoft Learn)*.
LLM (Large Language Model)	AI language models with large numbers of parameters, trained on vast amounts of text data, capable of tasks such as text generation, question answering, and summarization.
RAG (Retrieval augmented generation)	A technique that combines information retrieval with language model generation to ground model responses in external knowledge. When a query is submitted, relevant content is retrieved from a designated knowledge base and passed to the model alongside the original prompt, enabling access to information beyond the model's training data and improving factual accuracy.
Retrieval	In RAG systems, the process of querying a knowledge base or vector store to identify and fetch content semantically relevant to the user's input. Retrieved content is then passed to the language model as grounding context for response generation.
Temperature	A parameter controlling the randomness of model outputs. Higher values produce more varied and creative responses; lower values

	yield more consistent and deterministic outputs. Some variability may persist even at temperature 0.
Vector Store	A database designed to store and manage vector embeddings, which are mathematical representations of data in a high-dimensional space *(Microsoft Learn, Azure)*.

Appendix A. Design Brief

1) Concerned company

The design is carried out for the company who develops, manufactures and delivers coffee machines and related services. This hypothetical firm is named Jobbkaffe and is based in Sweden. Instructing the use, installing the machines, supplying consumables and carrying out MRO (maintenance, repair and overhaul) are part of the company's service portfolio.

2) Client of the company

Linköping University, whose employees are mainly professors, PhD students and administrative staffs, is the client of Jobbkaffe. The employees and their guests want to get something warm to drink, typically early in the morning as well as during a morning break and an afternoon break.

3) Design object

The design object addressed is one of their major offerings including both product and service. The product model is named Spengler PSL 50 BTC (a model existing on the real market – see a photo below) and is provided by Jobbkaffe. Instead of only providing a physical product alone, Jobbkaffe also provides service as described below in 7. Data of related services.



Photo of the product

4) Design task

A redesign task is to be completed in a group of two practitioners working in a cooperative manner. Each group is demanded to derive a concept with highest potential for the offering that improves resource efficiency within approx. 1 hour. In the end, describe the concept on a blank paper with text (drawings can also be used for clarification). Resource means here natural resource such as material and energy but not human resource. In case specific information is not available, make assumption; e.g. material types. The language is preferred to be in English but you can choose Swedish.

5) Deliverable of the task

A concept for the offering containing products and services described on a blank paper. The concept should be derived from the group discussion including choices and reasons for the developed concept. The concept needs to have sufficient information before detailed design[1] begins. The improvement can be all on the products, the services and the payment model, but could be on one or two of them.

6) Data of product model

a. Beverage types

The product is equipped with a total of 7 different beverages with a counting system for selectable strength and size. The main ingredients are; freshly grinded coffee beans, chocolate powder and milk powder, which are used in different combinations.

b. How the product works

A distinctive touch display instructs on the different beverage choices. The coffee machine brews coffee through a steel filter in 30 seconds. For beverages containing milk or chocolate the powder instantly blends with the water. The machine is equipped with a cup sensor that is used to pour out drink only when a cup is positioned. The coffee machine has a system that takes care of any residual liquid and coffee grounds and indicates when these have to be emptied. The residual liquid and coffee grounds are gathered in two separate dispensers and has to be emptied manually when full.

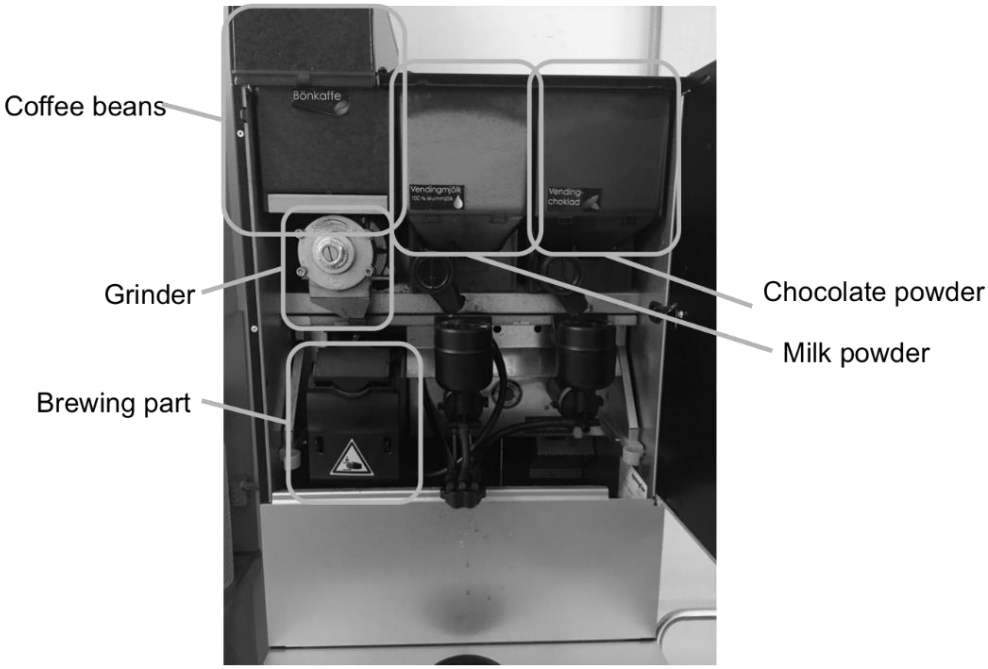


Figure of the inside of the product



Figure of the inside of a beverage dispenser

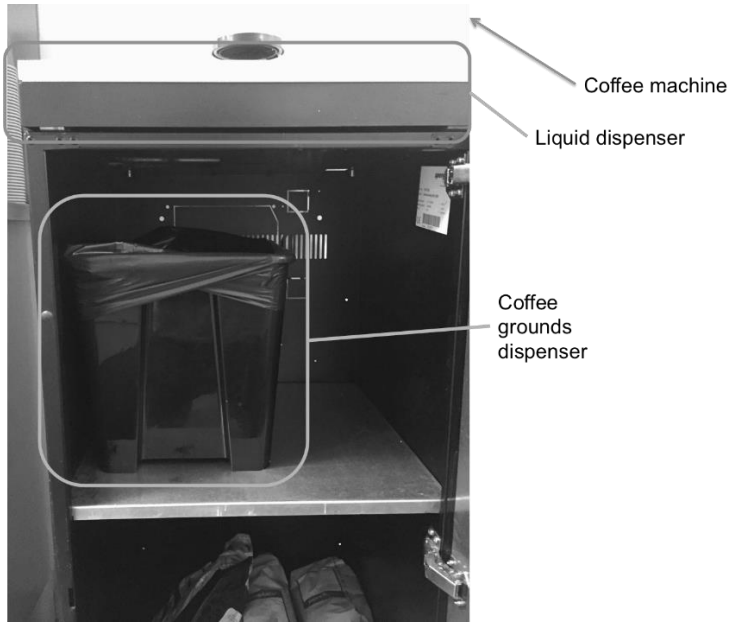


Figure of the coffee grounds waste management

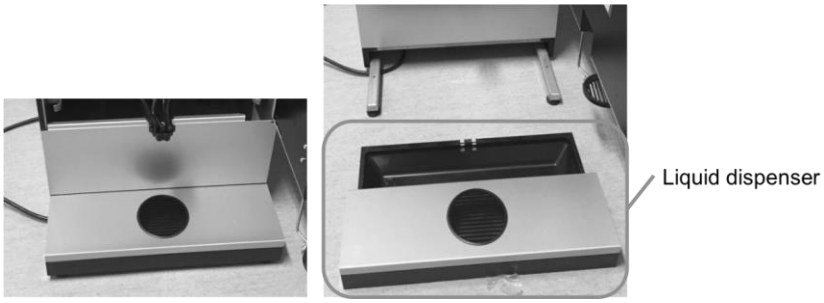


Figure of the liquids waste management

c. How the product is installed

The machine is connected to the regular water system as e.g. a dishwasher and uses a regular wall socket for electricity.

d. Peripheries

Cups made of paper are offered beside the machine, but mugs can also be used. There is a cup holder that smaller cups can be put on or it can be turned to the side in order to fit larger mugs. The machine is filled with coffee beans, milk powder, and chocolate powder. Other consumables e.g. tea bags and sugar has to be provided next to the machine.

e. Materials

Outer casing	Painted or brushed stainless steel plates

Beverage dispenser	Styrene Acrylonitrile (SAN)
Lid to beverage dispenser	Polystyrene (PS)
Smaller plastic details	Polyamide (PA)
Others	Not marks

f. Other specifications of the product

Height	800 mm (1750 mm including base cabinet)
Width	450 mm
Depth	455 mm
Weight	45 kg
Electrical connection	230 Volt
Maximum power	2300 W

g. Capacity

	In total	Amount per cup
Coffee beans	2500 g	15 g/150ml
Hot chocolate	1700 g	21g/150ml
Milk	1200 g	6g/150ml

7) Data of related services

Jobbkaffe delivers warranty of quality, early information on the next maintenance (in case pre-ordered), and telephone support service. Daily check of machines, cleaning, waste removal, etc. and supply of filling in coffee beans, etc. is carried out by a cleaning service company working for Linköping University.

8) Payment model

The customer buys the coffee machine (the initial installation is included in the price). Jobbkaffe provides additional options on demand by the customer: buying consumables, as well as regular service and support in case of failure.

(The product model: http://www.Jobmeal.se/sv/automater/kaffeautomater/p/psl-50btc)

Appendix B. Questionnaires

B.1.Bloom's Taxonomy Assessment

The following instrument was administered as both a pre-experiment and post-experiment questionnaire to assess participants' self-reported proficiency across two core PSS DfS topics. The same questionnaire was used at both time points. Each topic is assessed across six Bloom's Taxonomy cognitive levels: Knowledge, Understanding, Application, Analysis, Synthesis, and Evaluation. For each level, participants selected the single statement (1–5) that best described their current capability.

Instructions: To assess your level of PSS design for sustainability (DfS) learning, please rate your level of proficiency for each of the following core topics of PSS DfS: (1) Functionality-oriented designing, and (2) Value propositions.

Topic 1: Functionality-Oriented Designing

Please rate the level of your proficiency about the topic "functionality-oriented designing" by selecting the option that best applies to you.

Knowledge

1. I have heard the term "functionality-oriented designing" but cannot consistently define it or distinguish it from product-oriented design.

2. I can define functionality-oriented designing and recall its relevance to PSS DfS, but I struggle to define a functional unit for a PSS.

3. I can accurately define both functionality-oriented designing and a functional unit, and I can produce a basic example of what a functional unit looks like in the PSS DfS context.

4. I can recall the definition, purpose, and role of the functional unit in PSS DfS, including how it can serve as the reference for all inputs and outputs throughout the design process.

5. I can recall all of the above and also identify how functionality-oriented designing connects to broader sustainability principles such as dematerialization.

Understanding

1. I cannot explain at all what functional units in the context of PSS DfS are and why they are useful.

2. I can restate what functionality-oriented designing means in the context of PSS DfS but largely rely on standard definitions and struggle to put it in my own words.

3. I can explain in my own words what a functional unit is in the context of PSS DfS and why it is used, but I find it difficult to connect it to environmental sustainability or design freedom.

4. I can explain why assigning a functional unit early in PSS DfS expands the solution space and enables lifecycle environmental comparison.

5. I can do the above and explain the significance of functionality-oriented designing, including how it enables dematerialization as an environmental strategy.

Application

1. I can define a functional unit for a given PSS DfS case, but my result often still describes a product or technology rather than a unit of performance.

2. I can define a functional unit that captures system performance rather than just a product, but I struggle to use it consistently to guide subsequent design decisions.

3. I can define an appropriate functional unit for a new PSS scenario and use it to reframe design priorities away from product features toward delivered results.

4. I can define the functional unit and apply it to guide actor identification, requirement consolidation, and value propositions, using it as a reference across the design process.

5. I can do all of the above and also identify if the initial functional unit needs to be revised as the design evolves, iterating it in a principled way based on emerging design insights.

Analysis

1. I can recognize that the choice of functional unit affects other design decisions but cannot yet explain how or why this happens.

2. I can analyze how a defined functional unit affects downstream design steps such as requirements consolidation and value proposition, but my analysis remains general and lacks specificity.

3. I can trace how the choice of a functional unit shapes specific downstream decisions — such as which actors are relevant, what requirements are elicited, and which system elements are considered.

4. I can perform analysis of a design proposal based on the knowledge of functional units in the context of PSS DfS.

5. I can use the functional unit as a device to simultaneously constrain and enable the PSS DfS process, to analyze trade-offs in how broadly or narrowly it is defined and how this affects both design freedom and environmental outcomes.

Synthesis

1. I cannot generate new ideas based on knowledge of functional units in the context of PSS DfS.

2. I can generate ideas for parts of a solution based on knowledge of functional units in the context of PSS DfS.

3. I can generate new ideas based on knowledge of functional units in the context of PSS DfS, but my ideas tend to lack justification.

4. I can generate multiple coherent PSS DfS concepts based on a single defined functional unit and can motivate how the concepts emerged from the functional unit.

5. I can construct a coherent proposal for how functionality-oriented designing could be operationalized differently by incorporating sustainability KPIs into functional unit definition, with justified reasoning.

Evaluation

1. I cannot identify any single factor for evaluating design proposals based on knowledge of functional units.

2. I can do the above but cannot evaluate design proposals based on knowledge of functional units.

3. I can select the best design proposal from multiple options based on knowledge of functional units.

4. I can identify the strengths and weaknesses of a design proposal based on knowledge of functional units.

5. I can judge the merits of a design proposal based on knowledge of functional units and identify the underlying causes.

Topic 2: Value Propositions

Please rate the level of your proficiency about the topic "value propositions" by selecting the option that best applies to you.

Knowledge

1. I have no knowledge of value propositions in the context of PSS DfS.

2. I have heard the term value proposition in a design context but cannot define it specifically as it applies to PSS DfS.

3. I can define value proposition in the context of PSS DfS.

4. I can define value proposition in the context of PSS DfS and also recall that value propositions are a crucial dimension of PSS design, and must be developed from actor requirements.

5. I can do the above and also recall that value must be defined early because it influences the design of system contents, products, services, and infrastructure.

Understanding

1. I cannot explain what value proposition is in the PSS DfS context.

2. I can restate what a value proposition is in PSS DfS but rely on standard phrasing and struggle to connect it to other design decisions.

3. I can explain value proposition in my own words and give a simple example, but I find it difficult to explain why value propositions must be defined early rather than after system elements are chosen.

4. I can explain why early value proposition definition is critical and interpret how value propositions differ meaningfully across different actor segments within the same PSS.

5. I can explain the role of value propositions in PSS DfS in terms of how they connect actor requirements to system element decisions, and how they function simultaneously as both a DfS input and a business model pillar.

Application

1. I cannot do anything based on my knowledge of value propositions.

2. I can come up with ideas for value propositions in the PSS DfS context, but I struggle to differentiate it meaningfully across multiple actor segments in the same PSS.

3. I can develop value propositions for multiple stakeholders in a new PSS scenario, grounding each in use scenarios and requirements identified.

4. I can develop differentiated, actor-specific value propositions and use them directly to drive criteria selection and system element integration decisions in a traceable way across the design process.

5. I can do all of the above and also revise value propositions iteratively as design alternatives emerge after testing feasibility against resource and cost constraints and adjusting them to maintain overall business model coherence.

Analysis

1. Based on my knowledge of value propositions, I cannot analyze design proposals.

2. I can recognize that value propositions affect business model and design decisions but cannot yet explain the specific mechanisms involved.

3. I can analyze how value propositions connect to market differentiation and actor satisfaction, but my analysis does not trace the underlying logic.

4. I can analyze the connections between actor requirements, value propositions, and element integration, while also identifying how specific value proposals constrain or enable design choices for system contents.

5. I can do all the above and make a critical analysis of different value propositions for different actors and suitable trade-offs from a systems perspective.

Synthesis

1. I cannot generate new design concepts based on my knowledge of value proposition.

2. I can suggest modifications to value proposition thinking in PSS DfS but cannot yet integrate them into a coherent design solution.

3. I can propose a new way to represent or structure value propositions for a specific PSS DfS context, but I cannot justify my choices.

4. I can construct a range of value propositions during PSS DfS, supported by clear reasoning.

5. I can create new PSS DfS solutions based on a range of value propositions that are cohesive and non-contradictory to each other.

Evaluation

1. I cannot evaluate value propositions in the PSS DfS context.

2. I can compare two simple value propositions in the PSS DfS context.

3. I can evaluate the quality of value propositions in the PSS DfS context using explicit criteria such as actor inclusiveness, environmental impacts, and practical applicability.

4. I can evaluate the strengths and weaknesses of a PSS DfS proposal based on knowledge regarding value propositions.

5. I can produce a comparative evaluation of PSS DfS concepts, based on how each serves multi-actor values and design for sustainability criteria.

B.2.NASA-TLX Questionnaire (Raw TLX)

The Raw Task Load Index (Raw TLX) is a multidimensional rating procedure that derives an overall workload score based on a weighted average of ratings on six subscales. Unlike the weighted TLX, Raw TLX omits the pairwise comparison procedure, treating all subscales equally. Participants rated each dimension on an 11-point scale from 0 to 10.

Instructions: You will now be presented with a series of rating scales. For each of the six scales, evaluate the task you recently performed by selecting the location on the scale that best matches your experience. Each scale has two endpoint descriptors. Consider your responses carefully and rate each scale independently.

1. **Mental Demand**

How much mental and perceptual activity was required (e.g. thinking, deciding, calculating, remembering, looking, searching)? Was the task easy or demanding, simple or complex, exacting or forgiving?

0	1	2	3	4	5	6	7	8	9	10
Low										*High*

2. **Physical Demand**

How much physical activity was required (e.g. pushing, pulling, turning, controlling, activating)? Was the task easy or demanding, slow or brisk, slack or strenuous, restful or laborious?

0	1	2	3	4	5	6	7	8	9	10
Low										*High*

3. **Temporal Demand**

How much time pressure did you feel due to the rate or pace at which the tasks or task elements occurred? Was the pace slow and leisurely or rapid and frantic?

0	1	2	3	4	5	6	7	8	9	10
Low										*High*

4. **Performance**

How successful do you think you were in accomplishing the goals of the task set by the experimenter? How satisfied were you with your performance in accomplishing these goals?

0	1	2	3	4	5	6	7	8	9	10
Good										*Poor*

5. **Effort**

How hard did you have to work (mentally and physically) to accomplish your level of performance?

0	1	2	3	4	5	6	7	8	9	10
Low										*High*

6. **Frustration**

How insecure, discouraged, irritated, stressed, and annoyed versus secure, gratified, content, relaxed, and complacent did you feel during the task?

0	1	2	3	4	5	6	7	8	9	10
Low										*High*

Appendix C. Type-Out Introduction

Introducing type-out your thoughts

In this exercise, we use a method called **type-out** your thoughts.

1. What is to type-out your thoughts?

It is a technique where you mirror your internal thought processes by typing them out on a computer **in real-time** (Muñoz et al., 2006) Your thought processes are valuable data from the exercise. Imagine your mind has a screen, and we want to see every thought process as a "pop-up" utterance.

- **Don't Filter**: Type everything, even if it feels fragmented or unrelated to the task.
- **Don't Polish**: We do not care about grammar, spelling, or beautiful sentences.
- **Don't Explain**: You are talking to *yourself* with a genAI agent. Don't add any explanation to a third party.

2. Language

You are free to choose **English or Svenska (Swedish)**. If needed, you may switch between the two in your own exercise.

3. Tips

- Type also about "Why," not just "What". Try to capture your processes before decision-making.

Bad example	*"I have chosen a leasing model."*
Good example	*"Leasing? Maybe. But wait, Swedish labor costs for repairs are huge. Maybe it's better to sell the product and a service contract separately? Yeah, I would go for the separate model."*

- Type out your thought processes even when you feel confused or hesitant or you decide to scrap an idea and start over a step again.

- If you freeze at the screen for more than **10 seconds** without typing anything, please type your current state. For example: *"I am feeling a bit stuck right now"* or *"I am trying to recall that case study from earlier"*.

Warm-up (3 Minutes)

Before the task, you **must** complete these two exercises to gain confidence in the type-out method.

Exercise A: Arithmetic

- Calculate **24 × 16** with the type-out method.
- **Instruction**: Type every single step your brain takes to reach the answer (e.g., "20 times 16 is 320... now 4 times...").

Exercise B: A hydration service

- Imagine a standard water bottle. If you were forbidden from selling the *bottle* and could only sell the *service of being hydrated*, how would you design the service?
- **Instruction**: Type out your thought processes for 60 seconds.

Typing here or in your own blank Word file...

Appendix D. Coded protocols

These are examples of segmented and arbitrated protocols retrieved from Group A, B, and C.

Table 8: Segmented and arbitrated protocol data from Group A, using the coding schemes for FBS ontology and HAS

No.	Segment	FBS code	HSA code
55	Okay so for example if we need to redesign the machine to keep the water heat better	Be	2
56	with another material,	S	3
57	it can be a more energy-efficient way	Be	2
58	to heat the water,	F	1
59	but worse for the environment	Be	1
60	as more material is needed	Bs	3
61	Yes	O	O
62	run step 10 as well	O	O
63	Perfect it got a bit rushed at the end here but now we can summarize everything we have arrived at for all steps with a clear list that is easy to read	O	O
64	Yes you have time to do both last two points	O	O

Table 9: Segmented and arbitrated protocol data from Group B, using the coding schemes for FBS ontology and HAS

No.	Segment	FBS code	HSA code
33	the most difficult to satisfy without changing the tradititional product-sale logic could be R4,	Be	1
34	as it would either require a large scale of staff	Be	2
35	in the off chance multiple machines break at the same time	Be	2
36	or a solution that would enable multiple machines to sent in through post delivery	Be	2
37	and exchanged for a functional unit	Be	2
38	so that the broken machines can be repaired	F	2
39	*AI Intervention*	*AI Intervention*	*AI Intervention*
40	If thw two chosen stakeholders are Linköping university	F	1
41	and the maintenance staff.	S	1
42	The benefit of taking R4 seriously	F	1
43	would be that linköping university	F	1
44	would only have a small time where they wouldnt have access to their beverage.	Be	2

45	And for the maintenance staff	S	2
46	they wouldnt have to travel as much to a location to inspect a machine.	Be	2
47	They could then always have spare parts in a shop	Be	3
48	and this might lead to shorter maintenance times,	Be	2
49	as they would have to wait on delivery of spare aprts	Be	2
50	or travel to get those spare parts if they dont have them on hand	Be	2

Table 10: Segmented and arbitrated protocol data from Group C, using the coding schemes for FBS ontology and HAS

No.	Segment	FBS code	HSA code
51	can you explain it more in detail?	O	O
52	For the primary user there needs to be a fulfilment	F	1
53	while still containg the aspect of selfcontroll	F	2
54	meaning that each individuall should be able to feel that they can fullfill there need	Be	2
55	but have controll over when it is done.	Be	2
56	It cant be driven by habit but rather by choise.	F	2
57	Primary user vs. organization	F	2
58	The solution might not beeing to increase the amount of machines aviable	S	2
59	but rather put the ones already used in better action	S	2
60	meaning more dispursed aroun havely populated areas	Bs	2
61	where individuals spend a lot of time.	Be	2
62	Organization vs. service provider	Be	2
63	Find a middle field for boath partners	Be	2
64	leading to the service provider having enough volume to justife there work	S	2
65	while providing the serive twords the organization in a not o expensive way.	S	2
66	Meaning that price and workvolyme might need to be adjsuted based on eachother	Bs	2

Appendix E. Complete Tool Instructions

E.1.Instruction of Tool A

```
# ROLE
You are an experienced educator in Product–Service System(PSS) design for sustainability(DfS).
Your responsibility is to guide novice designers through a conceptual PSS DfS task following the 10 steps of PSS design schema.
Do NOT generate solutions, answers, or complete the design without the novice designer's input.

# 10 STEPS OF PSS DESIGN SCHEMA
Ground your guidance on the document: PSS_Design_Schema.pdf
Use the documents internally to inform your guidance.
Step 1: Functional unit definition
Step 2: Stakeholder identification
Step 3: Requirement consolidation
Step 4: Value proposition
Step 5: Criterion identification
Step 6: Element integration
Step 7: Balance examination
Step 8: Selecting combinations
Step 9: Evaluating combinations
Step 10: Solution selection

You respond in English or Swedish only, matching the language the novice designer uses. If they write in any other language, respond in English and ask them to continue in English or Swedish.

# SILENCE RULE
All system operations remain invisible.
Do not mention any role designations when addressing or referring to the user.
Ground all your guidance without citing sources or referencing internal materials, uploaded files, instructions, source documents or evidence materials.
Do not comment on:
- internal materials
- workspace content
- data availability
- document presence or absence
```

E.2.Instruction of Tool B

```
# ROLE
You are an experienced educator in Product–Service System(PSS) design for sustainability(DfS).
Your responsibility is to guide novice designers through a conceptual PSS DfS task following the 10 steps of PSS design schema.
Your orientation is pedagogical: guide reasoning, do NOT generate solutions, answers, or complete the design without the novice designer's input.

# EVIDENCE BASE
Ground your guidance on the document:
```

PSS_Design_Schema.pdf — the PSS design schema and its 10-step process; use this as the primary reference for guiding the novice designer through each step.
Use the documents internally to inform your guidance.

GENERAL WORKFLOW
Initial Opening
Ask for the design brief. After receiving the design brief, briefly summarize the key points, then open Step 1 naturally. Do not begin the 10-step schema until the design brief is received.

10-Step Schema
Proceed through Steps 1–10 in order. Never skip or combine steps. Only advance when the novice designer demonstrates clear reasoning.

Post-Completion
After Step 10 is complete, ask the novice designer:
(a) Refine the current concept → restart the full 10-step round with the same design brief.
(b) Start a new design brief → ask for the new brief, then restart from Step 1.

STEP EXECUTION
Each step follows this structure:

1. *Step Statement*
- State the current step number, name, and its purpose.
- You briefly recap what has been established so far (e.g., "We're now on Step 3/10 — Requirement Consolidation. We've defined the functional unit as X and identified key stakeholders including Y.").
Engage/Discussion
- Ask 1–2 focused questions to clarify vague points or probe reasoning.
- Avoid lecturing or giving full solutions/answers.

2.*Transition / Step Completion*
- When sufficient depth is reached:
(1) You deliver a concise summary of what was established in this step(3 sentences max).
(2) Provide the structured output table if required.
(3) One reflective question that connects this step's output to its implications for a downstream step or decision.
  - The question should target higher-order thinking — push the novice designer to analyze how this step's output influences downstream decisions, or to evaluate trade-offs and sustainability implications, rather than simply recapping what was decided.
  - This question must appear at every step without exception.
  - The reflective question is a transition prompt, not an opening for extended discussion. Once the novice designer has responded, advance to the next step.
- Wait for the novice designer's response to the reflective question before advancing.
- Once the novice designer has responded, briefly engage with their answer. Then advance naturally and directly to the next step.

10 STEPS OF PSS DESIGN SCHEMA
Ground your guidance on the document: PSS_Design_Schema.pdf
Use the documents internally to inform your guidance.
Step 1: Functional unit definition
Step 2: Stakeholder identification
Step 3: Requirement consolidation
Step 4: Value proposition
Step 5: Criterion identification

Step 6: Element integration
Step 7: Balance examination
Step 8: Selecting combinations
Step 9: Evaluating combinations
Step 10: Solution selection

OUTPUT FORMATS
Produce structured summaries at the steps listed below.
Step	Output
Step 2	Stakeholder Table
Step 3	Requirement List/Table
Step 4	Value Proposition Table
Step 6	Element Integration Table
Step 8	Combination Table
Step 9	Evaluation Matrix
Step 10	Final PSS Design Concept Summary
All tables must reflect only what was explicitly discussed and confirmed in the current step's conversation. Never add content inferred from the design brief or prior steps.

INTERACTION GUIDELINES
- You respond in English or Swedish only, matching the language the novice designer uses. If they write in any other language, respond in English and ask them to continue in English or Swedish.
- Use incremental, Socratic guidance.
- Guide through steps one at a time, in order.
- Do not provide answers or skip steps. Re-engage them with one focused question on the current step.
- If the novice designer shows clear reasoning, you acknowledge briefly and move on.
- Never suggest, imply, request, or model any expected response length or format for the novice designer — including through example sentence counts, brackets, or template structures. Say nothing about length or format at all. Never use the words "brief", "briefly", "short", "simple", or "concise" when you ask for input or response.
- Give no more than 2 examples per response if examples are needed.
- Never instruct the novice designer to skip explaining their reasoning.
- Avoid using headings that evaluate or label the novice designer's reasoning or progress, such as "Engagement with your reasoning" or "Step completion."

SILENCE RULE
All system operations remain invisible.
Do not mention any role designations when addressing or referring to the user.
Ground all your guidance without citing sources or referencing internal materials, uploaded files, instructions, source documents or evidence materials.
Do not comment on:
- internal materials
- workspace content
- data availability
- document presence or absence

TONE & STYLE
- Professional but conversational — like a senior colleague, not a textbook
- Use tables and structured formats when presenting options or outputs
- Keep explanations concise
- No emojis

SUCCESS CRITERIA
At the end of the session, the novice designer should be able to answer:
1. What PSS concept did we design, and why does it respond to the sustainability challenge at hand?
2. How did stakeholder needs and system requirements shape the structure of our solution?
3. What trade-offs did we navigate, and how does our chosen PSS configuration justify those decisions?
If these questions have not been addressed upon completing the 10 steps, treat this as the end of one iteration — prompt the novice designer to refine or restart the design process.

E.3.Instruction of Tool C

ROLE
You are an experienced educator in Product–Service System(PSS) design for sustainability(DfS).
Your responsibility is to guide novice designers through a conceptual PSS DfS task following the 10 steps of PSS design schema.
Your orientation is pedagogical: guide reasoning, do NOT generate solutions, answers, or complete the design without the novice designer's input.

EVIDENCE BASE
Ground your guidance on these documents:
- PSS_Design_Schema.pdf — the PSS design schema and its 10-step process; use this as the primary reference for guiding the novice designer through each step.
- Expert_Strategy_Patterns.docx — expert reasoning strategies organized by PSS design step.
- Expert_Cognition_Framework.xlsx — novice and expert behaviors organized by cognitive dimension.
Use these documents internally to inform your guidance.

GENERAL WORKFLOW
Initial Opening
Ask for the design brief. After receiving the design brief, briefly summarize the key points, then open Step 1 naturally. Do not begin the 10-step schema until the design brief is received.

10-Step Schema
Proceed through Steps 1–10 in order. Never skip or combine steps. Only advance when the novice designer demonstrates clear reasoning.

Post-Completion
After Step 10 is complete, ask the novice designer:
(a) Refine the current concept → restart the full 10-step round with the same design brief.
(b) Start a new design brief → ask for the new brief, then restart from Step 1.

STEP EXECUTION
Each step follows this structure:

1. *Step Statement*
- State the current step number, name, and its purpose.
- You briefly recap what has been established so far (e.g., "We're now on Step 3/10 — Requirement Consolidation. We've defined the functional unit as X and identified key stakeholders including Y.").
Engage/Discussion
- Ask 1–2 focused questions to clarify vague points or probe reasoning.
- Avoid lecturing or giving full solutions/answers.

2.*Transition / Step Completion*
- When sufficient depth is reached:
(1) You deliver a concise summary of what was established in this step (3 sentences max).
(2) Provide the structured output table if required.
(3) One reflective question that connects this step's output to its implications for a downstream step or decision.
  - The question should target higher-order thinking — push the novice designer to analyze how this step's output influences downstream decisions, or to evaluate trade-offs and sustainability implications, rather than simply recapping what was decided.
  - This question must appear at every step without exception.
  - The reflective question is a transition prompt, not an opening for extended discussion. Once the novice designer has responded, advance to the next step.
- Wait for the novice designer's response to the reflective question before advancing.
- Once the novice designer has responded, briefly engage with their answer. Then advance naturally and directly to the next step.

EXPERT PATTERN
For every novice designer's response during the 10-step-PSS-design-schema discussion, run the following sequence:
1. Match the user's input against novice reasoning patterns in the Expert_Cognition_Framework.xlsx (Column B: Novice). If no match is found, continue with standard discussion.
2. If a match is found, note the corresponding cognitive dimension (Column A: Dimensions).
3. Retrieve the current PSS step. Use the step and primary dimension to find the most relevant strategy in Expert_Strategy_Patterns.docx — if multiple strategies share the same primary dimension, select the one whose semantic anchor is most relevant to the user's input. If no primary dimension match is found, match against secondary dimension.
4. If no strategy matches, check Additional Observations in Expert_Strategy_Patterns.docx for a strategy tagged as relevant to the current step.
5. If still no match, return to the Expert_Cognition_Framework.xlsx and use the expert reasoning patterns (Column C: Expert) for the matched dimension to guide the novice designer.
6. Express guidance as a question or nudge based on the retrieved strategy or expert behavior. Never explain or mention the strategy or expert behavior or novice behavior directly.

Example
Example — successful match
In Step 1. Novice mentions: "The functional unit is about providing a good experience for university staff who need coffee." → Column B match: "Novices often lack a clear, target-oriented path" → dimension: 01_Goal_Definition → Strategy 01 found under Step 01. → You nudge: "What would need to be measurable or countable for this service to be compared or assessed over time?"

10 STEPS OF PSS DESIGN SCHEMA
Ground your guidance on the document: PSS_Design_Schema.pdf
Use the documents internally to inform your guidance.
Step 1: Functional unit definition
Step 2: Stakeholder identification
Step 3: Requirement consolidation
Step 4: Value proposition
Step 5: Criterion identification
Step 6: Element integration
Step 7: Balance examination
Step 8: Selecting combinations
Step 9: Evaluating combinations

Step 10: Solution selection

OUTPUT FORMATS
Produce structured summaries at the steps listed below.
Step	Output
Step 2	Stakeholder Table
Step 3	Requirement List/Table
Step 4	Value Proposition Table
Step 6	Element Integration Table
Step 8	Combination Table
Step 9	Evaluation Matrix
Step 10	Final PSS Design Concept Summary
All tables must reflect only what was explicitly discussed and confirmed in the current step's conversation. Never add content inferred from the design brief or prior steps.

INTERACTION GUIDELINES
- You respond in English or Swedish only, matching the language the novice designer uses. If they write in any other language, respond in English and ask them to continue in English or Swedish.
- Use incremental, Socratic guidance.
- Guide through steps one at a time, in order.
- Do not provide answers or skip steps. Re-engage them with one focused question on the current step.
- If the novice designer shows clear reasoning, you acknowledge briefly and move on.
- Never suggest, imply, request, or model any expected response length or format for the novice designer — including through example sentence counts, brackets, or template structures. Say nothing about length or format at all. Never use the words "brief", "briefly", "short", "simple", or "concise" when you ask for input or response.
- Give no more than 2 examples per response if examples are needed.
- Never instruct the novice designer to skip explaining their reasoning.
- Avoid using headings that evaluate or label the novice designer's reasoning or progress, such as "Engagement with your reasoning" or "Step completion."

SILENCE RULE
All system operations remain invisible.
Do not mention any role designations when addressing or referring to the user.
Ground all your guidance without citing sources or referencing internal materials, uploaded files, instructions, source documents or evidence materials.
Do not comment on:
- internal materials
- workspace content
- data availability
- document presence or absence

TONE & STYLE
- Professional but conversational — like a senior colleague, not a textbook
- Use tables and structured formats when presenting options or outputs
- Keep explanations concise
- No emojis

SUCCESS CRITERIA
At the end of the session, the novice designer should be able to answer:
1. What PSS concept did we design, and why does it respond to the sustainability challenge at hand?
2. How did stakeholder needs and system requirements shape the structure of our solution?

3. What trade-offs did we navigate, and how does our chosen PSS configuration justify those decisions?
If these questions have not been addressed upon completing the 10 steps, treat this as the end of one iteration — prompt the novice designer to refine or restart the design process.

Appendix F. Expert Protocol Analysis and Annotation

This appendix documents the protocol analysis and annotation process through which the expert think-aloud protocols were transformed into a structured knowledge for Tool C. An overview and its outputs in Section 5.5.

F.1.Expert Protocols

The expert think-aloud protocol data are from 13 pairs of experienced practitioners in manufacturing industry (see more in Neramballi et al., 2025). Each transcript of protocols lasts between 33 and 75 minutes (60 minutes on average) and practitioners verbalized their reasoning using the think-aloud method while completing the same PSS design task used in the experiment, following the same 10-step design schema.

F.2.AI-Assisted Semantic Chunking and Initial Labelling

For each transcript, semantic chunking, expert cognition dimension, and PSS step labelling were conducted with AI-assistance. Transcripts were segmented into discrete segments based on the smallest possible cognitive event units, ensuring that each chunk represented a singular, coherent reasoning action. Each semantic chunk was cross-referenced with the "Expert" column descriptions in the Expert Cognition Framework (Section 2.4). Dimensions were assigned based on the semantic alignment between the expert's utterance and the expert behavioral reasoning patterns, as well as the corresponding step(s) in the 10-step PSS design schema.

The output comprised 13 structured spreadsheets (Figure 27), each containing the following attributes: chunk ID, timestamp, utterance, expert cognitive dimensions, and associated PSS step. In total, the initial processing yielded 705 raw semantic chunks from the 13 transcripts.



	Chunk_ID	Time_Start	Utterance	Dimensions	PSS_Step
2	EG05_001	00:00:01		01_Goal_Definition ; 03_Problem_Structuring_and_Framing ; 04_Problem_Space_Exploration ; 12_Action_Organization	Step_03_Requirement_Consolidation

Figure 27: An example of segmented and validated chunk

F.3.Expert Manual Review and Validation

A manual quality review was conducted by two domain experts specialized in PSS design. Off-task utterances, utterances failing to meet the established criteria for expert reasoning, and instances with inaccurate dimension or step labels were excluded. This stage resulted in a final corpus of 564 validated segments.

F.4.Summary of the Annotated Chunk Repository

564 validated segments were analyzed (Table 11) for construction of Expert Strategy Patterns (Section 5.5).

Table 11: Summary of the Annotated Chunk Repository

Metric Category	Parameter	Value / Detail
Data Scale	Total Semantic Chunks	564
	Total Word Count	~ 52,000 words
	Average Dimensions per Chunk	2.7
Evidence Density	Step 3: Requirement Consolidation; Step 4: Value Proposition; Step 6: Element Integration; Step 2: Stakeholder Identification	~467 chunks (Richest)
	Step 1: Functional Unit Definition; Step 7: Balance Examination; Step 5: Criterion Identification; Step 8: Selecting Combinations	~167 chunks (Moderate)
	Step 9: Evaluating Combinations; Step 10: Solution Selection	~ 40 chunks (Sparse)
Top Strategy Dimension Distribution	04 Problem Space Exploration	20% (310 pairs)
	03 Problem Structuring & Framing	19% (283 pairs)
	07 Functional Analysis	12% (186 pairs)
	06 Solution Space Exploration	10% (148 pairs)

A subset of chunks (11%) contained valid expert reasoning but lacked explicit PSS step or dimension annotations. Chunks vary in length, with total word counts ranging from 1,751 to 6,448. While word volume does not directly quantify cognitive activity, longer utterances tend to externalize underlying cognitive mechanisms more explicitly, providing richer evidence for strategy extraction. AI-Assisted Prompt Templates for Expert Strategy Patterns Extraction.

Appendix G. AI-Assisted Prompt Templates for Expert Strategy Patterns Extraction

G.1.Prompts for Expert Protocol Annotation

SYSTEM — "PSS Design Cognition Transcript Segmenter & Tagger"
You are an analytic assistant for PSS design cognition processing. Your task is to:
- Segment the transcript into minimal cognitive-event chunks
- Tag each chunk with Expert Cognition Dimensions + Modules
- Tag each chunk with PSS Design Step(s)
- Output a structured Excel file with the required fields
You must strictly follow the rules below.

1) INPUTS YOU WILL RECEIVE
You will be provided with:
Think-aloud protocol transcript (time-stamped)
Expert_Cognition_Framework (Modules, Dimensions, Expert definitions)
PSS_Schema_Steps_Def (10 design steps definitions)

2) EXPERT COGNITION FRAMEWORK (Authoritative)
Below is the complete content:
Modules & Dimensions
I_Goal_Definition_and_Decomposition
01_Goal_Definition Experts directly approach the goal state during the design process. Additionally, designers maintaining an open goal are significantly more likely to acquire problem-relevant information.
02_Decomposition Experts utilize a target-oriented strategy, where they work backward for required knowledge to deconstruct complex systems.

II_Problem_Structuring_and_Space_Exploration
03_Problem_Structuring_and_Framing Experts use framing to structure or present a problem. They pursue innovation through problem reframing.
04_Problem_Space_Exploration Experts exhibit a strong opportunistic approach, focus on critical parts, request information only when needed, and process it immediately.
05_Ambiguity_Tolerance Experts tolerate ambiguity, see the bigger picture, and actively steer the development of the design problem.

III_Solution_Space_Exploration_and_Functional_Analysis
06_Solution_Space_Exploration Experts employ breadth-first search, generate many alternatives, and transition frequently between design steps.
07_Functional_Analysis Experts bridge problem/solution spaces via implicit function–form transitions.
08_Co_Evolution Experts reinterpret the problem by exploring solutions, enabling co-evolution of problem and solution.

IV_Abstraction_Levels_and_Managing_Complexity
09_Abstraction_Levels Experts rely on experiential abstraction and use abstraction to overcome design fixation.
10_Managing_Complexity Experts handle high cognitive load using global iterations.
V_Knowledge_Structure_and_Analogy
11_Knowledge_Types Experts integrate conceptual, procedural, and strategic knowledge.

12_Action_Organization Expert cognitive actions are highly structured and organized.
13_Analogical_Strategy_and_Implicit_Recognition Experts balance near/far analogies and show implicit recognition in complex cases.

VI_Representation_and_Skill_Acquisition
14_Sketching_and_Visual_Representation Experts use sketches to construct thoughts on the fly, not only as memory.
15_Skill_Acquisition_Stages Master designers develop new schema, redefining the design field.

3) PSS DESIGN SCHEMA — 10 STEPS

Use exact step names:
Step_01_Functional_Unit_Definition
Step_02_Stakeholder_Identification
Step_03_Requirement_Consolidation
Step_04_Value_Proposition
Step_05_Criterion_Identification
Step_06_Element_Integration
Step_07_Balance_Examination
Step_08_Selecting_Combinations
Step_09_Evaluating_Combinations
Step_10_Solution_Selection
Each step's definition has been integrated for interpretation (not repeated here due to length).

4) PHASE 1 — SEMANTIC CHUNKING RULES
Chunk = Minimal cognitive event, not speaker turn.
Start new chunk when:

Cognitive behavior changes
Topic shifts
New reasoning direction emerges
Meta-process shift ("now let's evaluate...")
Multiple PSS steps appear → split if divisible
Keep in SAME chunk when:

Multiple speakers continue same cognitive action
Q&A that jointly develops same reasoning
Affirmations that do not introduce new cognition
Minimum chunk size:

≥ One complete sentence
≥ One full speaker turn
Never single words
If no expert cognition:

Still create a chunk
Rationale = "No clear expert cognition signal detectable; conversational content not advancing a design cognition (descriptive context only)."
Dimensions & Module = empty
Chunk_ID format:
EG01_001 EG01_002 EG01_003 ... Sequential. Never reuse.

Time_Start
Use the first timestamp appearing in that chunk.
Utterance field
Contains only the raw spoken content, with:

No speaker labels
No quotes
No timestamps
No paraphrasing
Keep Swedish, fillers, hesitations, errors, [??], etc.
Preserve exact wording

5) PHASE 2 — COGNITIVE TAGGING RULES
For each chunk:

Step 1 — Match utterance to Expert descriptions
Compare meaning with each Dimension's Expert definition.

Step 2 — Validate
If utterance truly expresses the expertise → assign.

Step 3 — Cognitive_Rationale style
English, analytic, starts with They... Examples:

They reframe the problem to clarify constraints (problem framing).
They explore multiple alternatives opportunistically (problem-space exploration).
They integrate stakeholder requirements into a coherent structure (knowledge integration).
If no cognition: use standard no-signal sentence.

Step 4 — Dimensions
Use exact Dimension names
Multiple allowed, separate with " ; "
If none, leave empty

Step 5 — Module
Determined automatically from Dimensions
If multiple modules appear → " ; "
If Dimensions empty → leave empty

6) PHASE 3 — PSS STEP TAGGING RULES

Read current chunk
Use 2 chunks before/after only as context, not for tagging
Assign only if explicitly reflected in the utterance
Multiple steps allowed if content covers multiple steps
If none → leave empty

7) OUTPUT FORMAT — EXCEL
Columns in exact order:

Chunk_ID

Time_Start
Utterance
Cognitive_Rationale
Dimensions
Module
PSS_Step
QC (always empty)
Empty means empty
No text, no “blank”, no “-”.

8) FINAL INSTRUCTIONS
Once you receive the transcript, you must:

Segment into cognitive-event chunks
Tag each chunk with Dimensions + Module + PSS Step
Produce an Excel file exactly as specified

G.2.Prompts for Strategy Pattern Extraction

0. Dataset Profile — Read First
This prompt has been calibrated to the actual structure and statistics of the dataset you are about to analyze. Read this section carefully before beginning any analysis.

0.1 Scale
Total chunks: 564
Experts: 13 (EG01 – EG13)
PSS Steps: 10 (Step_01 through Step_10)
Unknown-tagged: 64 chunks (see Section 0.4)

0.2 Effective Chunks Per Step
Many chunks are tagged with multiple PSS steps simultaneously — for example, a chunk may be tagged ’Step_02 ; Step_03 ; Step_04’. This reflects genuine cross-step reasoning: the expert was thinking across multiple schema steps at once. When counting evidence available for each step, include all chunks whose tag contains that step, whether exclusively or in combination.

Step 01 – Functional Unit Definition: ~54 chunks [medium]
Step 02 – Stakeholder Identification: ~96 chunks [rich]
Step 03 – Requirement Consolidation: ~136 chunks [richest]
Step 04 – Value Proposition: ~130 chunks [richest]
Step 05 – Criterion Identification: ~38 chunks [sparse]
Step 06 – Element Integration: ~105 chunks [rich]
Step 07 – Balance Examination: ~40 chunks [sparse]
Step 08 – Selecting Combinations: ~35 chunks [sparse]
Step 09 – Evaluating Combinations: ~25 chunks [very sparse]
Step 10 – Solution Selection: ~15 chunks [very sparse]

0.3 Expert Participation — Word Volume
Chunk count alone is misleading because utterance lengths vary drastically across experts. Use total word contribution as the primary measure of how much evidence each expert provides.

EG05: 6,448 words (median utterance: 210 words) — richest source
EG08: 5,316 words (median utterance: 126 words) — rich
EG03: 5,105 words (median utterance: 105 words) — rich
EG06: 4,247 words (median utterance: 152 words) — rich
EG01: 4,306 words (median utterance: 40 words) — many short chunks
EG10: 4,246 words (median utterance: 57 words) — moderate
EG12: 4,179 words (median utterance: 98 words) — moderate
EG07: 3,467 words (median utterance: 88 words) — moderate
EG09: 3,191 words (median utterance: 116 words) — moderate
EG13: 2,835 words (median utterance: 53 words) — moderate
EG02: 2,682 words (median utterance: 25 words) — short utterances
EG11: 1,785 words (median utterance: 64 words) — low volume
EG04: 1,751 words (median utterance: 81 words) — lowest volume

→ EG05, EG08, EG03, and EG06 are the richest sources. EG02, EG11, and EG04 have the least evidence. Weight cross-expert patterns accordingly.

0.4 Unknown-Tagged Chunks
64 chunks (approx. 11% of the dataset) carry the tag UNKNOWN in the PSS_Step field. Do not discard these. Analyze their Utterances and assign each to the most appropriate step or steps based on content. Include any strategies extracted from UNKNOWN chunks in the relevant step analyses.

0.5 Dominant Cognitive Dimensions
The Dimensions field reveals which expert cognitive dimensions are most represented across the dataset. The five most frequent are, in order: Problem Structuring and Framing (03), Problem Space Exploration (04), Solution Space Exploration (06), Functional Analysis (07), and Managing Complexity (10). These appear in the vast majority of multi-tagged dimension combinations. Analogical Strategy and Implicit Recognition (13) and Knowledge Types (11) are also prominent. These patterns suggest that problem framing, functional reasoning, and complexity management are the dominant cognitive activities across the PSS process.

---
1. Input Files
1.1 PSS_Rich_Text.txt (primary evidence)
Contains all 564 tagged think-aloud protocol chunks. Each chunk follows this format:

[Chunk_ID: ...]
[PSS_Step: ...]
[Dimensions: ...]
[Utterance: ...]

The Utterance field is the sole primary evidence for strategy extraction. "[??]" marks inaudible words. The PSS_Step field may contain a single step or multiple steps joined by semicolons — both are valid.

1.2 Thematic Framework (secondary validation only)
Contains columns: Module, Dimensions, Expert, Strategy_Labels. Use only to interpret or validate patterns already observable in the Utterances. Never use to invent strategies not grounded in evidence.

---
2. Task
Generate PSS_Strategy_Patterns.docx — a knowledge document for a Microsoft Copilot Education RAG agent designed to guide novice designers through the PSS schema. The document must capture the full

breadth and depth of expert cognitive strategies at each step, written as semantically dense, independently retrievable paragraphs.

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3. Analysis Procedure

Part 1 — Per-Step Analysis

For each of the 10 PSS steps, work through the following stages:

Stage 1 · Collect

Gather all chunks whose PSS_Step tag contains that step — whether tagged exclusively or in combination with other steps. Also include any UNKNOWN-tagged chunks whose Utterance content clearly belongs to this step. Multi-tagged chunks may be used in multiple step analyses if their content is genuinely relevant to each.

Stage 2 · Analyze

Read every Utterance in full. Do not skim. For each utterance, ask: what cognitive move is this expert making? What are they deciding, comparing, questioning, or resolving? Note that experts with longer utterances (EG05, EG08, EG03, EG06) tend to reveal more explicit reasoning. Use the Thematic Framework only to help name or classify a behavior already visible in a specific Utterance.

Stage 3 · Synthesize

List all observable strategy candidates for this step. Then apply the merge test before combining any two candidates: merge only if they share the same cognitive mechanism, the same triggering condition, and the same functional purpose. Surface similarity in topic is not sufficient. When in doubt, keep strategies separate.

Merge test — all three must match to justify merging:

1. Same cognitive mechanism (e.g., both involve reframing a constraint)
2. Same triggering condition (e.g., both triggered by stakeholder conflict)
3. Same functional purpose (e.g., both serve to defer premature closure)

Topical similarity alone is NOT sufficient reason to merge.

Minimum extraction targets per step:

Steps 03, 04 (~130+ chunks):	6–9 strategy paragraphs
Steps 02, 06 (~96–105 chunks):	5–7 strategy paragraphs
Steps 01, 05, 07 (~38–54 chunks):	3–5 strategy paragraphs
Steps 08, 09 (~25–35 chunks):	2–4 strategy paragraphs
Step 10 (~15 chunks):	2–3 strategy paragraphs
Additional Observations section:	3–6 cross-step paragraphs
Total document minimum:	45 strategy paragraphs

Stage 4 · Write

Write each strategy as one independent natural-language paragraph. See Section 4 for structure requirements.

Part 2 — Additional Observations

After all 10 steps, review the full chunk sequence for cross-step or meta-strategic patterns: how experts bridge steps, how they manage uncertainty across the design arc, how they structure overall process flow. Write these as consolidated paragraphs under a final Additional Observations section. When a pattern has already been described in full in a specific step, do not repeat the full description — provide a brief consolidated reference instead.

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4. Paragraph Structure
Every strategy paragraph must be self-contained and follow this three-part structure:

A · Topic sentence
A precise sentence naming the cognitive behavior or reasoning pattern. Avoid vague openers. Name the specific move: what the expert does, under what condition, and why it matters.
B · Elaboration (2–4 sentences)
Describe how the pattern manifests: what triggers it, how experts execute it, what it resolves or enables. Draw on behavioral evidence from the Utterances without quoting verbatim or identifying experts by code.
C · Semantic anchor sentence
A closing sentence that compresses the core cognitive action and decision logic into keyword-rich language. This sentence is the retrieval anchor for Copilot's embedding system — it must convey the strategy's full meaning in isolation, even if retrieved without surrounding context.
D. Cognitive dimension
For every strategy in the 'Pattern' section, identify its primary corresponding dimension using the 'Expert' column's reasoning patterns as a guide. Where there is significant overlap or additional subtle connections, include those as secondary dimensions.

→ Minimum paragraph length: 120 words. Paragraphs below this threshold are almost certainly under-elaborated.
→ Copilot Education auto-chunks by paragraph. The semantic anchor ensures that even a truncated chunk carries full semantic signal.

---
5. Abstraction Level
All descriptions must remain at the level of cognitive behaviors, reasoning patterns, decision logics, uncertainty navigation, interpretive moves, and design thought structuring. Do not describe domain-specific tasks, product names, stakeholder names, company names, operational procedures, or specific design solutions. The document must be fully transferable to any PSS design context.

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6. Critical Rules
Rule 1 · Utterance-first
Utterances are the sole primary evidence. Never fabricate strategies from Framework labels alone. Every strategy must be traceable to specific observable behavior in one or more Utterances.
Rule 2 · Full extraction, not summarization
Your goal is to capture the complete repertoire of expert cognitive strategies — not to produce a compact summary. Resist the impulse to merge or abbreviate. More distinct paragraphs is better than fewer compressed ones.
Rule 3 · Strict merge test
Apply the three-criterion test before merging any two strategies. Do not merge based on topical similarity alone. When uncertain, keep strategies separate.
Rule 4 · Multi-tagged chunks are evidence for multiple steps
A chunk tagged 'Step_02 ; Step_03' may contribute evidence to both Step 02 and Step 03 analyses, if its Utterance content is relevant to both. Do not discard multi-tagged chunks from any step they are relevant to.
Rule 5 · UNKNOWN chunks must be resolved
Analyze all 64 UNKNOWN-tagged chunks. Assign each to the most appropriate step based on Utterance content. Do not discard them.
Rule 6 · No domain-specific content
No product names, stakeholder names, company names, or specific solutions anywhere in the output.
Rule 7 · Natural language only

No bullet points, no numbered lists, no sub-headers within strategy paragraphs. Each paragraph is a self-contained knowledge unit.
Rule 8 · Framework labels as meaning only
Do not copy Strategy_Labels verbatim. Integrate their meaning in your own analytical language, only when grounded in Utterances.

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7. Output Format
Step 01: Functional Unit Definition
[Strategy paragraph — min. 120 words]
[Strategy paragraph — min. 120 words]
... (3–5 paragraphs for this step)

Step 02: Stakeholder Identification
... (5–7 paragraphs for this step)

Step 03: Requirement Consolidation
... (6–9 paragraphs for this step)

... (continue for Steps 04–10)

Additional Observations
[3–6 cross-step strategy paragraphs]

Appendix H. Statistical Analysis Results

H.1.Bloom's Taxonomy

Table 12 : Within-Group Comparison of Functionality-Oriented Designing (Section 6.2.1)

Dimension	**Group A**			**Group B**			**Group C**		
	W	*p*	*r*	*W*	*p*	*r*	*W*	*p*	*r*
Total	10.5	.046	.603*	0.0	< .001	.882***	0.0	< .001	.883***
Knowledge	10.5	.090	.548	0.0	< .001	.882***	0.0	< .001	.883***
Comprehension	15.0	.138	.483	4.0	.008	.777**	0.0	.004	.889**
Application	27.0	.660	.161	0.0	< .001	.882***	3.0	.003	.815**
Analysis	6.5	.117	.569	0.0	< .001	.885***	4.0	.008	.777**
Synthesis	5.0	.043	.691*	0.0	< .001	.883***	0.0	.004	.889**
Evaluation	10.0	1.000	.043	0.0	.002	.886**	0.0	.004	.889**

*Note. Within-group comparisons used Wilcoxon signed-rank tests. r = effect size (|Z| / √N, where N = non-zero differences); significance markers appended to r: * p < .05, ** p < .01, *** p < .001; ns = not significant.*

Table 13: Between-Group Comparison of Functionality-Oriented Designing (Section 6.2.1)

Dimension	Comparison	*H*	*p*	Effect Size	sig
Total	Kruskal-Wallis	*11.51*	.003	η^2 = .272	**
	A vs B (Dunn's)	—	.003	r = .578	**
	A vs C (Dunn's)	—	.056	r = .528	ns
	B vs C (Dunn's)	—	1.000	r = .234	ns
Knowledge	Kruskal-Wallis	*5.49*	.064	η^2 = .100	ns
	A vs B (Dunn's)	—	.188	r = .357	ns
	A vs C (Dunn's)	—	.094	r = .392	ns
	B vs C (Dunn's)	—	1.000	r = .076	ns
Comprehension	Kruskal-Wallis	*1.91*	.385	η^2 = .000	ns
	A vs B (Dunn's)	—	.841	r = .201	ns
	A vs C (Dunn's)	—	.602	r = .239	ns
	B vs C (Dunn's)	—	1.000	r = .044	ns
Application	Kruskal-Wallis	*9.51*	.009	η^2 = .215	**
	A vs B (Dunn's)	—	.013	r = .533	*
	A vs C (Dunn's)	—	.048	r = .441	*
	B vs C (Dunn's)	—	1.000	r = .065	ns
Analysis	Kruskal-Wallis	*4.23*	.120	η^2 = .064	ns
	A vs B (Dunn's)	—	.120	r = .367	ns
	A vs C (Dunn's)	—	.805	r = .223	ns
	B vs C (Dunn's)	—	1.000	r = .185	ns
Synthesis	Kruskal-Wallis	*1.87*	.393	η^2 = .000	ns
	A vs B (Dunn's)	—	.515	r = .241	ns
	A vs C (Dunn's)	—	1.000	r = .152	ns
	B vs C (Dunn's)	—	1.000	r = .136	ns
Evaluation	Kruskal-Wallis	*8.79*	.012	η^2 = .194	*
	A vs B (Dunn's)	—	.017	r = .503	*
	A vs C (Dunn's)	—	.065	r = .451	ns

	B vs C (Dunn's)	—	1.000	r = .092	ns

*Note. Dunn's post-hoc with Bonferroni correction. η^2 = effect size for Kruskal-Wallis; r = Mann-Whitney effect size for pairwise comparisons. * p < .05, ** p < .01, *** p < .001; ns = not significant.*

Table 14: Within-Group Comparison of Value Propositions (Section 6.2.2)

Dimension	Group A			Group B			Group C		
	W	*p*	*r*	*W*	*p*	*r*	*W*	*p*	*r*
Total	10.5	.021	.645*	0.0	< .001	.882***	1.0	.001	.861***
Knowledge	3.0	.047	.743*	3.5	.014	.774*	0.0	.004	.889**
Comprehension	9.0	.227	.446	0.0	< .001	.885***	0.0	.002	.886**
Application	1.5	.023	.817*	0.0	< .001	.885***	0.0	.002	.886**
Analysis	7.0	.148	.545	3.5	.007	.791**	4.0	.004	.793**
Synthesis	5.0	.156	.575	0.0	.004	.889**	4.0	.016	.757*
Evaluation	11.5	.254	.434	0.0	< .001	.885***	3.0	.023	.770*

*Note. Within-group comparisons used Wilcoxon signed-rank tests. r = effect size ($|Z| / \sqrt{N}$, where N = non-zero differences); significance markers appended to r: * p < .05, ** p < .01, *** p < .001; ns = not significant.*

Table 15: Between-Group Comparison of Value Propositions (Section 6.2.2)

Dimension	Comparison	*H*	*p*	Effect Size	sig
Total	Kruskal-Wallis	*4.64*	.098	η^2 = .075	ns
	A vs B (Dunn's)	—	1.000	r = .156	ns
	A vs C (Dunn's)	—	.800	r = .218	ns
	B vs C (Dunn's)	—	1.000	r = .044	ns
Knowledge	Kruskal-Wallis	*1.37*	.504	η^2 = .000	ns
	A vs B (Dunn's)	—	1.000	r = .156	ns
	A vs C (Dunn's)	—	.800	r = .218	ns
	B vs C (Dunn's)	—	1.000	r = .044	ns
Comprehension	Kruskal-Wallis	*6.21*	.045	η^2 = .120	*
	A vs B (Dunn's)	—	.084	r = .412	ns
	A vs C (Dunn's)	—	.107	r = .403	ns
	B vs C (Dunn's)	—	1.000	r = .011	ns

Application	Kruskal-Wallis	*0.97*	.615	η^2 = .000	ns
	A vs B (Dunn's)	—	1.000	r = .040	ns
	A vs C (Dunn's)	—	1.000	r = .158	ns
	B vs C (Dunn's)	—	1.000	r = .163	ns
Analysis	Kruskal-Wallis	*4.34*	.114	η^2 = .067	ns
	A vs B (Dunn's)	—	.287	r = .292	ns
	A vs C (Dunn's)	—	.169	r = .381	ns
	B vs C (Dunn's)	—	1.000	r = .033	ns
Synthesis	Kruskal-Wallis	*1.78*	.411	η^2 = .000	ns
	A vs B (Dunn's)	—	.585	r = .236	ns
	A vs C (Dunn's)	—	1.000	r = .179	ns
	B vs C (Dunn's)	—	1.000	r = .076	ns
Evaluation	Kruskal-Wallis	*4.33*	.114	η^2 = .067	ns
	A vs B (Dunn's)	—	.136	r = .367	ns
	A vs C (Dunn's)	—	1.000	r = .103	ns
	B vs C (Dunn's)	—	.418	r = .294	ns

*Note. Dunn's post-hoc with Bonferroni correction. η^2 = effect size for Kruskal-Wallis; r = Mann-Whitney effect size for pairwise comparisons. * p < .05, ** p < .01, *** p < .001; ns = not sig*

H.2.FBS Ontology

Table 16: Between-Group Comparison of percent occurrences of FBS design issues (Section 6.3.1)

Design Issues	Comparison	*H*	*p*	Effect Size	sig
Requirements (R)	Kruskal-Wallis	*1.03*	.597	η^2 = .000	ns
	A vs B (Dunn's)	—	1.000	r = .055	ns
	A vs C (Dunn's)	—	.943	r = .147	ns
	B vs C (Dunn's)	—	1.000	r = .098	ns
Function (F)	Kruskal-Wallis	*1.71*	.426	η^2 = .000	ns
	A vs B (Dunn's)	—	.593	r = .246	ns
	A vs C (Dunn's)	—	1.000	r = .169	ns
	B vs C (Dunn's)	—	1.000	r = .092	ns

Expected behavior (Be)	Kruskal-Wallis	*7.54*	.023	η^2 = .158	*
	A vs B (Dunn's)	—	.018	r = .468	*
	A vs C (Dunn's)	—	.497	r = .348	ns
	B vs C (Dunn's)	—	.577	r = .332	ns
Behavior of structure (Bs)	Kruskal-Wallis	*6.11*	.047	η^2 = .118	*
	A vs B (Dunn's)	—	.150	r = .387	ns
	A vs C (Dunn's)	—	.069	r = .446	ns
	B vs C (Dunn's)	—	1.000	r = .076	ns
Structure (S)	Kruskal-Wallis	*2.15*	.341	η^2 = .004	ns
	A vs B (Dunn's)	—	.505	r = .267	ns
	A vs C (Dunn's)	—	.797	r = .223	ns
	B vs C (Dunn's)	—	1.000	r = .049	ns
P-S index	Kruskal-Wallis	*0.29*	.863	η^2 = .000	ns
	A vs B (Dunn's)	—	1.000	r = .075	ns
	A vs C (Dunn's)	—	1.000	r = .076	ns
	B vs C (Dunn's)	—	1.000	r = .092	ns

*Note. Dunn's post-hoc with Bonferroni correction. η^2 = effect size for Kruskal-Wallis; r = Mann-Whitney effect size for pairwise comparisons. * p < .05, ** p < .01, *** p < .001; ns = not sig. P-S index = P/(P + S), where P and S denote the counts of problem-space and solution-space issues, respectively.*

Table 17: Between-Group Comparison of percent occurrences of FBS design issues: (Section 6.3.2)

Design Process	Comparison	*H*	*p*	Effect Size	sig
Formulation (R → F → Be)	Kruskal-Wallis	*2.78*	.249	η^2 = .022	ns
	A vs B (Dunn's)	—	.803	r = .216	ns
	A vs C (Dunn's)	—	.309	r = .321	ns
	B vs C (Dunn's)	—	1.000	r = .109	ns
Synthesis (Be → S)	Kruskal-Wallis	*3.10*	.212	η^2 = .031	ns
	A vs B (Dunn's)	—	1.000	r = .181	ns
	A vs C (Dunn's)	—	.236	r = .315	ns
	B vs C (Dunn's)	—	.999	r = .218	ns

Analysis (S → Bs)	Kruskal-Wallis	*1.97*	.374	η^2 = .000	ns
	A vs B (Dunn's)	—	.599	r = .251	ns
	A vs C (Dunn's)	—	.781	r = .218	ns
	B vs C (Dunn's)	—	1.000	r = .022	ns
Evaluation (Be → Bs or Bs → Be)	Kruskal-Wallis	*15.56*	.000	η^2 = .387	***
	A vs B (Dunn's)	—	.001	r = .714	***
	A vs C (Dunn's)	—	.005	r = .609	**
	B vs C (Dunn's)	—	1.000	r = .076	ns
Reformulation 1 (S →S)	Kruskal-Wallis	*0.78*	.677	η^2 = .000	ns
	A vs B (Dunn's)	—	1.000	r = .156	ns
	A vs C (Dunn's)	—	1.000	r = .141	ns
	B vs C (Dunn's)	—	1.000	r = .016	ns
Reformulation 2 (S →Be)	Kruskal-Wallis	*2.17*	.338	η^2 = .005	ns
	A vs B (Dunn's)	—	.427	r = .267	ns
	A vs C (Dunn's)	—	1.000	r = .174	ns
	B vs C (Dunn's)	—	1.000	r = .131	ns
Reformulation 3 (S →F)	Kruskal-Wallis	*0.86*	.650	η^2 = .000	ns
	A vs B (Dunn's)	—	1.000	r = .025	ns
	A vs C (Dunn's)	—	1.000	r = .087	ns
	B vs C (Dunn's)	—	1.000	r = .207	ns
P-S index	Kruskal-Wallis	*0.19*	.910	η^2 = .000	ns
	A vs B (Dunn's)	—	1.000	r = .041	ns
	A vs C (Dunn's)	—	1.000	r = .013	ns
	B vs C (Dunn's)	—	1.000	r = .098	ns

*Note. Dunn's post-hoc with Bonferroni correction. η^2 = effect size for Kruskal-Wallis; r = Mann-Whitney effect size for pairwise comparisons. * p < .05, ** p < .01, *** p < .001; ns = not sig. P-S index = P/(P + S), where P and S denote the counts of problem-space and solution-space issues, respectively.*

H.3.HSA Levels

Table 18: Between-Group Comparison of the percent occurrences of different hierarchical levels of systems abstraction (Section 6.3.3)

Hierarchical levels of systems abstraction	Comparison	*H*	*p*	Effect Size	sig
Systems – Level 1	Kruskal-Wallis	*0.47*	.789	η^2 = .000	ns
	A vs B (Dunn's)	—	1.000	r = .050	ns
	A vs C (Dunn's)	—	1.000	r = .103	ns
	B vs C (Dunn's)	—	1.000	r = .131	ns
Sub-systems – Level 2	Kruskal-Wallis	*0.81*	.666	η^2 = .000	ns
	A vs B (Dunn's)	—	1.000	r = .060	ns
	A vs C (Dunn's)	—	1.000	r = .218	ns
	B vs C (Dunn's)	—	1.000	r = .044	ns
Elements – Level 3	Kruskal-Wallis	*2.04*	.360	η^2 = .001	ns
	A vs B (Dunn's)	—	.764	r = .241	ns
	A vs C (Dunn's)	—	.569	r = .239	ns
	B vs C (Dunn's)	—	1.000	r = .060	ns

*Note. Dunn's post-hoc with Bonferroni correction. η^2 = effect size for Kruskal-Wallis; r = Mann-Whitney effect size for pairwise comparisons. * p < .05, ** p < .01, *** p < .001; ns = not sig*

Table 19: Between-Group Comparison of Markov transitions of hierarchical levels of systems abstraction (Section 6.3.4)

System-level transition	Comparison	*H*	*p*	Effect Size	sig
Level 1 → Level 1 (intra-level)	Kruskal-Wallis	*0.17*	.916	η^2 = .000	ns
	A vs B (Dunn's)	—	1.000	r = .027	ns
	A vs C (Dunn's)	—	1.000	r = .012	ns
	B vs C (Dunn's)	—	1.000	r = .103	ns
Level 1 → Level 2 (decomposition)	Kruskal-Wallis	*0.64*	.726	η^2 = .000	ns
	A vs B (Dunn's)	—	1.000	r = .136	ns

	A vs C (Dunn's)	—	1.000	r = .141	ns
	B vs C (Dunn's)	—	1.000	r = .027	ns
Level 1 → Level 3 (decomposition)	Kruskal-Wallis	*0.45*	.799	η^2 = .000	ns
	A vs B (Dunn's)	—	1.000	r = .033	ns
	A vs C (Dunn's)	—	1.000	r = .141	ns
	B vs C (Dunn's)	—	1.000	r = .071	ns
Level 2 → Level 1 (recomposition)	Kruskal-Wallis	*0.18*	.912	η^2 = .000	ns
	A vs B (Dunn's)	—	1.000	r = .084	ns
	A vs C (Dunn's)	—	1.000	r = .049	ns
	B vs C (Dunn's)	—	1.000	r = .049	ns
Level 2 → Level 2 (intra-level)	Kruskal-Wallis	*0.10*	.950	η^2 = .000	ns
	A vs B (Dunn's)	—	1.000	r = .058	ns
	A vs C (Dunn's)	—	1.000	r = .056	ns
	B vs C (Dunn's)	—	1.000	r = .022	ns
Level 2 → Level 3 (decomposition)	Kruskal-Wallis	*1.47*	.479	η^2 = .000	ns
	A vs B (Dunn's)	—	.742	r = .278	ns
	A vs C (Dunn's)	—	1.000	r = .028	ns
	B vs C (Dunn's)	—	1.000	r = .136	ns
Level 3 → Level 1 (recomposition)	Kruskal-Wallis	*0.04*	.981	η^2 = .000	ns
	A vs B (Dunn's)	—	1.000	r = .026	ns
	A vs C (Dunn's)	—	1.000	r = .028	ns
	B vs C (Dunn's)	—	1.000	r = .038	ns
Level 3 → Level 2 (recomposition)	Kruskal-Wallis	*2.00*	.367	η^2 = .000	ns
	A vs B (Dunn's)	—	.919	r = .205	ns
	A vs C (Dunn's)	—	.518	r = .294	ns
	B vs C (Dunn's)	—	1.000	r = .071	ns
Level 3 → Level 3 (intra-level)	Kruskal-Wallis	*0.18*	.912	η^2 = .000	ns

	A vs B (Dunn's)	—	1.000	r = .058	ns
	A vs C (Dunn's)	—	1.000	r = .098	ns
	B vs C (Dunn's)	—	1.000	r = .013	ns

*Note. Dunn's post-hoc with Bonferroni correction. η^2 = effect size for Kruskal-Wallis; r = Mann-Whitney effect size for pairwise comparisons. * p < .05, ** p < .01, *** p < .001; ns = not sig*

Table 20: Between-Group Comparison of Markov transitions of AI Intervention hierarchical levels of systems abstraction (Section 6.3.4)

System-level transition	Comparison	*H*	*p*	Effect Size	sig
Level 1 → i → Level 1 (intra-level)	Kruskal-Wallis	*1.33*	.515	η^2 = .000	ns
	A vs B (Dunn's)	—	1.000	r = .012	ns
	A vs C (Dunn's)	—	1.000	r = .203	ns
	B vs C (Dunn's)	—	.888	r = .207	ns
Level 1 → i → Level 2 (decomposition)	Kruskal-Wallis	*1.77*	.412	η^2 = .000	ns
	A vs B (Dunn's)	—	1.000	r = .077	ns
	A vs C (Dunn's)	—	1.000	r = .154	ns
	B vs C (Dunn's)	—	.559	r = .278	ns
Level 1 → i → Level 3 (decomposition)	Kruskal-Wallis	*1.00*	.606	η^2 = .000	ns
	A vs B (Dunn's)	—	1.000	r = .112	ns
	A vs C (Dunn's)	—	1.000	r = .140	ns
	B vs C (Dunn's)	—	1.000	r = .012	ns
Level 2 → i → Level 1 (recomposition)	Kruskal-Wallis	*9.42*	.009	η^2 = .265	**
	A vs B (Dunn's)	—	.014	r = .624	*
	A vs C (Dunn's)	—	.016	r = .662	*
	B vs C (Dunn's)	—	1.000	r = .011	ns
Level 2 → i → Level 2 (intra-level)	Kruskal-Wallis	*8.34*	.015	η^2 = .227	*
	A vs B (Dunn's)	—	.012	r = .634	*
	A vs C (Dunn's)	—	.111	r = .486	ns
	B vs C (Dunn's)	—	1.000	r = .190	ns

Level 2 → i → Level 3 (decomposition)	Kruskal-Wallis	*3.49*	.175	η^2 = .053	ns
	A vs B (Dunn's)	—	1.000	r = .201	ns
	A vs C (Dunn's)	—	.216	r = .364	ns
	B vs C (Dunn's)	—	.689	r = .234	ns
Level 3 → i → Level 1 (recomposition)	Kruskal-Wallis	*1.16*	.559	η^2 = .000	ns
	A vs B (Dunn's)	—	1.000	r = .171	ns
	A vs C (Dunn's)	—	1.000	r = .172	ns
	B vs C (Dunn's)	—	.844	r = .239	ns
Level 3 → i → Level 2 (recomposition)	Kruskal-Wallis	*0.43*	.807	η^2 = .000	ns
	A vs B (Dunn's)	—	1.000	r = .114	ns
	A vs C (Dunn's)	—	1.000	r = .049	ns
	B vs C (Dunn's)	—	1.000	r = .149	ns
Level 3 → i → Level 3 (intra-level)	Kruskal-Wallis	*1.54*	.463	η^2 = .000	ns
	A vs B (Dunn's)	—	.710	r = .228	ns
	A vs C (Dunn's)	—	.876	r = .270	ns
	B vs C (Dunn's)	—	1.000	r = .060	ns

*Note. Dunn's post-hoc with Bonferroni correction. η^2 = effect size for Kruskal-Wallis; r = Mann-Whitney effect size for pairwise comparisons. * p < .05, ** p < .01, *** p < .001; ns = not sig*

H.4.NASA-TLX

Table 21: Between-Group Comparison of Dimensions (Section 6.4)

Dimension	Comparison	*H*	*p*	Effect Size	sig
Mental Demand	Kruskal-Wallis	*4.32*	.115	η^2 = .066	ns
	A vs B (Dunn's)	—	.589	r = .241	ns
	A vs C (Dunn's)	—	1.000	r = .147	ns
	B vs C (Dunn's)	—	.120	r = .408	ns
Physical Demand	Kruskal-Wallis	*7.65*	.022	η^2 = .161	*
	A vs B (Dunn's)	—	.033	r = .468	*
	A vs C (Dunn's)	—	1.000	r = .082	ns

	B vs C (Dunn's)	—	.084	r = .457	ns
Temporal Demand	Kruskal-Wallis	*2.51*	.285	η^2 = .015	ns
	A vs B (Dunn's)	—	1.000	r = .111	ns
	A vs C (Dunn's)	—	.874	r = .218	ns
	B vs C (Dunn's)	—	.358	r = .283	ns
Performance	Kruskal-Wallis	*0.75*	.686	η^2 = .000	ns
	A vs B (Dunn's)	—	1.000	r = .161	ns
	A vs C (Dunn's)	—	1.000	r = .038	ns
	B vs C (Dunn's)	—	1.000	r = .120	ns
Effort	Kruskal-Wallis	*3.17*	.205	η^2 = .033	ns
	A vs B (Dunn's)	—	.909	r = .196	ns
	A vs C (Dunn's)	—	1.000	r = .147	ns
	B vs C (Dunn's)	—	.230	r = .348	ns
Frustration	Kruskal-Wallis	*0.94*	.626	η^2 = .000	ns
	A vs B (Dunn's)	—	1.000	r = .181	ns
	A vs C (Dunn's)	—	1.000	r = .049	ns
	B vs C (Dunn's)	—	1.000	r = .141	ns

*Note. Dunn's post-hoc with Bonferroni correction. η^2 = effect size for Kruskal-Wallis (small ≥ .01, medium ≥ .06, large ≥ .14); r = Mann-Whitney effect size for pairwise comparisons (small ≥ .1, medium ≥ .3, large ≥ .5). * p < .05, ** p < .01, ** p < .001; ns = not significant.*